W1 Practical (In Depth)

> Evidence File can be found in Politemall.

Structure of a Generic MBR (Pulled from Lecture Slides)

Structure of a Generic MBR

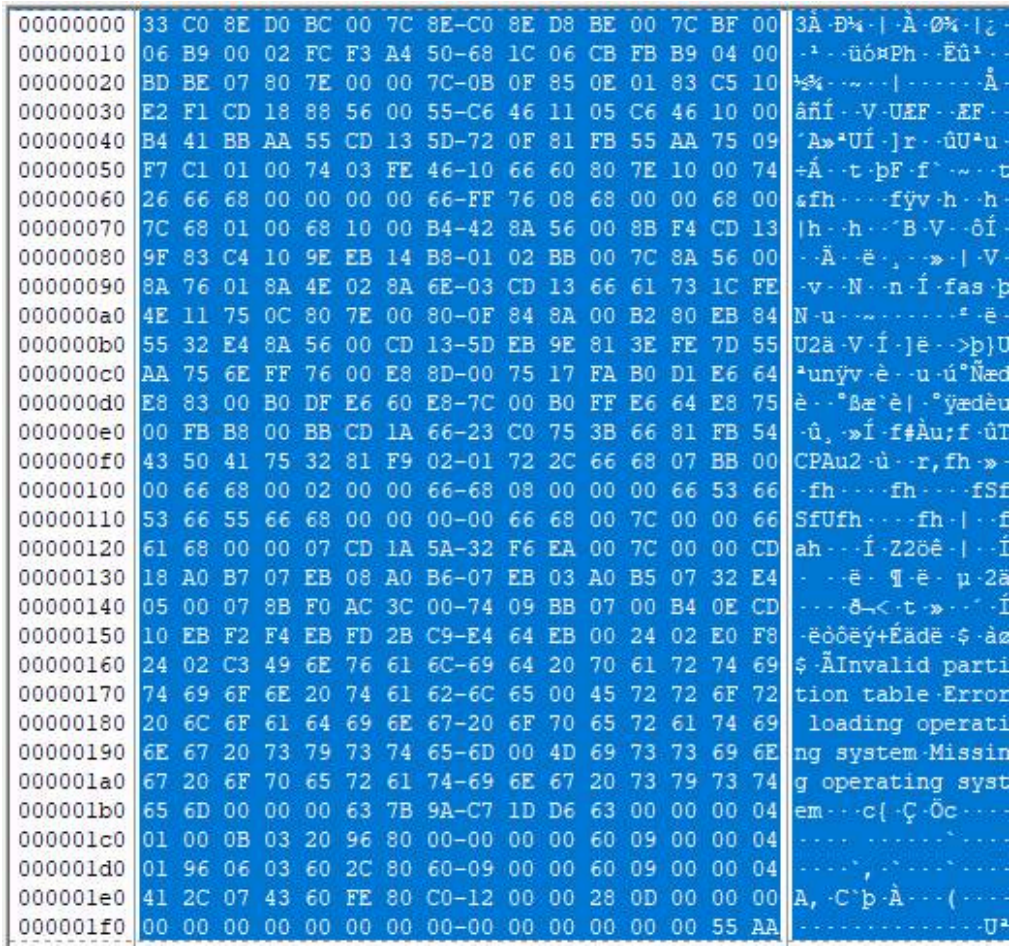
Offset	Description	Size in bytes
0x000	Bootstrap Code Area	446
0x1BE	Partition entry #1	16
0x1CE	Partition entry #2	16
0x1DE	Partition entry #3	16
0x1EE	Partition entry #4	16
0x1FE	0x55	1
0x1FF	0xAA	1

How to read the hex properly?

I will be using 0x1BE to describe it.

- 0x - Nice way to write hex code
- 1B - look for Row 1B in the MBR. Usually it will be like 0000001b0 (in FTK Imager).
- E - Look for Column E. (Remember hex is 16 char AND STARTS from 0). So it is 0-9, then A-F.

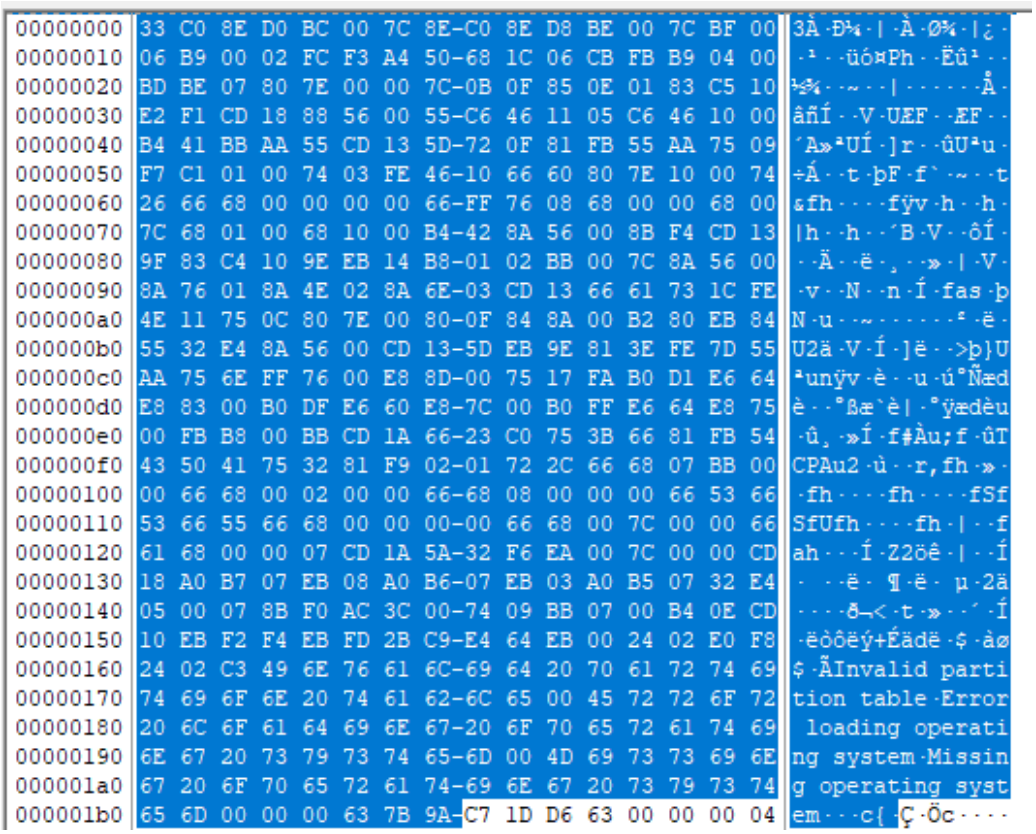
MBR (E01 File - How to Identify)



```
00000000 33 C0 8E D0 BC 00 7C 8E-C0 8E D8 BE 00 7C BF 00 3À·Ð¼·|·À·Ø¾·|¿·
00000010 06 B9 00 02 FC F3 A4 50-68 1C 06 CB FB B9 04 00 ·¹··üó¤Ph··Ëû¹··
00000020 BD BE 07 80 7E 00 00 7C-0B 0F 85 0E 01 83 C5 10 ½¾·€~··|·····ƒÅ·
00000030 E2 F1 CD 18 88 56 00 55-C6 46 11 05 C6 46 10 00 âñÍ··V·UÆF··ÆF··
00000040 B4 41 BB AA 55 CD 13 5D-72 0F 81 FB 55 AA 75 09 ´A»ªUÍ·]r··ûUªu·
00000050 F7 C1 01 00 74 03 FE 46-10 66 60 80 7E 10 00 74 ÷Á··t·þF·f`·~···t
00000060 26 66 68 00 00 00 00 66-FF 76 08 68 00 00 68 00 &fh····fÿv·h··h·
00000070 7C 68 01 00 68 10 00 B4-42 8A 56 00 8B F4 CD 13 |h··h··´B·V··ôÍ·
00000080 9F 83 C4 10 9E EB 14 B8-01 02 BB 00 7C 8A 56 00 ·ƒÄ··ë·¸··»·|·V·
00000090 8A 76 01 8A 4E 02 8A 6E-03 CD 13 66 61 73 1C FE ·v·N··n·Í·fas·þ
000000a0 4E 11 75 0C 80 7E 00 80-0F 84 8A 00 B2 80 EB 84 N·u··€~·····ª·ë·
000000b0 55 32 E4 8A 56 00 CD 13-5D EB 9E 81 3E FE 7D 55 U2ä·V·Í·]ë··>þ}U
000000c0 AA 75 6E FF 76 00 E8 8D-00 75 17 FA B0 D1 E6 64 ªunÿv·è··u·ú°Ñæd
000000d0 E8 83 00 B0 DF E6 60 E8-7C 00 B0 FF E6 64 E8 75 è··°ßæ`è|·°ÿædèu
000000e0 00 FB B8 00 BB CD 1A 66-23 C0 75 3B 66 81 FB 54 ·û¸·»Í·f#Àu;f·ûT
000000f0 43 50 41 75 32 81 F9 02-01 72 2C 66 68 07 BB 00 CPAu2·ù··r,fh·»·
00000100 00 66 68 00 02 00 00 66-68 08 00 00 00 66 53 66 ·fh····fh····fSf
00000110 53 66 55 66 68 00 00 00-00 66 68 00 7C 00 00 66 SfUfh····fh·|··f
00000120 61 68 00 00 07 CD 1A 5A-32 F6 EA 00 7C 00 00 CD ah···Í·Z2öê·|··Í
00000130 18 A0 B7 07 EB 08 A0 B6-07 EB 03 A0 B5 07 32 E4 ··ë·¶·ë·µ·2ä
00000140 05 00 07 8B F0 AC 3C 00-74 09 BB 07 00 B4 0E CD ····ð¬<·t·»··´·Í
00000150 10 EB F2 F4 EB FD 2B C9-E4 64 EB 00 24 02 E0 F8 ·ëòôëý+Éädë·$·àø
00000160 24 02 C3 49 6E 76 61 6C-69 64 20 70 61 72 74 69 $·ÃInvalid parti
00000170 74 69 6F 6E 20 74 61 62-6C 65 00 45 72 72 6F 72 tion table·Error
00000180 20 6C 6F 61 64 69 6E 67-20 6F 70 65 72 61 74 69 loading operati
00000190 6E 67 20 73 79 73 74 65-6D 00 4D 69 73 73 69 6E ng system·Missin
000001a0 67 20 6F 70 65 72 61 74-69 6E 67 20 73 79 73 74 g operating syst
000001b0 65 6D 00 00 00 63 7B 9A-C7 1D D6 63 00 00 00 04 em···c{·Ç·Öc····
000001c0 01 00 0B 03 20 96 80 00-00 00 00 60 09 00 00 04 ···· ·····`····
000001d0 01 96 06 03 60 2C 80 60-09 00 00 60 09 00 00 04 ····`,·`···`····
000001e0 41 2C 07 43 60 FE 80 C0-12 00 00 28 0D 00 00 00 A,·C`þ·À···(····
000001f0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 55 AA ··············U²
```

MBR will start from 0x00 to 0x1FF (512 bytes) - always ends @ 55 AA (Boot Signature).

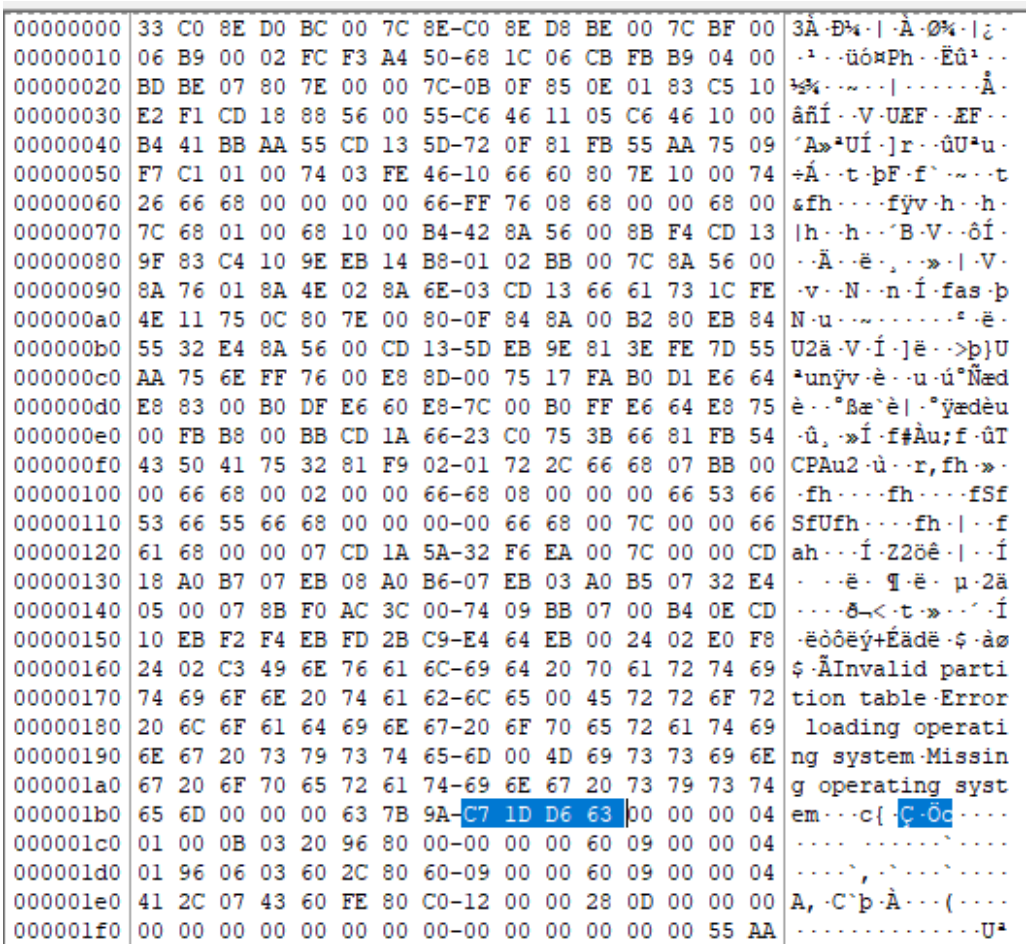
Bootstrap Code Area (440 bytes)



```
00000000  33 C0 8E D0 BC 00 7C 8E-C0 8E D8 BE 00 7C BF 00  3À·Ð¼·|·À·Ø¾·|¿·
00000010  06 B9 00 02 FC F3 A4 50-68 1C 06 CB FB B9 04 00  ·¹··üó¤Ph··ËÛ¹··
00000020  BD BE 07 80 7E 00 00 7C-0B 0F 85 0E 01 83 C5 10  ½¾·€~··|····…·ƒÅ·
00000030  E2 F1 CD 18 88 56 00 55-C6 46 11 05 C6 46 10 00  âñÍ··V·UÆF··ÆF··
00000040  B4 41 BB AA 55 CD 13 5D-72 0F 81 FB 55 AA 75 09  ´A»ªUÍ·]r··ûUªu·
00000050  F7 C1 01 00 74 03 FE 46-10 66 60 80 7E 10 00 74  ÷Á··t·þF·f`·~··t
00000060  26 66 68 00 00 00 00 66-FF 76 08 68 00 00 68 00  &fh····fÿv·h··h·
00000070  7C 68 01 00 68 10 00 B4-42 8A 56 00 8B F4 CD 13  |h··h··´B·V··ôÍ·
00000080  9F 83 C4 10 9E EB 14 B8-01 02 BB 00 7C 8A 56 00  ·ƒÄ··ë·¸··»·|·V·
00000090  8A 76 01 8A 4E 02 8A 6E-03 CD 13 66 61 73 1C FE  ·v··N··n·Í·fas·þ
000000a0  4E 11 75 0C 80 7E 00 80-0F 84 8A 00 B2 80 EB 84  N·u··€·····„·²·ë·
000000b0  55 32 E4 8A 56 00 CD 13-5D EB 9E 81 3E FE 7D 55  U2ä·V·Í·]ë··>þ}U
000000c0  AA 75 6E FF 76 00 E8 8D-00 75 17 FA B0 D1 E6 64  ªunÿv·è··u·ú°Ñæd
000000d0  E8 83 00 B0 DF E6 60 E8-7C 00 B0 FF E6 64 E8 75  è··°ßæ`è|·°ÿædèu
000000e0  00 FB B8 00 BB CD 1A 66-23 C0 75 3B 66 81 FB 54  ·û¸·»Í·f#Àu;f·ûT
000000f0  43 50 41 75 32 81 F9 02-01 72 2C 66 68 07 BB 00  CPAu2·ù··r,fh·»·
00000100  00 66 68 00 02 00 00 66-68 08 00 00 00 66 53 66  ·fh····fh····fSf
00000110  53 66 55 66 68 00 00 00-00 66 68 00 7C 00 00 66  SfUfh····fh·|··f
00000120  61 68 00 00 07 CD 1A 5A-32 F6 EA 00 7C 00 00 CD  ah···Í·Z2öê·|··Í
00000130  18 A0 B7 07 EB 08 A0 B6-07 EB 03 A0 B5 07 32 E4  ·  ··ë· ¶·ë· µ·2ä
00000140  05 00 07 8B F0 AC 3C 00-74 09 BB 07 00 B4 0E CD  ····‹ð¬<·t·»··´·Í
00000150  10 EB F2 F4 EB FD 2B C9-E4 64 EB 00 24 02 E0 F8  ·ëòôëý+Éädë·$·àø
00000160  24 02 C3 49 6E 76 61 6C-69 64 20 70 61 72 74 69  $·ÃInvalid parti
00000170  74 69 6F 6E 20 74 61 62-6C 65 00 45 72 72 6F 72  tion table·Error
00000180  20 6C 6F 61 64 69 6E 67-20 6F 70 65 72 61 74 69   loading operati
00000190  6E 67 20 73 79 73 74 65-6D 00 4D 69 73 73 69 6E  ng system·Missin
000001a0  67 20 6F 70 65 72 61 74-69 6E 67 20 73 79 73 74  g operating syst
000001b0  65 6D 00 00 00 63 7B 9A-C7 1D D6 63 00 00 00 04  em···c{·Ç·Öc····
```

Bootstrap Code Area spans from 0x00 to 0x1B7. It is 440 bytes.

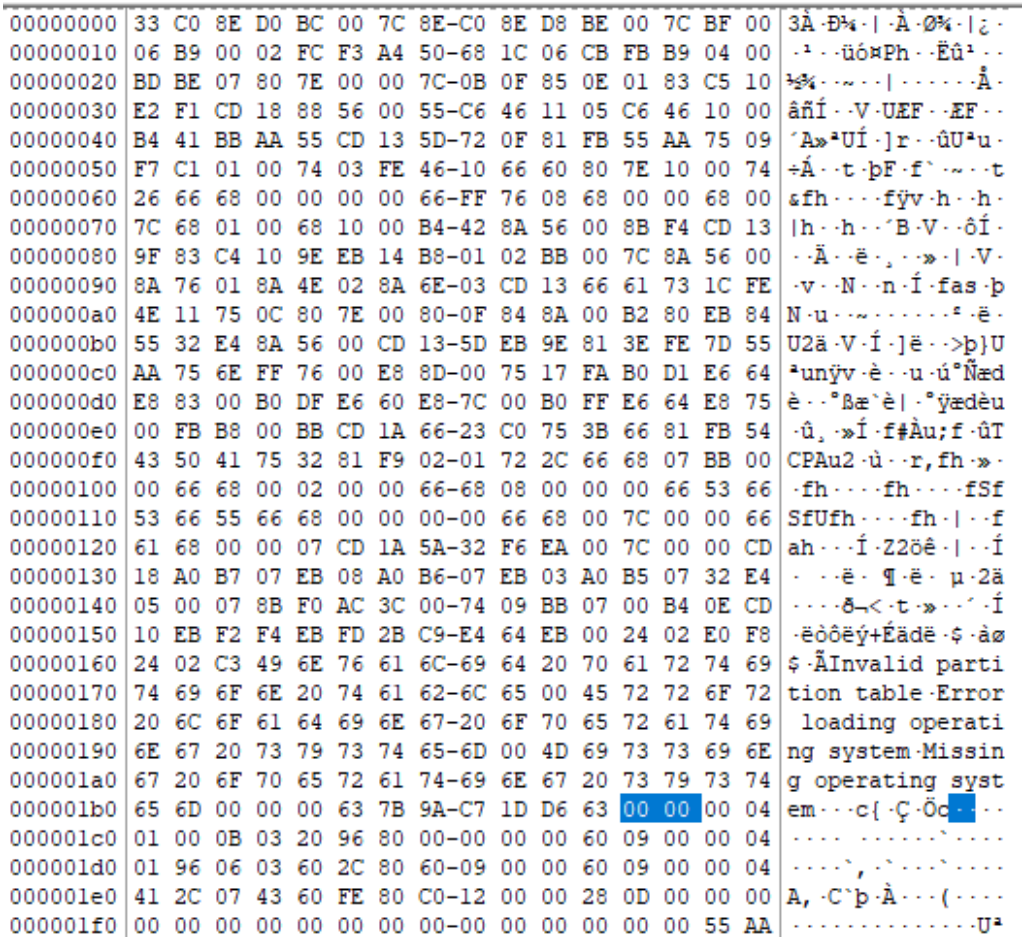
Disk ID (4 bytes)



```
00000000 33 C0 8E D0 BC 00 7C 8E-C0 8E D8 BE 00 7C BF 00 3À·Ð¼·|·À·Ø¾·|¿·
00000010 06 B9 00 02 FC F3 A4 50-68 1C 06 CB FB B9 04 00 ·¹··üó¤Ph··ËÛ¹··
00000020 BD BE 07 80 7E 00 00 7C-0B 0F 85 0E 01 83 C5 10 ½¾·€~··|·····ƒÅ·
00000030 E2 F1 CD 18 88 56 00 55-C6 46 11 05 C6 46 10 00 âñÍ··V·UÆF··ÆF··
00000040 B4 41 BB AA 55 CD 13 5D-72 0F 81 FB 55 AA 75 09 ´A»ªUÍ·]r··ûUªu·
00000050 F7 C1 01 00 74 03 FE 46-10 66 60 80 7E 10 00 74 ÷Á··t·þF·f`·~··t
00000060 26 66 68 00 00 00 00 66-FF 76 08 68 00 00 68 00 &fh····fÿv·h··h·
00000070 7C 68 01 00 68 10 00 B4-42 8A 56 00 8B F4 CD 13 |h··h··´BŠV·‹ôÍ·
00000080 9F 83 C4 10 9E EB 14 B8-01 02 BB 00 7C 8A 56 00 ·ƒÄ··ë·¸··»·|·V·
00000090 8A 76 01 8A 4E 02 8A 6E-03 CD 13 66 61 73 1C FE ·v··N··n·Í·fas·þ
000000a0 4E 11 75 0C 80 7E 00 80-0F 84 8A 00 B2 80 EB 84 N·u··€~·····ª·ë·
000000b0 55 32 E4 8A 56 00 CD 13-5D EB 9E 81 3E FE 7D 55 U2ä·V·Í·]ë··>þ}U
000000c0 AA 75 6E FF 76 00 E8 8D-00 75 17 FA B0 D1 E6 64 ªunÿv·è··u·ú°Ñæd
000000d0 E8 83 00 B0 DF E6 60 E8-7C 00 B0 FF E6 64 E8 75 è··°ßæ`è|·°ÿædèu
000000e0 00 FB B8 00 BB CD 1A 66-23 C0 75 3B 66 81 FB 54 ·û¸·»Í·f#Àu;f·ûT
000000f0 43 50 41 75 32 81 F9 02-01 72 2C 66 68 07 BB 00 CPAu2·ù··r,fh·»·
00000100 00 66 68 00 02 00 00 66-68 08 00 00 00 66 53 66 ·fh····fh····fSf
00000110 53 66 55 66 68 00 00 00-00 66 68 00 7C 00 00 66 SfUfh····fh·|··f
00000120 61 68 00 00 07 CD 1A 5A-32 F6 EA 00 7C 00 00 CD ah···Í·Z2öê·|··Í
00000130 18 A0 B7 07 EB 08 A0 B6-07 EB 03 A0 B5 07 32 E4 ·  ··ë· ¶·ë· µ·2ä
00000140 05 00 07 8B F0 AC 3C 00-74 09 BB 07 00 B4 0E CD ····ð¬<·t·»··´·Í
00000150 10 EB F2 F4 EB FD 2B C9-E4 64 EB 00 24 02 E0 F8 ·ëòôëý+Éädë·$·àø
00000160 24 02 C3 49 6E 76 61 6C-69 64 20 70 61 72 74 69 $·ÃInvalid parti
00000170 74 69 6F 6E 20 74 61 62-6C 65 00 45 72 72 6F 72 tion table·Error
00000180 20 6C 6F 61 64 69 6E 67-20 6F 70 65 72 61 74 69  loading operati
00000190 6E 67 20 73 79 73 74 65-6D 00 4D 69 73 73 69 6E ng system·Missin
000001a0 67 20 6F 70 65 72 61 74-69 6E 67 20 73 79 73 74 g operating syst
000001b0 65 6D 00 00 00 63 7B 9A-C7 1D D6 63 00 00 00 04 em···c{·Ç·Öc····
000001c0 01 00 0B 03 20 96 80 00-00 00 00 60 09 00 00 04 ···· ·····`····
000001d0 01 96 06 03 60 2C 80 60-09 00 00 60 09 00 00 04 ····`,·`···`····
000001e0 41 2C 07 43 60 FE 80 C0-12 00 00 28 0D 00 00 00 A,·C`þ·À···(····
000001f0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 55 AA ··············Uª
```

Disk ID spans from 0x1B8 to 0x1BB. It is 4 bytes.

Reserved (2 bytes)



```
00000000 33 C0 8E D0 BC 00 7C 8E-C0 8E D8 BE 00 7C BF 00 3À·Ð¼·|·À·Ø¾·|¿·
00000010 06 B9 00 02 FC F3 A4 50-68 1C 06 CB FB B9 04 00 ·¹··üó¤Ph··ËÛ¹··
00000020 BD BE 07 80 7E 00 00 7C-0B 0F 85 0E 01 83 C5 10 ½¾·€~··|·····ƒÅ·
00000030 E2 F1 CD 18 88 56 00 55-C6 46 11 05 C6 46 10 00 âñÍ··V·UÆF··ÆF··
00000040 B4 41 BB AA 55 CD 13 5D-72 0F 81 FB 55 AA 75 09 ´A»ªUÍ·]r··ûUªu·
00000050 F7 C1 01 00 74 03 FE 46-10 66 60 80 7E 10 00 74 ÷Á··t·þF·f`·~··t
00000060 26 66 68 00 00 00 00 66-FF 76 08 68 00 00 68 00 &fh····fÿv·h··h·
00000070 7C 68 01 00 68 10 00 B4-42 8A 56 00 8B F4 CD 13 |h··h··´BŠV·‹ôÍ·
00000080 9F 83 C4 10 9E EB 14 B8-01 02 BB 00 7C 8A 56 00 ·ƒÄ··ë·¸··»·|·V·
00000090 8A 76 01 8A 4E 02 8A 6E-03 CD 13 66 61 73 1C FE ·v·ŠN·Šn·Í·fas·þ
000000a0 4E 11 75 0C 80 7E 00 80-0F 84 8A 00 B2 80 EB 84 N·u··€~·····Š·ë·
000000b0 55 32 E4 8A 56 00 CD 13-5D EB 9E 81 3E FE 7D 55 U2äŠV·Í·]ë··>þ}U
000000c0 AA 75 6E FF 76 00 E8 8D-00 75 17 FA B0 D1 E6 64 ªunÿv·è··u·ú°Ñæd
000000d0 E8 83 00 B0 DF E6 60 E8-7C 00 B0 FF E6 64 E8 75 è··°ßæ`è|·°ÿædèu
000000e0 00 FB B8 00 BB CD 1A 66-23 C0 75 3B 66 81 FB 54 ·û¸·»Í·f#Àu;f·ûT
000000f0 43 50 41 75 32 81 F9 02-01 72 2C 66 68 07 BB 00 CPAu2·ù··r,fh·»·
00000100 00 66 68 00 02 00 00 66-68 08 00 00 00 66 53 66 ·fh····fh····fSf
00000110 53 66 55 66 68 00 00 00-00 66 68 00 7C 00 00 66 SfUfh····fh·|··f
00000120 61 68 00 00 07 CD 1A 5A-32 F6 EA 00 7C 00 00 CD ah···Í·Z2öê·|··Í
00000130 18 A0 B7 07 EB 08 A0 B6-07 EB 03 A0 B5 07 32 E4 · ·ë· ¶·ë· µ·2ä
00000140 05 00 07 8B F0 AC 3C 00-74 09 BB 07 00 B4 0E CD ····ð¬<·t·»··´·Í
00000150 10 EB F2 F4 EB FD 2B C9-E4 64 EB 00 24 02 E0 F8 ·ëòôëý+Éädë·$·àø
00000160 24 02 C3 49 6E 76 61 6C-69 64 20 70 61 72 74 69 $·ÃInvalid parti
00000170 74 69 6F 6E 20 74 61 62-6C 65 00 45 72 72 6F 72 tion table·Error
00000180 20 6C 6F 61 64 69 6E 67-20 6F 70 65 72 61 74 69  loading operati
00000190 6E 67 20 73 79 73 74 65-6D 00 4D 69 73 73 69 6E ng system·Missin
000001a0 67 20 6F 70 65 72 61 74-69 6E 67 20 73 79 73 74 g operating syst
000001b0 65 6D 00 00 00 63 7B 9A-C7 1D D6 63 00 00 00 04 em···c{·Ç·Öc····
000001c0 01 00 0B 03 20 96 80 00-00 00 00 60 09 00 00 04 ···· ·····`····
000001d0 01 96 06 03 60 2C 80 60-09 00 00 60 09 00 00 04 ····`,·`···`····
000001e0 41 2C 07 43 60 FE 80 C0-12 00 00 28 0D 00 00 00 A,·C`þ·À···(····
000001f0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 55 AA ··············Uª
```

Reserved bytes span from 0x1BC to 0x1BD.

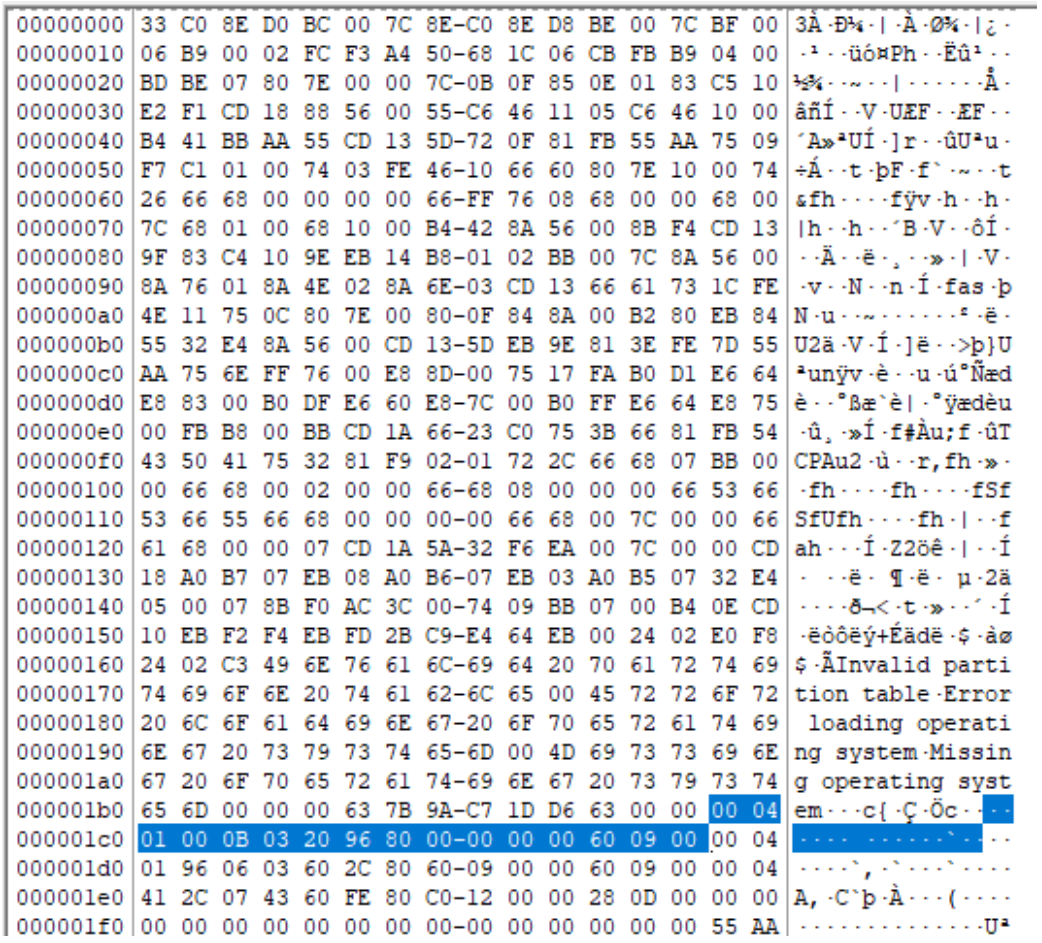
Reading Partition Entries

A Sample Partition Entry

Offset	Description	Size in bytes
0x0	0x80 Active or 0x00 Inactive	1
0x1	CHS address of the 1st sector in partition	3
0x4	Partition Type e.g. 0x04 means it is a FAT16 partition	1
0x5	CHS address of last sector in the partition	3
0x8	LBA of the 1st sector in the partition	4
0xC	Number of sectors in the partition	4

Partition Entry #1 (16 bytes)

> For future documents, I will be referring to Partition Entries as PE.



```
00000000  33 C0 8E D0 BC 00 7C 8E-C0 8E D8 BE 00 7C BF 00  3À·Ð¼·|·À·Ø¾·|¿·
00000010  06 B9 00 02 FC F3 A4 50-68 1C 06 CB FB B9 04 00  ·¹··üó¤Ph··Ëû¹··
00000020  BD BE 07 80 7E 00 00 7C-0B 0F 85 0E 01 83 C5 10  ½¾·~··|·····ÅÅ·
00000030  E2 F1 CD 18 88 56 00 55-C6 46 11 05 C6 46 10 00  âñÍ··V·UÆF··ÆF··
00000040  B4 41 BB AA 55 CD 13 5D-72 0F 81 FB 55 AA 75 09  ´A»ªUÍ·]r··ûUªu·
00000050  F7 C1 01 00 74 03 FE 46-10 66 60 80 7E 10 00 74  ÷Á··t·þF·f`·~··t
00000060  26 66 68 00 00 00 00 66-FF 76 08 68 00 00 68 00  &fh····fÿv·h··h·
00000070  7C 68 01 00 68 10 00 B4-42 8A 56 00 8B F4 CD 13  |h··h··´B·V··ôÍ·
00000080  9F 83 C4 10 9E EB 14 B8-01 02 BB 00 7C 8A 56 00  ··Ä··ë·¸··»·|·V·
00000090  8A 76 01 8A 4E 02 8A 6E-03 CD 13 66 61 73 1C FE  ·v··N··n·Í·fas·þ
000000a0  4E 11 75 0C 80 7E 00 80-0F 84 8A 00 B2 80 EB 84  N·u··~·····ª·ë·
000000b0  55 32 E4 8A 56 00 CD 13-5D EB 9E 81 3E FE 7D 55  U2ä·V·Í·]ë··>þ}U
000000c0  AA 75 6E FF 76 00 E8 8D-00 75 17 FA B0 D1 E6 64  ªunÿv·è··u·ú°Ñæd
000000d0  E8 83 00 B0 DF E6 60 E8-7C 00 B0 FF E6 64 E8 75  è··°ßæ`è|·°ÿædèu
000000e0  00 FB B8 00 BB CD 1A 66-23 C0 75 3B 66 81 FB 54  ·û¸·»Í·f#Àu;f·ûT
000000f0  43 50 41 75 32 81 F9 02-01 72 2C 66 68 07 BB 00  CPAu2·ù··r,fh·»·
00000100  00 66 68 00 02 00 00 66-68 08 00 00 00 66 53 66  ·fh····fh····fSf
00000110  53 66 55 66 68 00 00 00-00 66 68 00 7C 00 00 66  SfUfh····fh·|··f
00000120  61 68 00 00 07 CD 1A 5A-32 F6 EA 00 7C 00 00 CD  ah···Í·Z2öê·|··Í
00000130  18 A0 B7 07 EB 08 A0 B6-07 EB 03 A0 B5 07 32 E4  · ··ë· ¶·ë· µ·2ä
00000140  05 00 07 8B F0 AC 3C 00-74 09 BB 07 00 B4 0E CD  ····ð¬<·t·»··´·Í
00000150  10 EB F2 F4 EB FD 2B C9-E4 64 EB 00 24 02 E0 F8  ·ëòôëý+Éädë·$·àø
00000160  24 02 C3 49 6E 76 61 6C-69 64 20 70 61 72 74 69  $·ÃInvalid parti
00000170  74 69 6F 6E 20 74 61 62-6C 65 00 45 72 72 6F 72  tion table·Error
00000180  20 6C 6F 61 64 69 6E 67-20 6F 70 65 72 61 74 69   loading operati
00000190  6E 67 20 73 79 73 74 65-6D 00 4D 69 73 73 69 6E  ng system·Missin
000001a0  67 20 6F 70 65 72 61 74-69 6E 67 20 73 79 73 74  g operating syst
000001b0  65 6D 00 00 00 63 7B 9A-C7 1D D6 63 00 00 00 04  em···c{·Ç·Öc····
000001c0  01 00 0B 03 20 96 80 00-00 00 00 60 09 00 00 04  ···· ·····`····
000001d0  01 96 06 03 60 2C 80 60-09 00 00 60 09 00 00 04  ····`,·`···`····
000001e0  41 2C 07 43 60 FE 80 C0-12 00 00 28 0D 00 00 00  A,·C`þ·À···(····
000001f0  00 00 00 00 00 00 00 00-00 00 00 00 00 00 55 AA  ··············Uª
```

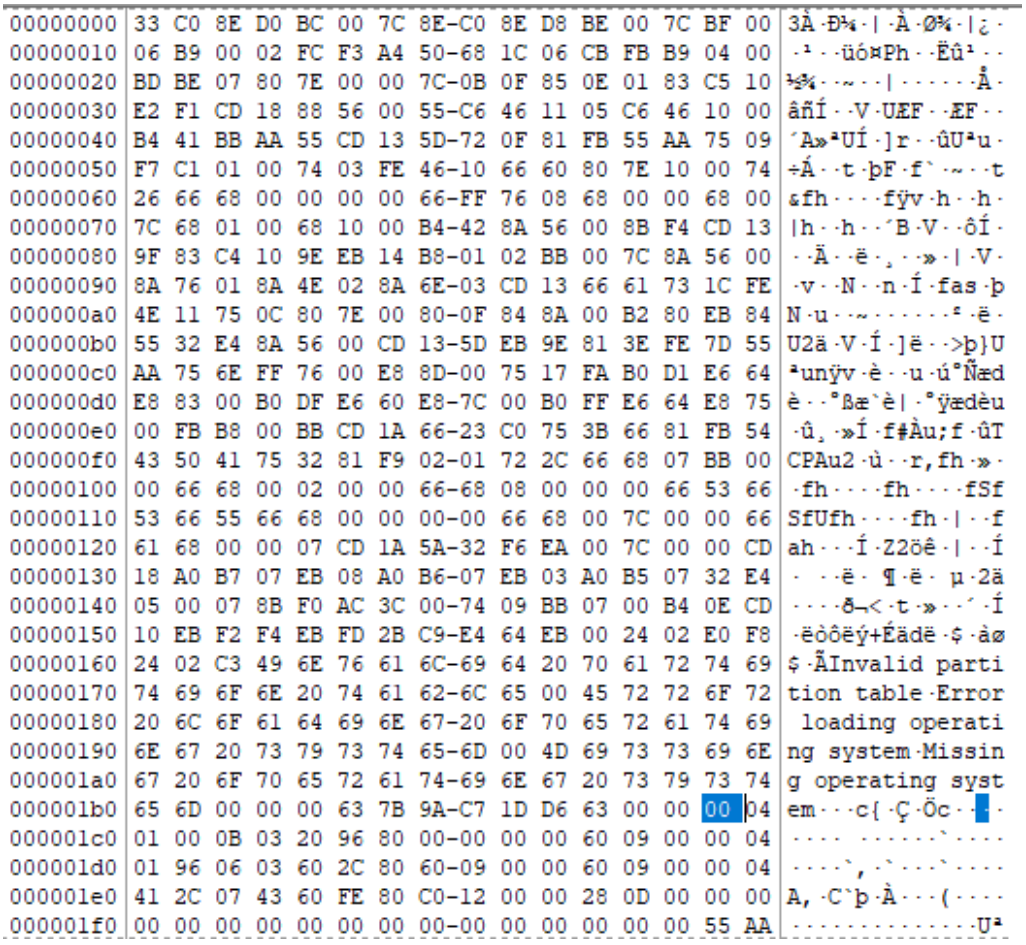
Partition Entry #1 (PE1) spans from 0x1BE to 0x1CD.

> NOTE: The above Sample Partition Entries 0x00 DIFFERS from the 0x00 in the START of the MBR.

Analyzing Partition Entry

0x00 - 1 byte

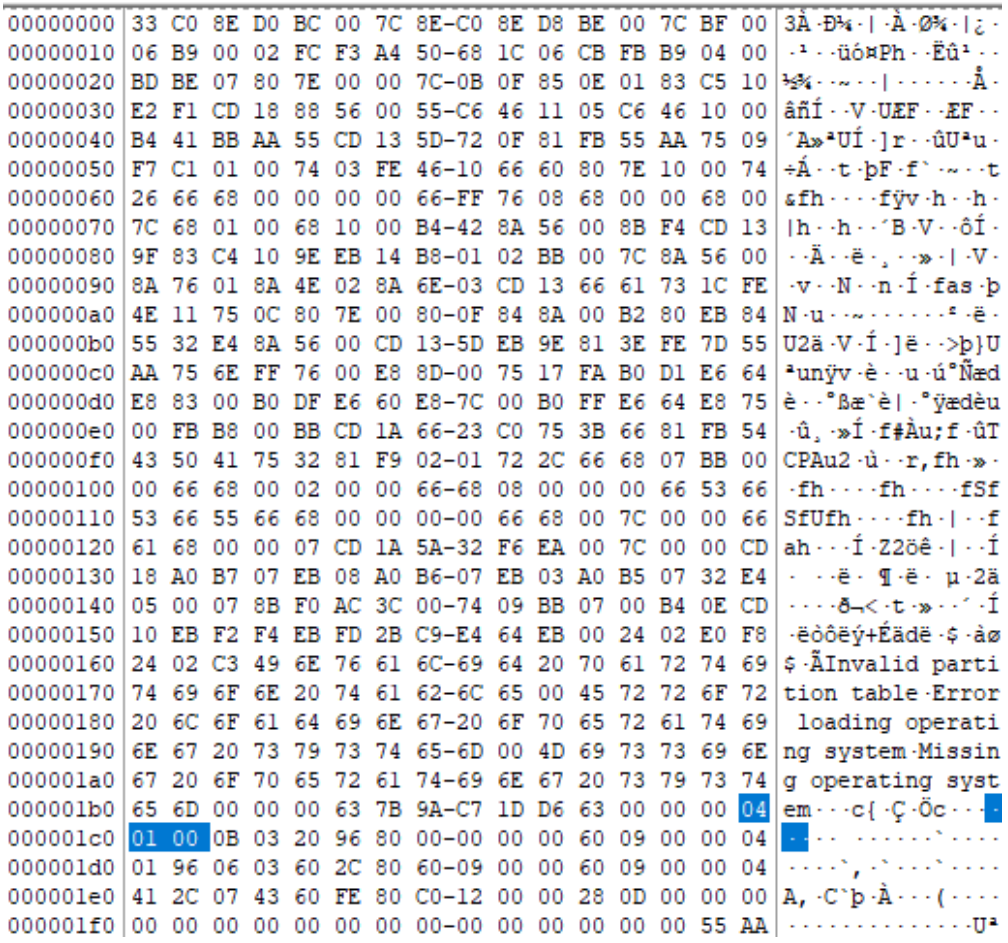
In this partition, 0x00 starts from 0x1BE (MBR).



```
00000000 33 C0 8E D0 BC 00 7C 8E-C0 8E D8 BE 00 7C BF 00  3À·Ð¼·|·À·Ø¾·|¿·
00000010 06 B9 00 02 FC F3 A4 50-68 1C 06 CB FB B9 04 00  ·¹··üó¤Ph··ËÛ¹··
00000020 BD BE 07 80 7E 00 00 7C-0B 0F 85 0E 01 83 C5 10  ½¾·€~··|·····ƒÅ·
00000030 E2 F1 CD 18 88 56 00 55-C6 46 11 05 C6 46 10 00  âñÍ··V·UÆF··ÆF··
00000040 B4 41 BB AA 55 CD 13 5D-72 0F 81 FB 55 AA 75 09  ´A»ªUÍ·]r··ûUªu·
00000050 F7 C1 01 00 74 03 FE 46-10 66 60 80 7E 10 00 74  ÷Á··t·þF·f`·~··t
00000060 26 66 68 00 00 00 00 66-FF 76 08 68 00 00 68 00  &fh····fÿv·h··h·
00000070 7C 68 01 00 68 10 00 B4-42 8A 56 00 8B F4 CD 13  |h··h··´B·V··ôÍ·
00000080 9F 83 C4 10 9E EB 14 B8-01 02 BB 00 7C 8A 56 00  ··Ä··ë·¸··»·|·V·
00000090 8A 76 01 8A 4E 02 8A 6E-03 CD 13 66 61 73 1C FE  ·v··N··n·Í·fas·þ
000000a0 4E 11 75 0C 80 7E 00 80-0F 84 8A 00 B2 80 EB 84  N·u··€~·····ª·ë·
000000b0 55 32 E4 8A 56 00 CD 13-5D EB 9E 81 3E FE 7D 55  U2ä·V·Í·]ë··>þ}U
000000c0 AA 75 6E FF 76 00 E8 8D-00 75 17 FA B0 D1 E6 64  ªunÿv·è··u·ú°Ñæd
000000d0 E8 83 00 B0 DF E6 60 E8-7C 00 B0 FF E6 64 E8 75  è··°ßæ`è|·°ÿædèu
000000e0 00 FB B8 00 BB CD 1A 66-23 C0 75 3B 66 81 FB 54  ·û¸·»Í·f#Àu;f·ûT
000000f0 43 50 41 75 32 81 F9 02-01 72 2C 66 68 07 BB 00  CPAu2·ù··r,fh·»·
00000100 00 66 68 00 02 00 00 66-68 08 00 00 00 66 53 66  ·fh····fh····fSf
00000110 53 66 55 66 68 00 00 00-00 66 68 00 7C 00 00 66  SfUfh····fh·|··f
00000120 61 68 00 00 07 CD 1A 5A-32 F6 EA 00 7C 00 00 CD  ah···Í·Z2öê·|··Í
00000130 18 A0 B7 07 EB 08 A0 B6-07 EB 03 A0 B5 07 32 E4  · ··ë· ¶·ë· µ·2ä
00000140 05 00 07 8B F0 AC 3C 00-74 09 BB 07 00 B4 0E CD  ····ð¬<·t·»··´·Í
00000150 10 EB F2 F4 EB FD 2B C9-E4 64 EB 00 24 02 E0 F8  ·ëòôëý+Éädë·$·àø
00000160 24 02 C3 49 6E 76 61 6C-69 64 20 70 61 72 74 69  $·ÃInvalid parti
00000170 74 69 6F 6E 20 74 61 62-6C 65 00 45 72 72 6F 72  tion table·Error
00000180 20 6C 6F 61 64 69 6E 67-20 6F 70 65 72 61 74 69   loading operati
00000190 6E 67 20 73 79 73 74 65-6D 00 4D 69 73 73 69 6E  ng system·Missin
000001a0 67 20 6F 70 65 72 61 74-69 6E 67 20 73 79 73 74  g operating syst
000001b0 65 6D 00 00 00 63 7B 9A-C7 1D D6 63 00 00 00 04  em···c{·Ç·Öc····
000001c0 01 00 0B 03 20 96 80 00-00 00 00 60 09 00 00 04  ···· ·····`····
000001d0 01 96 06 03 60 2C 80 60-09 00 00 60 09 00 00 04  ····`,·`···`····
000001e0 41 2C 07 43 60 FE 80 C0-12 00 00 28 0D 00 00 00  A,·C`þ·À···(····
000001f0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 55 AA  ··············U ª
```

Since it shows 00, it is an INACTIVE partition. If it shows 08, it will be active.

CHS Address (0x1) - 3 bytes



```
00000000 33 C0 8E D0 BC 00 7C 8E-C0 8E D8 BE 00 7C BF 00  3À·Ð¼·|·À·Ø¾·|¿·
00000010 06 B9 00 02 FC F3 A4 50-68 1C 06 CB FB B9 04 00  ·¹··üó¤Ph··Ëû¹··
00000020 BD BE 07 80 7E 00 00 7C-0B 0F 85 0E 01 83 C5 10  ½¾·€~··|·····ƒÅ·
00000030 E2 F1 CD 18 88 56 00 55-C6 46 11 05 C6 46 10 00  âñÍ··V·UÆF··ÆF··
00000040 B4 41 BB AA 55 CD 13 5D-72 0F 81 FB 55 AA 75 09  ´A»ªUÍ·]r··ûUªu·
00000050 F7 C1 01 00 74 03 FE 46-10 66 60 80 7E 10 00 74  ÷Á··t·þF·f`·~··t
00000060 26 66 68 00 00 00 00 66-FF 76 08 68 00 00 68 00  &fh····fÿv·h··h·
00000070 7C 68 01 00 68 10 00 B4-42 8A 56 00 8B F4 CD 13  |h··h··´B·V··ôÍ·
00000080 9F 83 C4 10 9E EB 14 B8-01 02 BB 00 7C 8A 56 00  ··Ä··ë·¸··»·|·V·
00000090 8A 76 01 8A 4E 02 8A 6E-03 CD 13 66 61 73 1C FE  ·v·N·n·Í·fas·þ
000000a0 4E 11 75 0C 80 7E 00 80-0F 84 8A 00 B2 80 EB 84  N·u··~·····ª·ë·
000000b0 55 32 E4 8A 56 00 CD 13-5D EB 9E 81 3E FE 7D 55  U2ä·V·Í·]ë··>þ}U
000000c0 AA 75 6E FF 76 00 E8 8D-00 75 17 FA B0 D1 E6 64  ªunÿv·è··u·ú°Ñæd
000000d0 E8 83 00 B0 DF E6 60 E8-7C 00 B0 FF E6 64 E8 75  è··°ßæ`è|·°ÿædèu
000000e0 00 FB B8 00 BB CD 1A 66-23 C0 75 3B 66 81 FB 54  ·û¸·»Í·f#Àu;f·ûT
000000f0 43 50 41 75 32 81 F9 02-01 72 2C 66 68 07 BB 00  CPAu2·ù··r,fh·»·
00000100 00 66 68 00 02 00 00 66-68 08 00 00 00 66 53 66  ·fh····fh····fSf
00000110 53 66 55 66 68 00 00 00-00 66 68 00 7C 00 00 66  SfUfh····fh·|··f
00000120 61 68 00 00 07 CD 1A 5A-32 F6 EA 00 7C 00 00 CD  ah···Í·Z2öê·|··Í
00000130 18 A0 B7 07 EB 08 A0 B6-07 EB 03 A0 B5 07 32 E4  · ··ë· ¶·ë· µ·2ä
00000140 05 00 07 8B F0 AC 3C 00-74 09 BB 07 00 B4 0E CD  ····ð¬<·t·»··´·Í
00000150 10 EB F2 F4 EB FD 2B C9-E4 64 EB 00 24 02 E0 F8  ·ëòôëý+Éädë·$·àø
00000160 24 02 C3 49 6E 76 61 6C-69 64 20 70 61 72 74 69  $·ÃInvalid parti
00000170 74 69 6F 6E 20 74 61 62-6C 65 00 45 72 72 6F 72  tion table·Error
00000180 20 6C 6F 61 64 69 6E 67-20 6F 70 65 72 61 74 69   loading operati
00000190 6E 67 20 73 79 73 74 65-6D 00 4D 69 73 73 69 6E  ng system·Missin
000001a0 67 20 6F 70 65 72 61 74-69 6E 67 20 73 79 73 74  g operating syst
000001b0 65 6D 00 00 00 63 7B 9A-C7 1D D6 63 00 00 00 04  em···c{·Ç·Öc····
000001c0 01 00 0B 03 20 96 80 00-00 00 00 60 09 00 00 04  ···· ·····`····
000001d0 01 96 06 03 60 2C 80 60-09 00 00 60 09 00 00 04  ····`,·`···`····
000001e0 41 2C 07 43 60 FE 80 C0-12 00 00 28 0D 00 00 00  A,·C`þ·À···(····
000001f0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 55 AA  ··············U ª
```

The CHS Address starts from 0x1 in Partition OR 0x1BF (MBR) and ends at 0x1C1 (MBR). It is 3 bytes.

Purpose: **CHS** stands for **Cylinder-Head-Sector**. It is an old-school geometry mapping method used by the BIOS to find the physical location of the partition's very first sector on the disk platter.

Partition Type (0x4) - 1 byte

> Below are the types of Partitions.

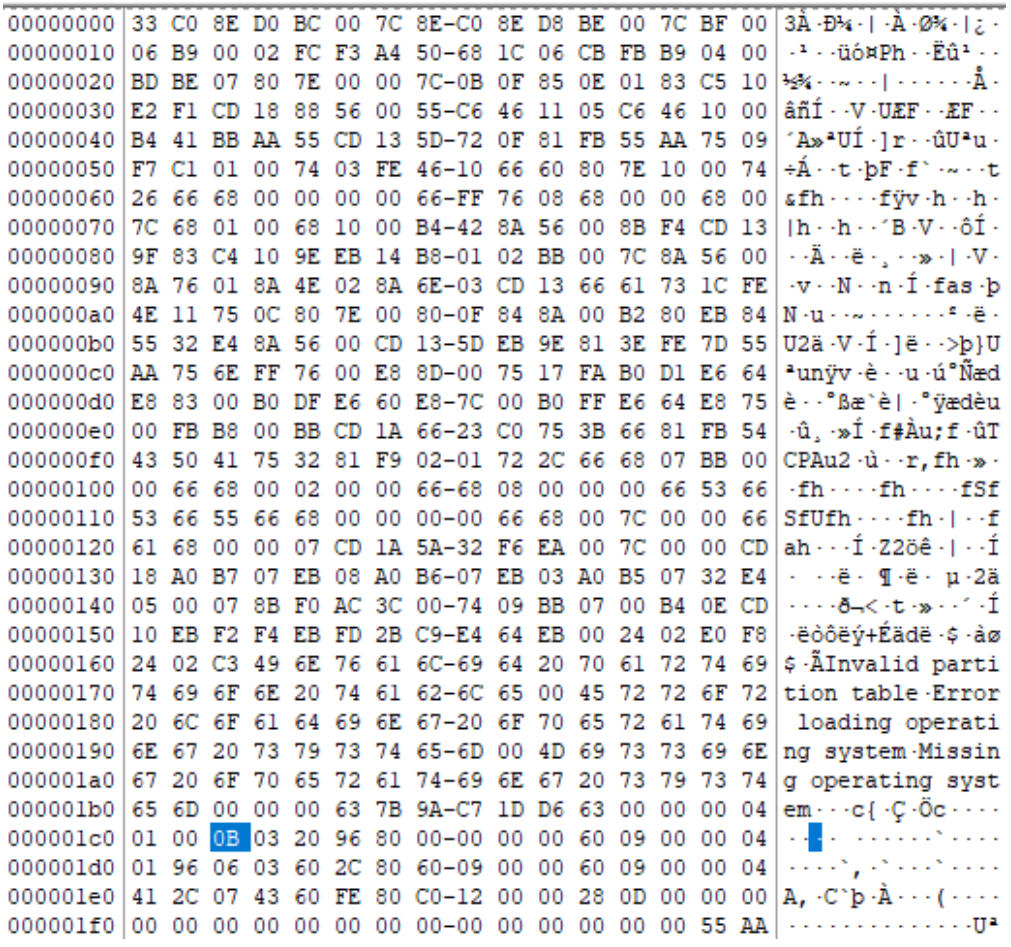
Types of Partitions

Some common types of partition are

- 0x01 FAT12
- 0x04 FAT16
- 0x05 FAT16
- 0X06 FAT16B
- 0x07 NTFS, exFAT
- 0x0B FAT32 with CHS addressing
- 0x0C FAT32 with LBA addressing
- 0x0E FAT16B with LBA
- 0x82 Linux Swap Space
- 0x83 Linux file system Ext2, Ext3
- 0x8E Linux LVM
- 0xAF Apple HFS, HFS Plus

Please read the full set of types of partition at https://en.wikipedia.org/wiki/Partition_type

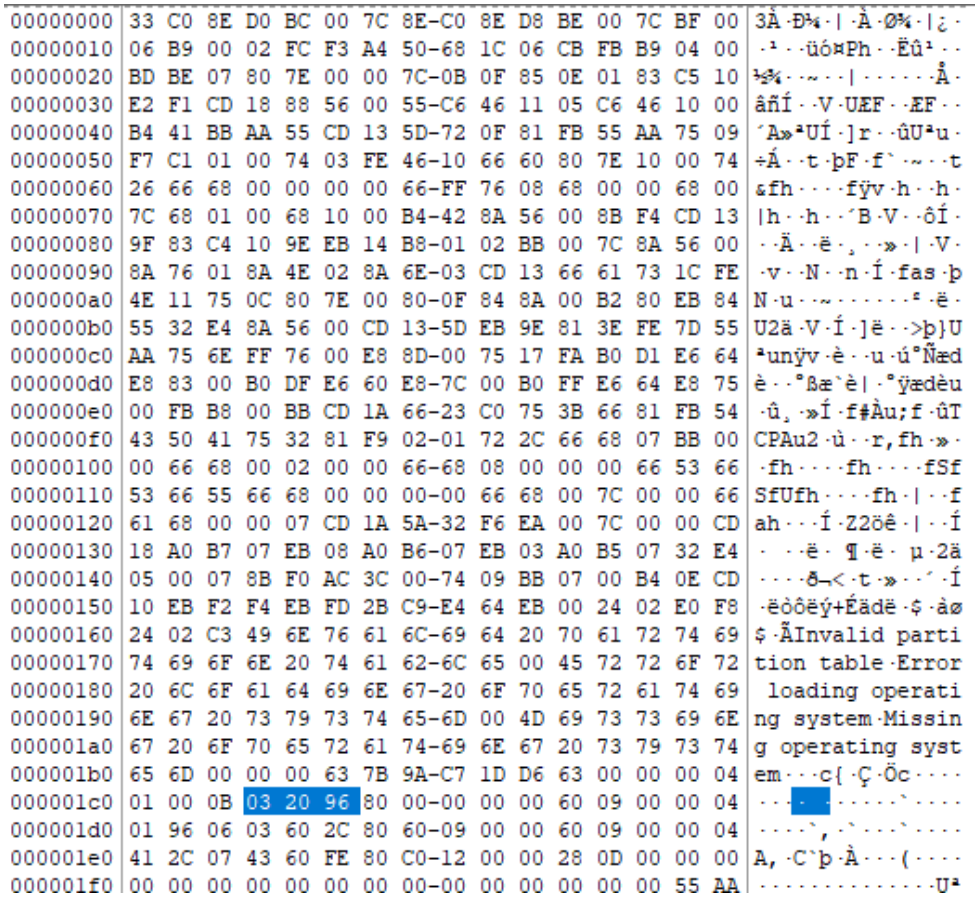
Below is E01 (#PE1) Partition Type.



```
00000000  33 C0 8E D0 BC 00 7C 8E-C0 8E D8 BE 00 7C BF 00  3À·Ð¼·|·À·Ø¾·|¿·
00000010  06 B9 00 02 FC F3 A4 50-68 1C 06 CB FB B9 04 00  ·¹··üó¤Ph··Ëû¹··
00000020  BD BE 07 80 7E 00 00 7C-0B 0F 85 0E 01 83 C5 10  ½¾·€~··|·····ƒÅ·
00000030  E2 F1 CD 18 88 56 00 55-C6 46 11 05 C6 46 10 00  âñÍ··V·UÆF··ÆF··
00000040  B4 41 BB AA 55 CD 13 5D-72 0F 81 FB 55 AA 75 09  ´A»ªUÍ·]r··ûUªu·
00000050  F7 C1 01 00 74 03 FE 46-10 66 60 80 7E 10 00 74  ÷Á··t·þF·f`·~··t
00000060  26 66 68 00 00 00 00 66-FF 76 08 68 00 00 68 00  &fh····fÿv·h··h·
00000070  7C 68 01 00 68 10 00 B4-42 8A 56 00 8B F4 CD 13  |h··h··´BŠV·‹ôÍ·
00000080  9F 83 C4 10 9E EB 14 B8-01 02 BB 00 7C 8A 56 00  ··Ä··ë·¸··»·|·V·
00000090  8A 76 01 8A 4E 02 8A 6E-03 CD 13 66 61 73 1C FE  ·v··N··n·Í·fas·þ
000000a0  4E 11 75 0C 80 7E 00 80-0F 84 8A 00 B2 80 EB 84  N·u··€~·····Š·ë·
000000b0  55 32 E4 8A 56 00 CD 13-5D EB 9E 81 3E FE 7D 55  U2ä·V·Í·]ë··>þ}U
000000c0  AA 75 6E FF 76 00 E8 8D-00 75 17 FA B0 D1 E6 64  ªunÿv·è··u·ú°Ñæd
000000d0  E8 83 00 B0 DF E6 60 E8-7C 00 B0 FF E6 64 E8 75  è··°ßæ`è|·°ÿædèu
000000e0  00 FB B8 00 BB CD 1A 66-23 C0 75 3B 66 81 FB 54  ·û¸·»Í·f#Àu;f·ûT
000000f0  43 50 41 75 32 81 F9 02-01 72 2C 66 68 07 BB 00  CPAu2·ù··r,fh·»·
00000100  00 66 68 00 02 00 00 66-68 08 00 00 00 66 53 66  ·fh····fh····fSf
00000110  53 66 55 66 68 00 00 00-00 66 68 00 7C 00 00 66  SfUfh····fh·|··f
00000120  61 68 00 00 07 CD 1A 5A-32 F6 EA 00 7C 00 00 CD  ah···Í·Z2öê·|··Í
00000130  18 A0 B7 07 EB 08 A0 B6-07 EB 03 A0 B5 07 32 E4  · ·ë· ¶·ë· µ·2ä
00000140  05 00 07 8B F0 AC 3C 00-74 09 BB 07 00 B4 0E CD  ····ð¬<·t·»··´·Í
00000150  10 EB F2 F4 EB FD 2B C9-E4 64 EB 00 24 02 E0 F8  ·ëòôëý+Éädë·$·àø
00000160  24 02 C3 49 6E 76 61 6C-69 64 20 70 61 72 74 69  $·ÃInvalid parti
00000170  74 69 6F 6E 20 74 61 62-6C 65 00 45 72 72 6F 72  tion table·Error
00000180  20 6C 6F 61 64 69 6E 67-20 6F 70 65 72 61 74 69   loading operati
00000190  6E 67 20 73 79 73 74 65-6D 00 4D 69 73 73 69 6E  ng system·Missin
000001a0  67 20 6F 70 65 72 61 74-69 6E 67 20 73 79 73 74  g operating syst
000001b0  65 6D 00 00 00 63 7B 9A-C7 1D D6 63 00 00 00 04  em···c{·Ç·Öc····
000001c0  01 00 0B 03 20 96 80 00-00 00 00 60 09 00 00 04  ···· ·····`····
000001d0  01 96 06 03 60 2C 80 60-09 00 00 60 09 00 00 04  ····`,·`···`····
000001e0  41 2C 07 43 60 FE 80 C0-12 00 00 28 0D 00 00 00  A,·C`þ·À···(····
000001f0  00 00 00 00 00 00 00 00-00 00 00 00 00 00 55 AA  ··············Uª
```

For this image, PE1 partition type is @ 0x1C2 (MBR). It shows a 0B. Based on the above slide, it uses FAT32 with CHS Addressing.

CHS Address of the last sector (0x5) - 3 bytes



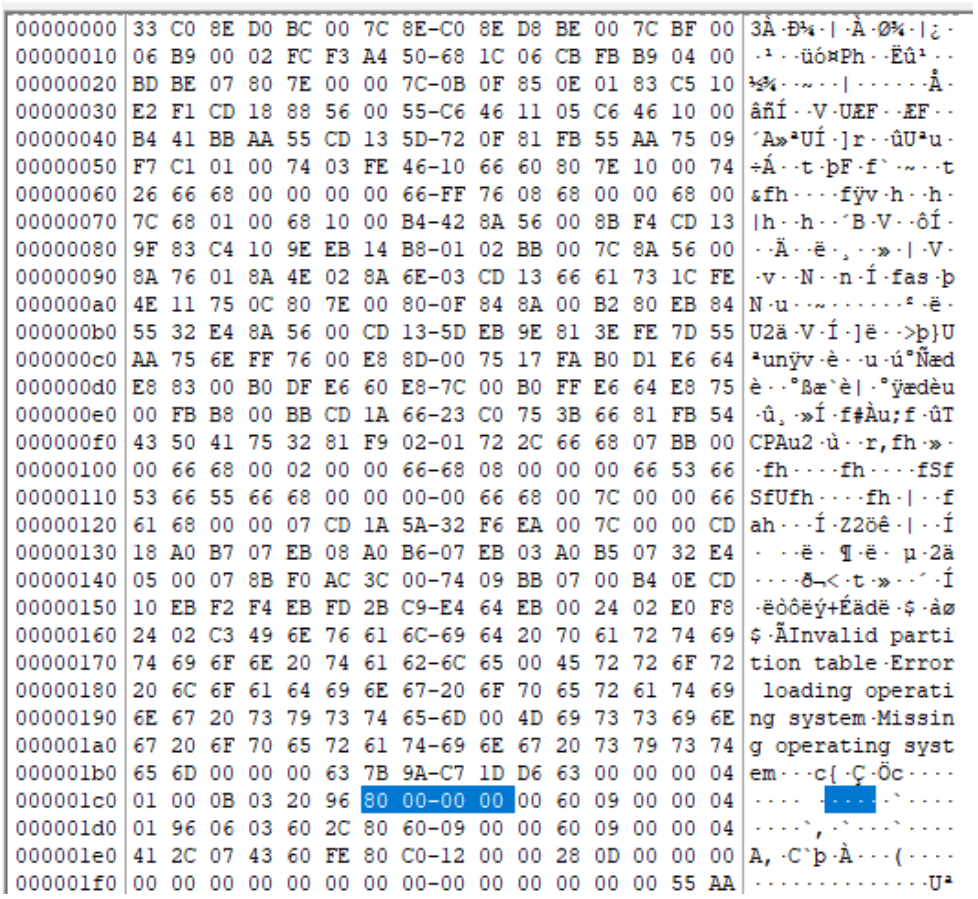
```
00000000  33 C0 8E D0 BC 00 7C 8E-C0 8E D8 BE 00 7C BF 00  3À·Ð¼·|·À·Ø¾·|¿·
00000010  06 B9 00 02 FC F3 A4 50-68 1C 06 CB FB B9 04 00  ·¹··üó¤Ph··Ëû¹··
00000020  BD BE 07 80 7E 00 00 7C-0B 0F 85 0E 01 83 C5 10  ½¾·€~··|·····ƒÅ·
00000030  E2 F1 CD 18 88 56 00 55-C6 46 11 05 C6 46 10 00  âñÍ··V·UÆF··ÆF··
00000040  B4 41 BB AA 55 CD 13 5D-72 0F 81 FB 55 AA 75 09  ´A»ªUÍ·]r··ûUªu·
00000050  F7 C1 01 00 74 03 FE 46-10 66 60 80 7E 10 00 74  ÷Á··t·þF·f`·~··t
00000060  26 66 68 00 00 00 00 66-FF 76 08 68 00 00 68 00  &fh····fÿv·h··h·
00000070  7C 68 01 00 68 10 00 B4-42 8A 56 00 8B F4 CD 13  |h··h··´B·V··ôÍ·
00000080  9F 83 C4 10 9E EB 14 B8-01 02 BB 00 7C 8A 56 00  ··Ä··ë·¸··»·|·V·
00000090  8A 76 01 8A 4E 02 8A 6E-03 CD 13 66 61 73 1C FE  ·v··N··n·Í·fas·þ
000000a0  4E 11 75 0C 80 7E 00 80-0F 84 8A 00 B2 80 EB 84  N·u··€·····ª·ë·
000000b0  55 32 E4 8A 56 00 CD 13-5D EB 9E 81 3E FE 7D 55  U2ä·V·Í·]ë··>þ}U
000000c0  AA 75 6E FF 76 00 E8 8D-00 75 17 FA B0 D1 E6 64  ªunÿv·è··u·ú°Ñæd
000000d0  E8 83 00 B0 DF E6 60 E8-7C 00 B0 FF E6 64 E8 75  è··°ßæ`è|·°ÿædèu
000000e0  00 FB B8 00 BB CD 1A 66-23 C0 75 3B 66 81 FB 54  ·û¸·»Í·f#Àu;f·ûT
000000f0  43 50 41 75 32 81 F9 02-01 72 2C 66 68 07 BB 00  CPAu2·ù··r,fh·»·
00000100  00 66 68 00 02 00 00 66-68 08 00 00 00 66 53 66  ·fh····fh····fSf
00000110  53 66 55 66 68 00 00 00-00 66 68 00 7C 00 00 66  SfUfh····fh·|··f
00000120  61 68 00 00 07 CD 1A 5A-32 F6 EA 00 7C 00 00 CD  ah···Í·Z2öê·|··Í
00000130  18 A0 B7 07 EB 08 A0 B6-07 EB 03 A0 B5 07 32 E4  · ··ë· ¶·ë· µ·2ä
00000140  05 00 07 8B F0 AC 3C 00-74 09 BB 07 00 B4 0E CD  ····ð¬<·t·»··´·Í
00000150  10 EB F2 F4 EB FD 2B C9-E4 64 EB 00 24 02 E0 F8  ·ëòôëý+Éädë·$·àø
00000160  24 02 C3 49 6E 76 61 6C-69 64 20 70 61 72 74 69  $·ÃInvalid parti
00000170  74 69 6F 6E 20 74 61 62-6C 65 00 45 72 72 6F 72  tion table·Error
00000180  20 6C 6F 61 64 69 6E 67-20 6F 70 65 72 61 74 69   loading operati
00000190  6E 67 20 73 79 73 74 65-6D 00 4D 69 73 73 69 6E  ng system·Missin
000001a0  67 20 6F 70 65 72 61 74-69 6E 67 20 73 79 73 74  g operating syst
000001b0  65 6D 00 00 00 63 7B 9A-C7 1D D6 63 00 00 00 04  em···c{·Ç·Öc····
000001c0  01 00 0B 03 20 96 80 00-00 00 00 60 09 00 00 04  ···· ·····`····
000001d0  01 96 06 03 60 2C 80 60-09 00 00 60 09 00 00 04  ····`,·`···`····
000001e0  41 2C 07 43 60 FE 80 C0-12 00 00 28 0D 00 00 00  A,·C`þ·À···(····
000001f0  00 00 00 00 00 00 00 00-00 00 00 00 00 00 55 AA  ··············U ª
```

Purpose: The CHS Address of the Last Sector tells legacy systems the exact physical cylinder, head, and sector where the partition ends on the disk platter.

The CHS Address of the Last Sector starts at 0x1C3 and ends at 0x1C5.

LBA of the 1st sector in the partition (0x8) - 4 bytes

> **Starting sector of the partition in decimal** - WK1 LAB (this is how u identify it)



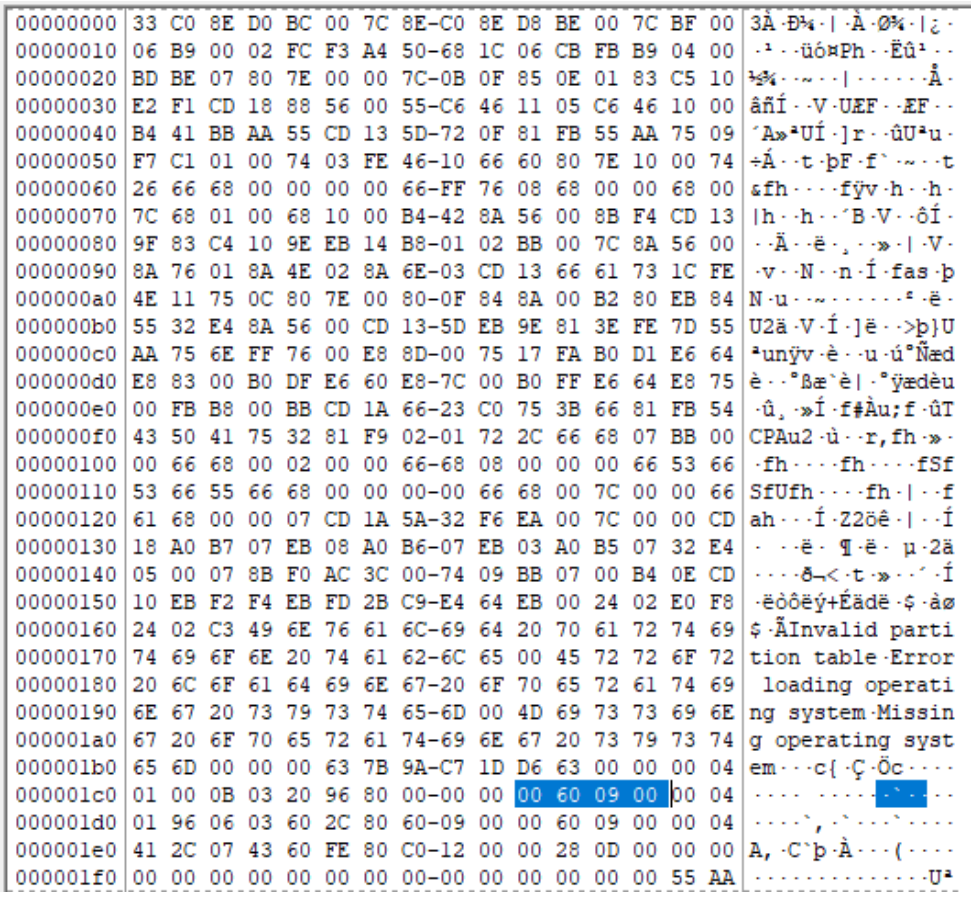
```
00000000 33 C0 8E D0 BC 00 7C 8E-C0 8E D8 BE 00 7C BF 00  3À·Ð¼·|·À·Ø¾·|¿·
00000010 06 B9 00 02 FC F3 A4 50-68 1C 06 CB FB B9 04 00  ·¹··üó¤Ph··Ëû¹··
00000020 BD BE 07 80 7E 00 00 7C-0B 0F 85 0E 01 83 C5 10  ½¾·€~··|·····ƒÅ·
00000030 E2 F1 CD 18 88 56 00 55-C6 46 11 05 C6 46 10 00  âñÍ··V·UÆF··ÆF··
00000040 B4 41 BB AA 55 CD 13 5D-72 0F 81 FB 55 AA 75 09  ´A»ªUÍ·]r··ûUªu·
00000050 F7 C1 01 00 74 03 FE 46-10 66 60 80 7E 10 00 74  ÷Á··t·þF·f`·~··t
00000060 26 66 68 00 00 00 00 66-FF 76 08 68 00 00 68 00  &fh····fÿv·h··h·
00000070 7C 68 01 00 68 10 00 B4-42 8A 56 00 8B F4 CD 13  |h··h··´B·V··ôÍ·
00000080 9F 83 C4 10 9E EB 14 B8-01 02 BB 00 7C 8A 56 00  ··Ä··ë·¸··»·|·V·
00000090 8A 76 01 8A 4E 02 8A 6E-03 CD 13 66 61 73 1C FE  ·v··N··n·Í·fas·þ
000000a0 4E 11 75 0C 80 7E 00 80-0F 84 8A 00 B2 80 EB 84  N·u··€·····ª·ë·
000000b0 55 32 E4 8A 56 00 CD 13-5D EB 9E 81 3E FE 7D 55  U2ä·V·Í·]ë··>þ}U
000000c0 AA 75 6E FF 76 00 E8 8D-00 75 17 FA B0 D1 E6 64  ªunÿv·è··u·ú°Ñæd
000000d0 E8 83 00 B0 DF E6 60 E8-7C 00 B0 FF E6 64 E8 75  è··°ßæ`è|·°ÿædèu
000000e0 00 FB B8 00 BB CD 1A 66-23 C0 75 3B 66 81 FB 54  ·û¸·»Í·f#Àu;f·ûT
000000f0 43 50 41 75 32 81 F9 02-01 72 2C 66 68 07 BB 00  CPAu2·ù··r,fh·»·
00000100 00 66 68 00 02 00 00 66-68 08 00 00 00 66 53 66  ·fh····fh····fSf
00000110 53 66 55 66 68 00 00 00-00 66 68 00 7C 00 00 66  SfUfh····fh·|··f
00000120 61 68 00 00 07 CD 1A 5A-32 F6 EA 00 7C 00 00 CD  ah···Í·Z2öê·|··Í
00000130 18 A0 B7 07 EB 08 A0 B6-07 EB 03 A0 B5 07 32 E4  · ··ë· ¶·ë· µ·2ä
00000140 05 00 07 8B F0 AC 3C 00-74 09 BB 07 00 B4 0E CD  ····ð¬<·t·»··´·Í
00000150 10 EB F2 F4 EB FD 2B C9-E4 64 EB 00 24 02 E0 F8  ·ëòôëý+Éädë·$·àø
00000160 24 02 C3 49 6E 76 61 6C-69 64 20 70 61 72 74 69  $·ÃInvalid parti
00000170 74 69 6F 6E 20 74 61 62-6C 65 00 45 72 72 6F 72  tion table·Error
00000180 20 6C 6F 61 64 69 6E 67-20 6F 70 65 72 61 74 69   loading operati
00000190 6E 67 20 73 79 73 74 65-6D 00 4D 69 73 73 69 6E  ng system·Missin
000001a0 67 20 6F 70 65 72 61 74-69 6E 67 20 73 79 73 74  g operating syst
000001b0 65 6D 00 00 00 63 7B 9A-C7 1D D6 63 00 00 00 04  em···c{·Ç·Öc····
000001c0 01 00 0B 03 20 96 80 00-00 00 00 60 09 00 00 04  ···· ·····`····
000001d0 01 96 06 03 60 2C 80 60-09 00 00 60 09 00 00 04  ····`,·`···`····
000001e0 41 2C 07 43 60 FE 80 C0-12 00 00 28 0D 00 00 00  A,·C`þ·À···(····
000001f0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 55 AA  ··············Uª
```

Purpose: The purpose of the **LBA (Logical Block Addressing) of the First Sector** is **to provide the exact, flat sector number where the partition actually begins**, allowing modern systems to immediately locate and load the Volume Boot Record (VBR)

For this, it starts at 0x1C6 (MBR) and ends at 0x1C9 (MBR).

Number of Sectors in the Partition (0xC) - 4 bytes

> **Size of the partition in sectors (in decimal)** - WK1 LAB (this is how u identify it)



```
00000000 33 C0 8E D0 BC 00 7C 8E-C0 8E D8 BE 00 7C BF 00 3À·Ð¼·|·À·Ø¾·|¿·
00000010 06 B9 00 02 FC F3 A4 50-68 1C 06 CB FB B9 04 00 ·¹··üó¤Ph··Ëû¹··
00000020 BD BE 07 80 7E 00 00 7C-0B 0F 85 0E 01 83 C5 10 ½¾·€~··|·····ƒÅ·
00000030 E2 F1 CD 18 88 56 00 55-C6 46 11 05 C6 46 10 00 âñÍ··V·UÆF··ÆF··
00000040 B4 41 BB AA 55 CD 13 5D-72 0F 81 FB 55 AA 75 09 ´A»ªUÍ·]r··ûUªu·
00000050 F7 C1 01 00 74 03 FE 46-10 66 60 80 7E 10 00 74 ÷Á··t·þF·f`·~··t
00000060 26 66 68 00 00 00 00 66-FF 76 08 68 00 00 68 00 &fh····fÿv·h··h·
00000070 7C 68 01 00 68 10 00 B4-42 8A 56 00 8B F4 CD 13 |h··h··´B·V··ôÍ·
00000080 9F 83 C4 10 9E EB 14 B8-01 02 BB 00 7C 8A 56 00 ·ƒÄ··ë·¸··»·|·V·
00000090 8A 76 01 8A 4E 02 8A 6E-03 CD 13 66 61 73 1C FE ·v··N··n·Í·fas·þ
000000a0 4E 11 75 0C 80 7E 00 80-0F 84 8A 00 B2 80 EB 84 N·u··€~·····ª·ë·
000000b0 55 32 E4 8A 56 00 CD 13-5D EB 9E 81 3E FE 7D 55 U2ä·V·Í·]ë··>þ}U
000000c0 AA 75 6E FF 76 00 E8 8D-00 75 17 FA B0 D1 E6 64 ªunÿv·è··u·ú°Ñæd
000000d0 E8 83 00 B0 DF E6 60 E8-7C 00 B0 FF E6 64 E8 75 è··°ßæ`è|·°ÿædèu
000000e0 00 FB B8 00 BB CD 1A 66-23 C0 75 3B 66 81 FB 54 ·û¸·»Í·f#Àu;f·ûT
000000f0 43 50 41 75 32 81 F9 02-01 72 2C 66 68 07 BB 00 CPAu2·ù··r,fh·»·
00000100 00 66 68 00 02 00 00 66-68 08 00 00 00 66 53 66 ·fh····fh····fSf
00000110 53 66 55 66 68 00 00 00-00 66 68 00 7C 00 00 66 SfUfh····fh·|··f
00000120 61 68 00 00 07 CD 1A 5A-32 F6 EA 00 7C 00 00 CD ah···Í·Z2öê·|··Í
00000130 18 A0 B7 07 EB 08 A0 B6-07 EB 03 A0 B5 07 32 E4 · ·ë· ¶·ë· µ·2ä
00000140 05 00 07 8B F0 AC 3C 00-74 09 BB 07 00 B4 0E CD ····ð¬<·t·»··´·Í
00000150 10 EB F2 F4 EB FD 2B C9-E4 64 EB 00 24 02 E0 F8 ·ëòôëý+Éädë·$·àø
00000160 24 02 C3 49 6E 76 61 6C-69 64 20 70 61 72 74 69 $·ÃInvalid parti
00000170 74 69 6F 6E 20 74 61 62-6C 65 00 45 72 72 6F 72 tion table·Error
00000180 20 6C 6F 61 64 69 6E 67-20 6F 70 65 72 61 74 69  loading operati
00000190 6E 67 20 73 79 73 74 65-6D 00 4D 69 73 73 69 6E ng system·Missin
000001a0 67 20 6F 70 65 72 61 74-69 6E 67 20 73 79 73 74 g operating syst
000001b0 65 6D 00 00 00 63 7B 9A-C7 1D D6 63 00 00 00 04 em···c{·Ç·Öc····
000001c0 01 00 0B 03 20 96 80 00-00 00 00 60 09 00 00 04 ···· ·····`·····
000001d0 01 96 06 03 60 2C 80 60-09 00 00 60 09 00 00 04 ····`,·`···`····
000001e0 41 2C 07 43 60 FE 80 C0-12 00 00 28 0D 00 00 00 A,·C`þ·À···(····
000001f0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 55 AA ··············Uª
```

Purpose: The purpose of the **Number of Sectors in the Partition** is to define the exact size and capacity of the partition by specifying the total number of data blocks it contains.

This is the last part of the partition that tells the number of sectors in the partition. It starts from 0x1CA to 0x1CD.

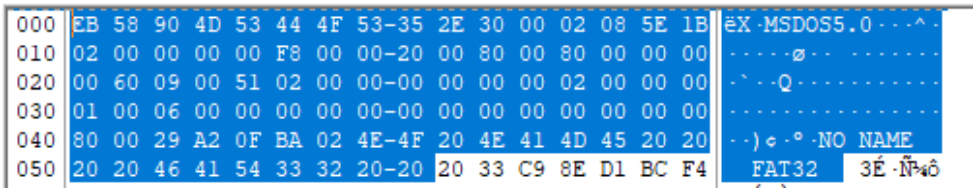
Overall (#PE1 - WK1 LAB)

Entry	Active / Inactive	Partition Type (e.g., FAT32)	Starting Sector of the Partition (in Decimal) **(LBA)**	Size of the partition in sector (in decimal) **(No. of Sectors)**
1	0x00, Inactive	0x0B, FAT32 with CHS Addressing	Hex - 0x00000008 (Little Endian) Decimal - 128	Hex - 0x00096000 Decimal - 614,400

			(use the Hex Value Integer in FTK to decode)	
2				
3				

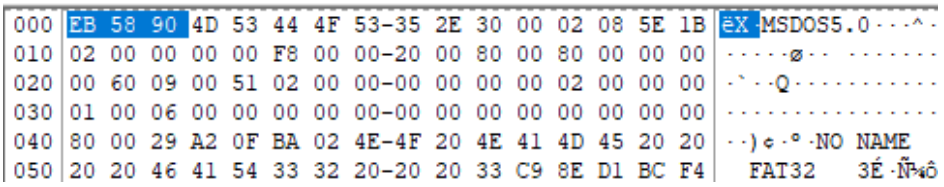
> For PE2 & PE3, try yourself 😃!

Understanding Volume Boot Record (VBR)



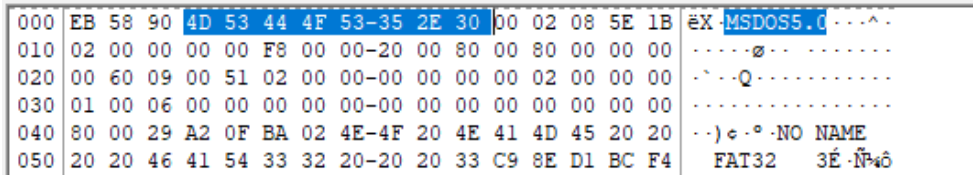
Analyzing the VBR.

Boot Jump Instruction (0x00 to 0x02) - 3 bytes



Purpose: It is an assembly language instruction (`JMP`) that tells the computer's CPU to skip over the upcoming metadata blocks and jump directly to the executable bootloader code further down the sector.

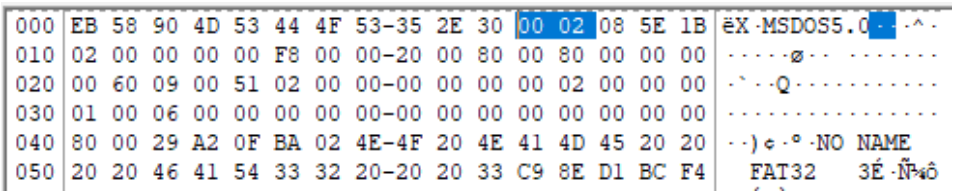
OEM / Name ID (0x03 to 0x0A) - 8 bytes



Purpose:
This contains an ASCII string identifying the operating system or tool that formatted the volume. In your image, it decodes to `MSDOS5.0`.

Bytes Per Sector (0x0B to 0x0C) - 2 bytes

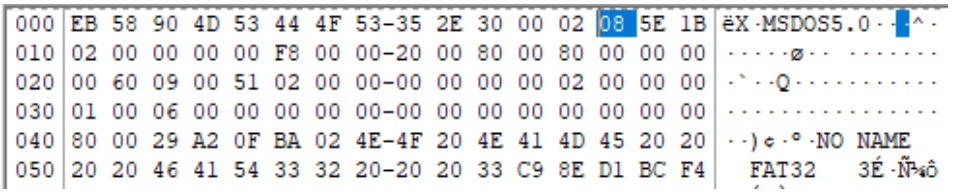
> In questions can be phrased as “What is the size (in bytes) OF a sector in the file system?”



Purpose: **Purpose:** Defines the physical size of a single sector. (Reversed in Little-Endian to `0x0200`, which equals **512 bytes**).

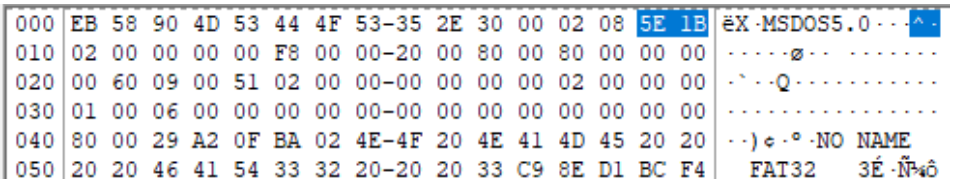
Sectors Per Cluster (0x0D) - 1 byte

> Question can be phrased as “What is the NUMBER OF SECTORS in a cluster in the file system?”



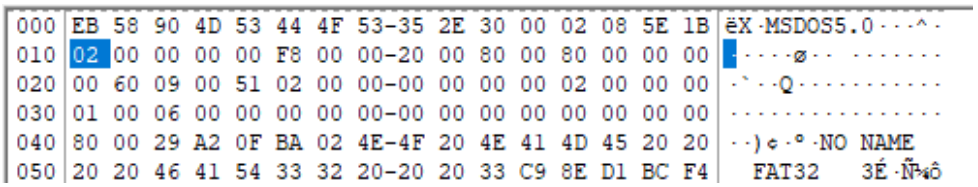
Purpose: Defines how many sectors are grouped together to form a single cluster (the fundamental unit of file storage allocation). In your image, it is exactly **8 sectors**.

Reserved Sector Count (0x0E to 0x0F) - 2 bytes



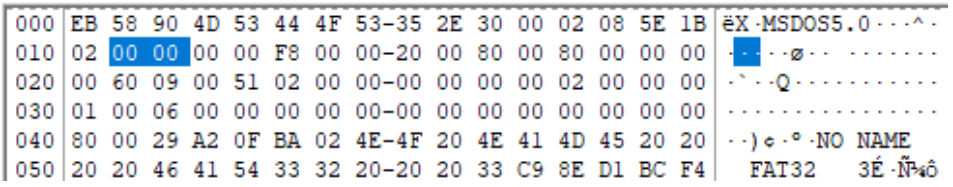
Purpose: **Purpose:** Specifies the number of sectors at the beginning of the volume that are reserved for the boot record and its backups. This dictates exactly where the first File Allocation Table (FAT) begins. (Reversed in Little-Endian to `0x1B5E`, which equals **7,006 sectors**).

Number of File Allocation Table (0x10) - 1 byte



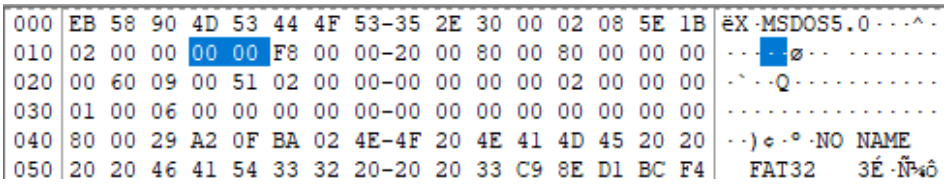
Purpose: Specifies how many copies of the File Allocation Table are maintained on the volume for redundancy. In your image, there are exactly **2 tables** (FAT1 and FAT2).

Maximum Root Directory Entries (0x11 to 0x12) - 2 bytes



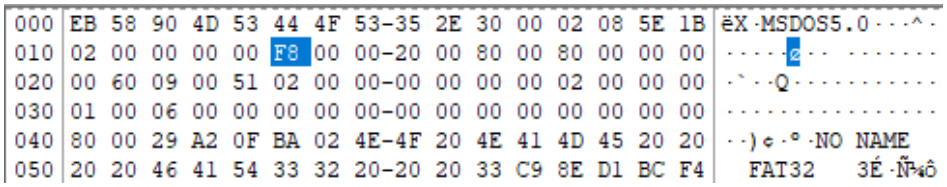
Purpose: **Purpose:** On older FAT12/FAT16 file systems, this defined how many files could fit in the root folder. For **FAT32, this must always be** `00 00` because the root directory is stored as a dynamic cluster chain instead.

Total Sectors (16-bit) (0x13 to 0x14) - 2 bytes



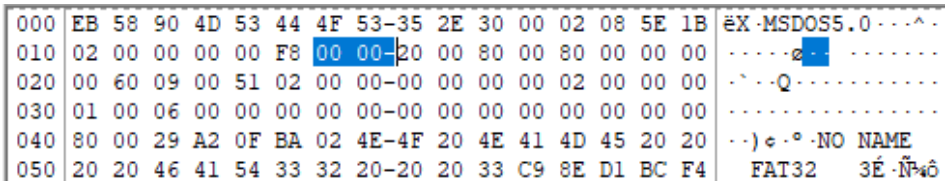
Purpose: Stores the total number of sectors on the volume if the count fits inside 2 bytes (under 65,535 sectors). For **FAT32, this is always** `00 00` because large modern volumes require the 32-bit field located further down at offset `0x20`.

Media Descriptor (0x15) - 1 byte



Purpose: Indicates the type of storage media being used. The value `F8` is the standard industry code for a fixed, non-removable hard disk drive.

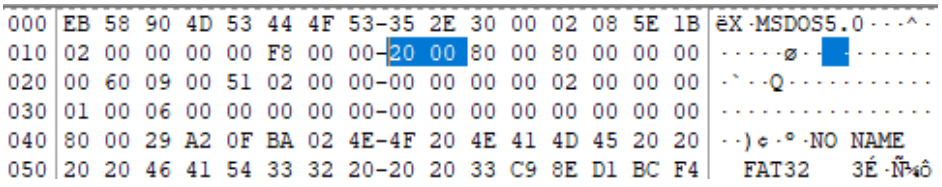
Sectors Per FAT (0x16 to 0x17) - 2 bytes



Purpose: Stores the size of a single File Allocation Table on legacy FAT12/FAT16 systems. For **FAT32, this is always** `00 00` because FAT32 tables are much larger and require the 32-bit field located at offset `0x24`.

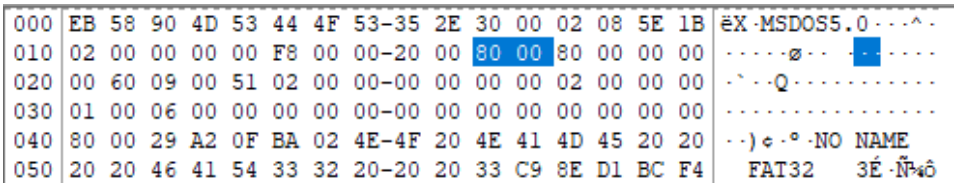
Drive Geometry (0x18 to 0x1B)

Sectors Per Track (0x18 to 0x19) - 2 bytes



Purpose: 32 sectors per track.

Number of Heads (0x1A to 0x1B) - 2 bytes

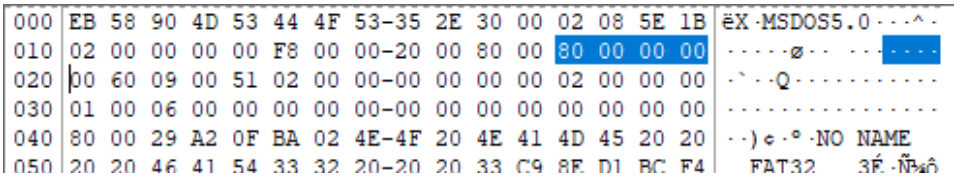


Purpose: This parameter defines the total number of physical magnetic read/write heads (sides of disk platters) available on the storage drive structure. The system multiplies this number by the sectors per track to

calculate how much data can be read without physically shifting the drive's mechanical disk actuator arm.

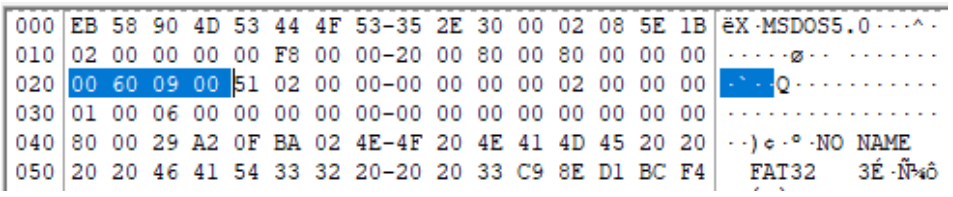
Sector Counts

Hidden Sectors (0x1C - 0x1F) - 4 bytes



Purpose:
This parameter specifies the exact number of physical disk sectors that precede the very first block of this particular partition. It acts as an absolute positioning guide, telling the operating system how many sectors it must skip from the absolute beginning of the entire physical drive (`Sector 0`) to properly locate this volume's start point.

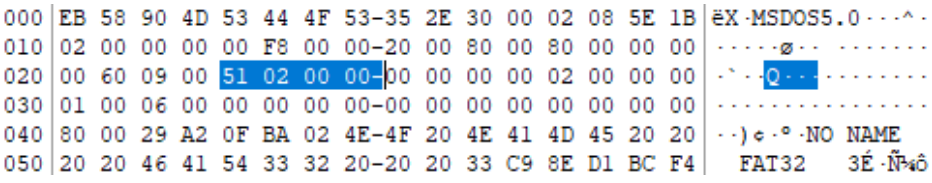
Total Sectors (0x20 - 0x23) - 4 bytes



Purpose: **614,400 total sectors** in this FAT32 partition.

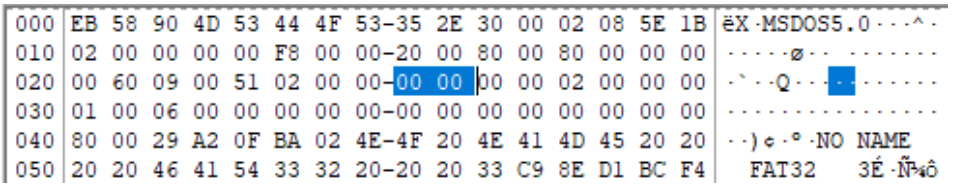
FAT32 Specific Controls (0x24 to 0x33)

Sectors Per FAT (0x24 to 0x27) - 4 bytes



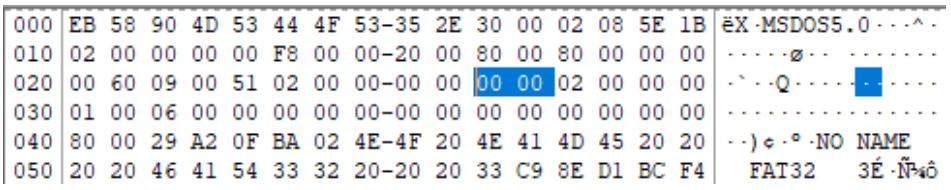
Purpose: **593 sectors** allocated for each FAT table copy.

Extended Flags (0x28 to 0x29) - 2 bytes



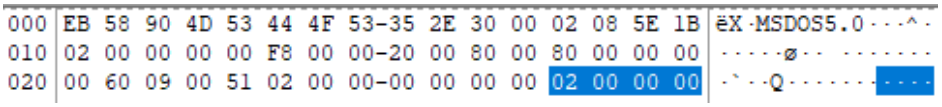
Purpose: Active FAT mirroring flags (standard setup).

File System Version (0x2A to 0x2B) - 2 bytes



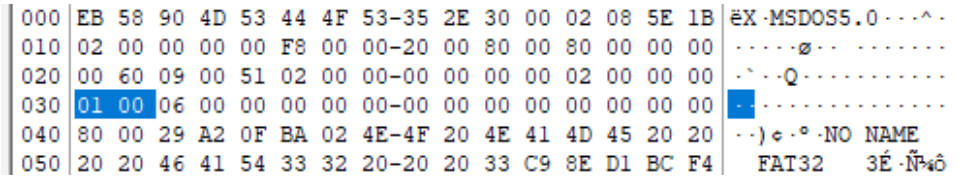
Purpose: Version 0.0.

Root Directory First Cluster (0x2C - 0x2F) - 4 bytes



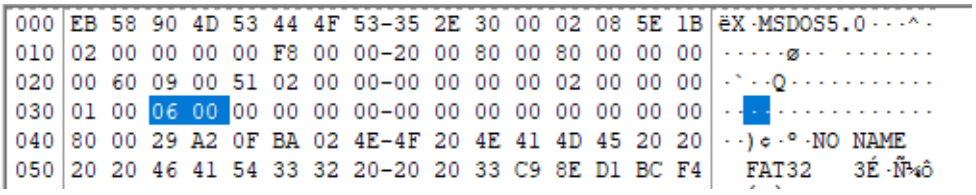
Purpose: Cluster 2 (where the root directory folder structure physically begins).

FSInfo Sector Number (0x30 - 0x31) - 2 bytes



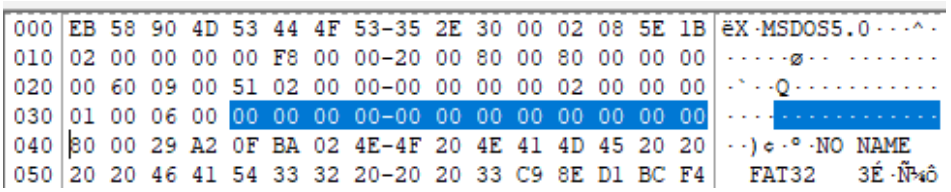
Purpose: Sector 1 contains the File System Information structural block.

Backup Boot Sector (0x32 - 0x33) - 2 bytes



Purpose: Sector 6 contains the mirror copy of this VBR for recovery.

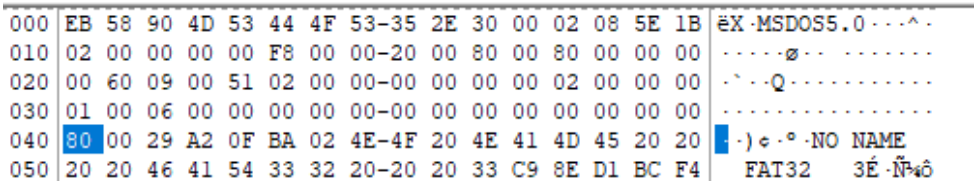
System Buffer (Reserved Space) (0x34 to 0x3F)



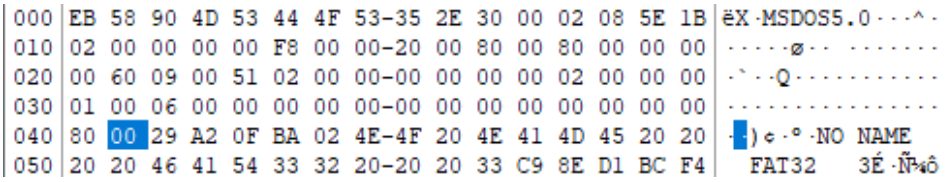
Purpose: Hardcoded blank buffer space reserved for future FAT expansion.

Volume Identifiers (Extended Boot Record)

Physical Drive Number (0x40) - 1 byte

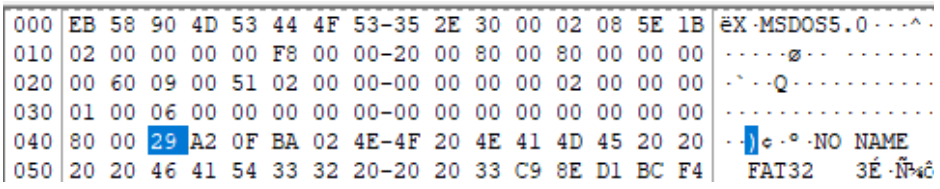


Reserved (NT) (0x41) - 1 byte



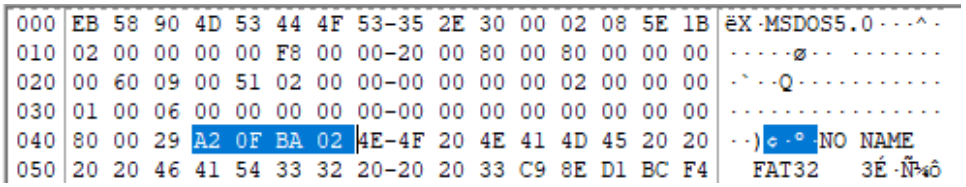
Purpose: Used by Windows NT systems for layout checks.

Extended Boot Signature (0x42) - 1 byte



Purpose: `0x29` (the key validation check byte verifying that a serial number and label follow).

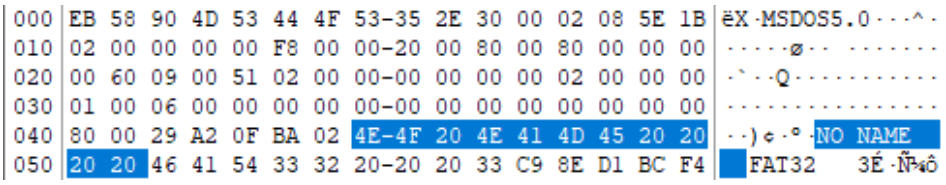
Volume Serial Number (0x43 - 0x46) - 4 bytes



```
000 EB 58 90 4D 53 44 4F 53-35 2E 30 00 02 08 5E 1B ëX·MSDOS5.0···^·
010 02 00 00 00 00 F8 00 00-20 00 80 00 80 00 00 00 ·····ø··  ·······
020 00 60 09 00 51 02 00 00-00 00 00 00 02 00 00 00 ·`··Q···········
030 01 00 06 00 00 00 00 00-00 00 00 00 00 00 00 00 ················
040 80 00 29 A2 0F BA 02 4E-4F 20 4E 41 4D 45 20 20 ··)¢·°·NO NAME
050 20 20 46 41 54 33 32 20-20 20 33 C9 8E D1 BC F4   FAT32   3É·Ñ¼ô
```

Purpose: The distinct drive identity ID code.

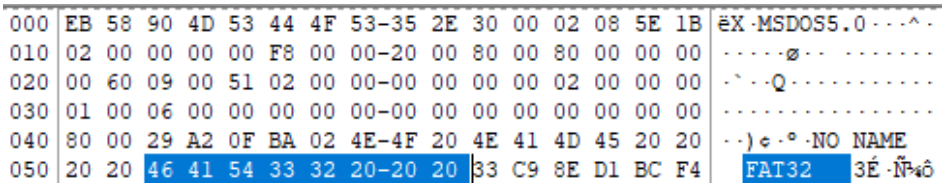
Volume Label (0x47 - 0x51) - 11 bytes



```
000 EB 58 90 4D 53 44 4F 53-35 2E 30 00 02 08 5E 1B ëX·MSDOS5.0···^·
010 02 00 00 00 00 F8 00 00-20 00 80 00 80 00 00 00 ·····ø··  ·······
020 00 60 09 00 51 02 00 00-00 00 00 00 02 00 00 00 ·`··Q···········
030 01 00 06 00 00 00 00 00-00 00 00 00 00 00 00 00 ················
040 80 00 29 A2 0F BA 02 4E-4F 20 4E 41 4D 45 20 20 ··)¢·°·NO NAME
050 20 20 46 41 54 33 32 20-20 20 33 C9 8E D1 BC F4   FAT32   3É·Ñ¼ô
```

Purpose*:* Decodes to the ASCII string `NO NAME` .

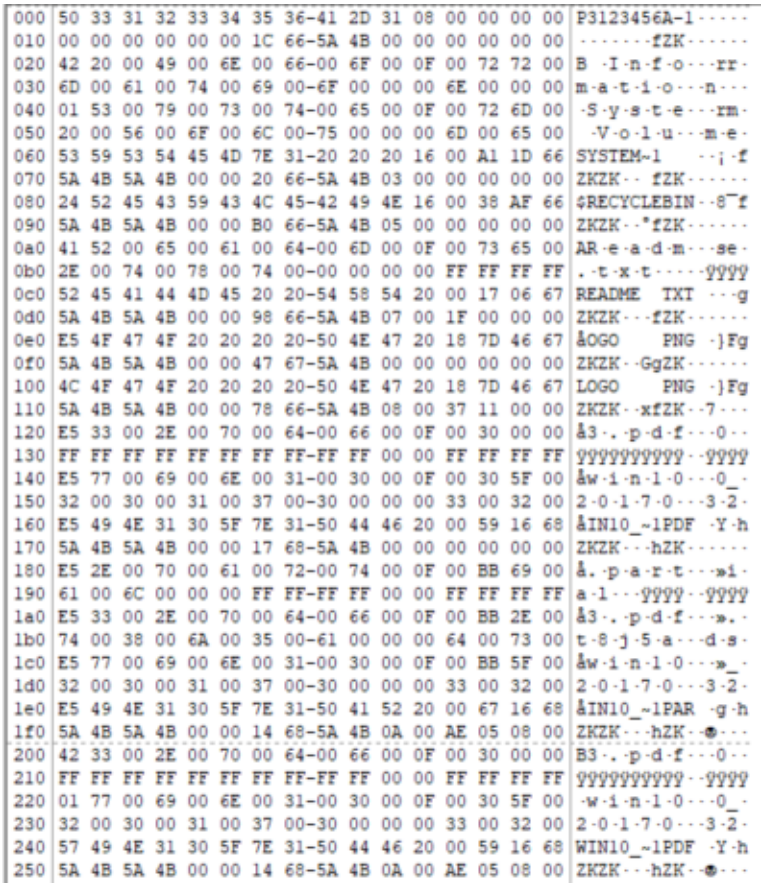
File System Type String (0x52 - 0x59) - 8 bytes



```
000 EB 58 90 4D 53 44 4F 53-35 2E 30 00 02 08 5E 1B ëX·MSDOS5.0···^·
010 02 00 00 00 00 F8 00 00-20 00 80 00 80 00 00 00 ·····ø··  ·······
020 00 60 09 00 51 02 00 00-00 00 00 00 02 00 00 00 ·`··Q···········
030 01 00 06 00 00 00 00 00-00 00 00 00 00 00 00 00 ················
040 80 00 29 A2 0F BA 02 4E-4F 20 4E 41 4D 45 20 20 ··)¢·°·NO NAME
050 20 20 46 41 54 33 32 20-20 20 33 C9 8E D1 BC F4   FAT32   3É·Ñ¼ô
```

Purpose: Decodes to the explicit ASCII string `FAT32` .

Root Folder



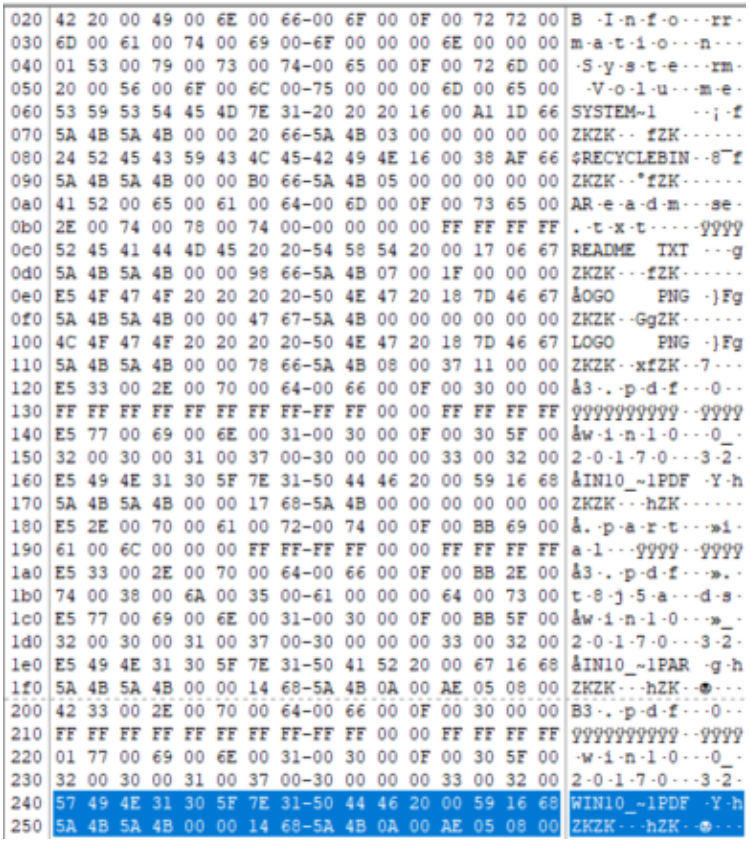
000	50	33	31	32	33	34	35	36-41	2D	31	08	00	00	00	00	P3123456A-1
010	00	00	00	00	00	00	1C	66-5A	4B	00	00	00	00	00	00	fZK
020	42	20	00	49	00	6E	00	66-00	6F	00	0F	00	72	72	00	B I n f o rr
030	6D	00	61	00	74	00	69	00-6F	00	00	00	6E	00	00	00	m a t i o n
040	01	53	00	79	00	73	00	74-00	65	00	0F	00	72	6D	00	S y s t e rm
050	20	00	56	00	6F	00	6C	00-75	00	00	00	6D	00	65	00	V o l u m e
060	53	59	53	54	45	4D	7E	31-20	20	20	16	00	A1	1D	66	SYSTEM~1 ¡ f
070	5A	4B	5A	4B	00	00	20	66-5A	4B	03	00	00	00	00	00	ZKZK fZK
080	24	52	45	43	59	43	4C	45-42	49	4E	16	00	38	AF	66	$RECYCLEBIN 8¯f
090	5A	4B	5A	4B	00	00	B0	66-5A	4B	05	00	00	00	00	00	ZKZK °fZK
0a0	41	52	00	65	00	61	00	64-00	6D	00	0F	00	73	65	00	AR e a d m se
0b0	2E	00	74	00	78	00	74	00-00	00	00	00	FF	FF	FF	FF	. t x t ÿÿÿÿ
0c0	52	45	41	44	4D	45	20	20-54	58	54	20	00	17	06	67	README TXT g
0d0	5A	4B	5A	4B	00	00	98	66-5A	4B	07	00	1F	00	00	00	ZKZK fZK
0e0	E5	4F	47	4F	20	20	20	20-50	4E	47	20	18	7D	46	67	åOGO PNG }Fg
0f0	5A	4B	5A	4B	00	00	47	67-5A	4B	00	00	00	00	00	00	ZKZK GgZK
100	4C	4F	47	4F	20	20	20	20-50	4E	47	20	18	7D	46	67	LOGO PNG }Fg
110	5A	4B	5A	4B	00	00	78	66-5A	4B	08	00	37	11	00	00	ZKZK xfZK 7
120	E5	33	00	2E	00	70	00	64-00	66	00	0F	00	30	00	00	å3 . p d f 0
130	FF	FF	FF	FF	FF	FF	FF	FF-FF	FF	00	00	FF	FF	FF	FF	ÿÿÿÿÿÿÿÿÿÿ ÿÿÿÿ
140	E5	77	00	69	00	6E	00	31-00	30	00	0F	00	30	5F	00	åw i n 1 0 0_
150	32	00	30	00	31	00	37	00-30	00	00	00	33	00	32	00	2 0 1 7 0 3 2
160	E5	49	4E	31	30	5F	7E	31-50	44	46	20	00	59	16	68	åIN10_~1PDF Y h
170	5A	4B	5A	4B	00	00	17	68-5A	4B	00	00	00	00	00	00	ZKZK hZK
180	E5	2E	00	70	00	61	00	72-00	74	00	0F	00	BB	69	00	å. p a r t »i
190	61	00	6C	00	00	00	FF	FF-FF	FF	00	00	FF	FF	FF	FF	a l ÿÿÿÿ ÿÿÿÿ
1a0	E5	33	00	2E	00	70	00	64-00	66	00	0F	00	BB	2E	00	å3 . p d f ».
1b0	74	00	38	00	6A	00	35	00-61	00	00	00	64	00	73	00	t 8 j 5 a d s
1c0	E5	77	00	69	00	6E	00	31-00	30	00	0F	00	BB	5F	00	åw i n 1 0 »_
1d0	32	00	30	00	31	00	37	00-30	00	00	00	33	00	32	00	2 0 1 7 0 3 2
1e0	E5	49	4E	31	30	5F	7E	31-50	41	52	20	00	67	16	68	åIN10_~1PAR g h
1f0	5A	4B	5A	4B	00	00	14	68-5A	4B	0A	00	AE	05	08	00	ZKZK hZK ®
200	42	33	00	2E	00	70	00	64-00	66	00	0F	00	30	00	00	B3 . p d f 0
210	FF	FF	FF	FF	FF	FF	FF	FF-FF	FF	00	00	FF	FF	FF	FF	ÿÿÿÿÿÿÿÿÿÿ ÿÿÿÿ
220	01	77	00	69	00	6E	00	31-00	30	00	0F	00	30	5F	00	w i n 1 0 0_
230	32	00	30	00	31	00	37	00-30	00	00	00	33	00	32	00	2 0 1 7 0 3 2
240	57	49	4E	31	30	5F	7E	31-50	44	46	20	00	59	16	68	WIN10_~1PDF Y h
250	5A	4B	5A	4B	00	00	14	68-5A	4B	0A	00	AE	05	08	00	ZKZK hZK ®

Finding Filenames

Click on viewing Pane → Find → type in file name (e.g., ‘win10’) → Find → Find next till error pops up

Finding Sizes (of File) & First Cluster

Example 1 - Win 10



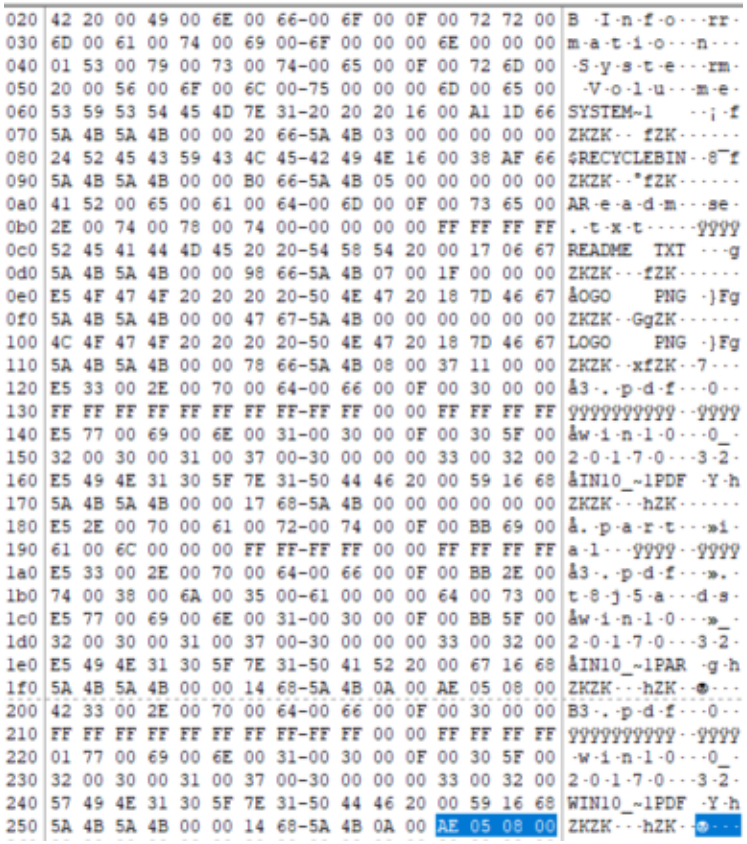
020	42	20	00	49	00	6E	00	66	00	6F	00	0F	00	72	72	00
030	6D	00	61	00	74	00	69	00	6F	00	00	00	6E	00	00	00
040	01	53	00	79	00	73	00	74	00	65	00	0F	00	72	6D	00
050	20	00	56	00	6F	00	6C	00	75	00	00	00	6D	00	65	00
060	53	59	53	54	45	4D	7E	31	20	20	20	16	00	A1	1D	66
070	5A	4B	5A	4B	00	00	20	66	5A	4B	03	00	00	00	00	00
080	24	52	45	43	59	43	4C	45	42	49	4E	16	00	38	AF	66
090	5A	4B	5A	4B	00	00	B0	66	5A	4B	05	00	00	00	00	00
0a0	41	52	00	65	00	61	00	64	00	6D	00	0F	00	73	65	00
0b0	2E	00	74	00	78	00	74	00	00	00	00	00	FF	FF	FF	FF
0c0	52	45	41	44	4D	45	20	20	54	58	54	20	00	17	06	67
0d0	5A	4B	5A	4B	00	00	98	66	5A	4B	07	00	1F	00	00	00
0e0	E5	4F	47	4F	20	20	20	20	50	4E	47	20	18	7D	46	67
0f0	5A	4B	5A	4B	00	00	47	67	5A	4B	00	00	00	00	00	00
100	4C	4F	47	4F	20	20	20	20	50	4E	47	20	18	7D	46	67
110	5A	4B	5A	4B	00	00	78	66	5A	4B	08	00	37	11	00	00
120	E5	33	00	2E	00	70	00	64	00	66	00	0F	00	30	00	00
130	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	00	00	FF	FF	FF	FF
140	E5	77	00	69	00	6E	00	31	00	30	00	0F	00	30	5F	00
150	32	00	30	00	31	00	37	00	30	00	00	00	33	00	32	00
160	E5	49	4E	31	30	5F	7E	31	50	44	46	20	00	59	16	68
170	5A	4B	5A	4B	00	00	17	68	5A	4B	00	00	00	00	00	00
180	E5	2E	00	70	00	61	00	72	00	74	00	0F	00	BB	69	00
190	61	00	6C	00	00	00	FF	FF	FF	FF	00	00	FF	FF	FF	FF
1a0	E5	33	00	2E	00	70	00	64	00	66	00	0F	00	BB	2E	00
1b0	74	00	38	00	6A	00	35	00	61	00	00	00	64	00	73	00
1c0	E5	77	00	69	00	6E	00	31	00	30	00	0F	00	BB	5F	00
1d0	32	00	30	00	31	00	37	00	30	00	00	00	33	00	32	00
1e0	E5	49	4E	31	30	5F	7E	31	50	41	52	20	00	67	16	68
1f0	5A	4B	5A	4B	00	00	14	68	5A	4B	0A	00	AE	05	08	00
200	42	33	00	2E	00	70	00	64	00	66	00	0F	00	30	00	00
210	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	00	00	FF	FF	FF	FF
220	01	77	00	69	00	6E	00	31	00	30	00	0F	00	30	5F	00
230	32	00	30	00	31	00	37	00	30	00	00	00	33	00	32	00
240	57	49	4E	31	30	5F	7E	31	50	44	46	20	00	59	16	68
250	5A	4B	5A	4B	00	00	14	68	5A	4B	0A	00	AE	05	08	00

> Once you are at the win10 row, you can find the entire file structure in that row & the row below.

From here row 240 - 0x0, while 250 - 0x10

Size

To cover this, it is very easy. It’s the last 4 bytes in row 250 (0x1C - 0x1F).



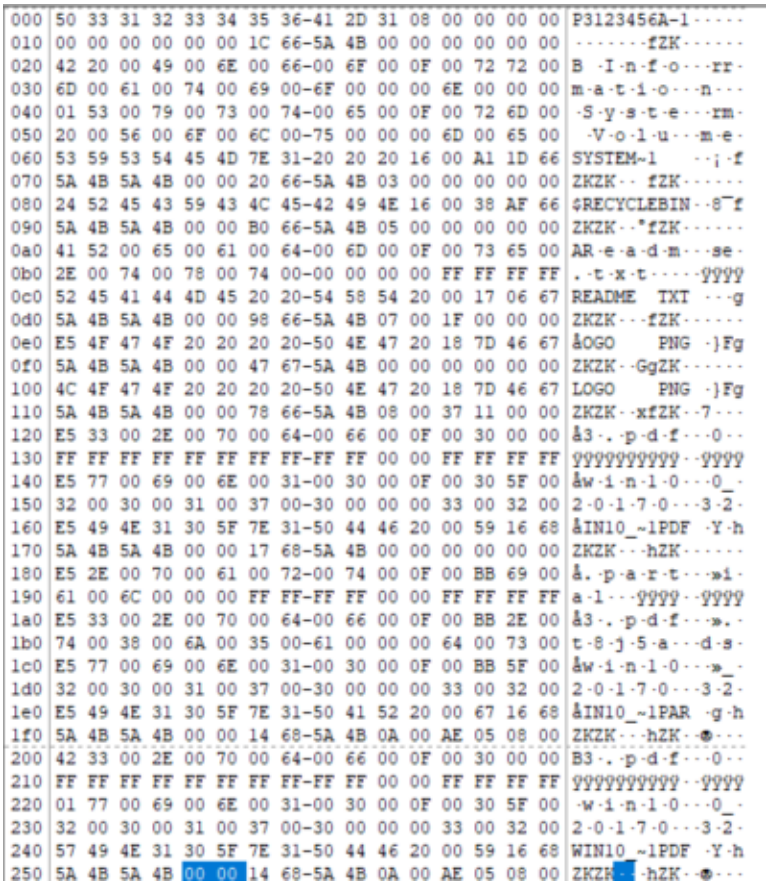
020	42	20	00	49	00	6E	00	66	00	6F	00	0F	00	72	72	00	B ·I·n·f·o···rr·
030	6D	00	61	00	74	00	69	00	6F	00	00	00	6E	00	00	00	m·a·t·i·o···n···
040	01	53	00	79	00	73	00	74	00	65	00	0F	00	72	6D	00	·S·y·s·t·e···rm·
050	20	00	56	00	6F	00	6C	00	75	00	00	00	6D	00	65	00	·V·o·l·u···m·e·
060	53	59	53	54	45	4D	7E	31	20	20	20	16	00	A1	1D	66	SYSTEM~1 ···¡·f
070	5A	4B	5A	4B	00	00	20	66	5A	4B	03	00	00	00	00	00	ZKZK·· fZK·····
080	24	52	45	43	59	43	4C	45	42	49	4E	16	00	38	AF	66	$RECYCLEBIN··8¯f
090	5A	4B	5A	4B	00	00	B0	66	5A	4B	05	00	00	00	00	00	ZKZK··°fZK·····
0a0	41	52	00	65	00	61	00	64	00	6D	00	0F	00	73	65	00	AR·e·a·d·m···se·
0b0	2E	00	74	00	78	00	74	00	00	00	00	00	FF	FF	FF	FF	.·t·x·t·····ÿÿÿÿ
0c0	52	45	41	44	4D	45	20	20	54	58	54	20	00	17	06	67	README TXT ···g
0d0	5A	4B	5A	4B	00	00	98	66	5A	4B	07	00	1F	00	00	00	ZKZK···fZK·····
0e0	E5	4F	47	4F	20	20	20	20	50	4E	47	20	18	7D	46	67	åOGO PNG ·}Fg
0f0	5A	4B	5A	4B	00	00	47	67	5A	4B	00	00	00	00	00	00	ZKZK··GgZK·····
100	4C	4F	47	4F	20	20	20	20	50	4E	47	20	18	7D	46	67	LOGO PNG ·}Fg
110	5A	4B	5A	4B	00	00	78	66	5A	4B	08	00	37	11	00	00	ZKZK··xfZK··7···
120	E5	33	00	2E	00	70	00	64	00	66	00	0F	00	30	00	00	å3·.·p·d·f···0··
130	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	00	00	FF	FF	FF	FF	ÿÿÿÿÿÿÿÿÿÿ··ÿÿÿÿ
140	E5	77	00	69	00	6E	00	31	00	30	00	0F	00	30	5F	00	åw·i·n·l·0···0_·
150	32	00	30	00	31	00	37	00	30	00	00	00	33	00	32	00	2·0·1·7·0···3·2·
160	E5	49	4E	31	30	5F	7E	31	50	44	46	20	00	59	16	68	åIN10_~1PDF ·Y·h
170	5A	4B	5A	4B	00	00	17	68	5A	4B	00	00	00	00	00	00	ZKZK···hZK·····
180	E5	2E	00	70	00	61	00	72	00	74	00	0F	00	BB	69	00	å.·p·a·r·t···»i·
190	61	00	6C	00	00	00	FF	FF	FF	FF	00	00	FF	FF	FF	FF	a·l···ÿÿÿÿ··ÿÿÿÿ
1a0	E5	33	00	2E	00	70	00	64	00	66	00	0F	00	BB	2E	00	å3·.·p·d·f···».·
1b0	74	00	38	00	6A	00	35	00	61	00	00	00	64	00	73	00	t·8·j·5·a···d·s·
1c0	E5	77	00	69	00	6E	00	31	00	30	00	0F	00	BB	5F	00	åw·i·n·l·0···»_·
1d0	32	00	30	00	31	00	37	00	30	00	00	00	33	00	32	00	2·0·1·7·0···3·2·
1e0	E5	49	4E	31	30	5F	7E	31	50	41	52	20	00	67	16	68	åIN10_~1PAR ·g·h
1f0	5A	4B	5A	4B	00	00	14	68	5A	4B	0A	00	AE	05	08	00	ZKZK···hZK·®···
200	42	33	00	2E	00	70	00	64	00	66	00	0F	00	30	00	00	B3·.·p·d·f···0··
210	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	00	00	FF	FF	FF	FF	ÿÿÿÿÿÿÿÿÿÿ··ÿÿÿÿ
220	01	77	00	69	00	6E	00	31	00	30	00	0F	00	30	5F	00	·w·i·n·l·0···0_·
230	32	00	30	00	31	00	37	00	30	00	00	00	33	00	32	00	2·0·1·7·0···3·2·
240	57	49	4E	31	30	5F	7E	31	50	44	46	20	00	59	16	68	WIN10_~1PDF ·Y·h
250	5A	4B	5A	4B	00	00	14	68	5A	4B	0A	00	AE	05	08	00	ZKZK···hZK·®···

Using the hex value interpreter in FTK Imager, you will see the value 525,742.

Cluster

In FAT32, starting clusters are split into two 16-bit parts. High Cluster Word and Low Cluster Word.

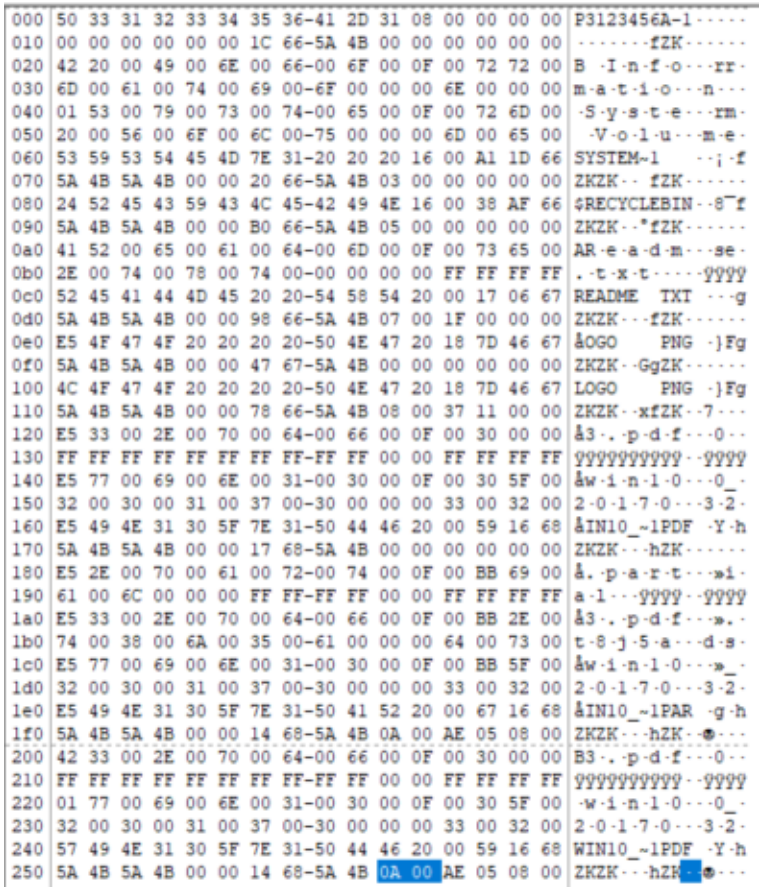
High Cluster Word (0x14 - 0x15)



000	50	33	31	32	33	34	35	36-41	2D	31	08	00	00	00	00	P3123456A-1·····
010	00	00	00	00	00	00	1C	66-5A	4B	00	00	00	00	00	00	·····fZK·····
020	42	20	00	49	00	6E	00	66-00	6F	00	0F	00	72	72	00	B ·I·n·f·o···rr·
030	6D	00	61	00	74	00	69	00-6F	00	00	00	6E	00	00	00	m·a·t·i·o···n···
040	01	53	00	79	00	73	00	74-00	65	00	0F	00	72	6D	00	·S·y·s·t·e···rm·
050	20	00	56	00	6F	00	6C	00-75	00	00	00	6D	00	65	00	·V·o·l·u···m·e·
060	53	59	53	54	45	4D	7E	31-20	20	20	16	00	A1	1D	66	SYSTEM~1 ···¡·f
070	5A	4B	5A	4B	00	00	20	66-5A	4B	03	00	00	00	00	00	ZKZK·· fZK·····
080	24	52	45	43	59	43	4C	45-42	49	4E	16	00	38	AF	66	$RECYCLEBIN··8¯f
090	5A	4B	5A	4B	00	00	B0	66-5A	4B	05	00	00	00	00	00	ZKZK··°fZK·····
0a0	41	52	00	65	00	61	00	64-00	6D	00	0F	00	73	65	00	AR·e·a·d·m···se·
0b0	2E	00	74	00	78	00	74	00-00	00	00	00	FF	FF	FF	FF	.·t·x·t·····ÿÿÿÿ
0c0	52	45	41	44	4D	45	20	20-54	58	54	20	00	17	06	67	README TXT ···g
0d0	5A	4B	5A	4B	00	00	98	66-5A	4B	07	00	1F	00	00	00	ZKZK···fZK·····
0e0	E5	4F	47	4F	20	20	20	20-50	4E	47	20	18	7D	46	67	åOGO PNG ·}Fg
0f0	5A	4B	5A	4B	00	00	47	67-5A	4B	00	00	00	00	00	00	ZKZK··GgZK·····
100	4C	4F	47	4F	20	20	20	20-50	4E	47	20	18	7D	46	67	LOGO PNG ·}Fg
110	5A	4B	5A	4B	00	00	78	66-5A	4B	08	00	37	11	00	00	ZKZK··xfZK··7···
120	E5	33	00	2E	00	70	00	64-00	66	00	0F	00	30	00	00	å3·.·p·d·f···0··
130	FF	FF	FF	FF	FF	FF	FF	FF-FF	FF	00	00	FF	FF	FF	FF	ÿÿÿÿÿÿÿÿÿÿ··ÿÿÿÿ
140	E5	77	00	69	00	6E	00	31-00	30	00	0F	00	30	5F	00	åw·i·n·l·0···0_·
150	32	00	30	00	31	00	37	00-30	00	00	00	33	00	32	00	2·0·1·7·0···3·2·
160	E5	49	4E	31	30	5F	7E	31-50	44	46	20	00	59	16	68	åIN10_~1PDF ·Y·h
170	5A	4B	5A	4B	00	00	17	68-5A	4B	00	00	00	00	00	00	ZKZK···hZK·····
180	E5	2E	00	70	00	61	00	72-00	74	00	0F	00	BB	69	00	å.·p·a·r·t···»i·
190	61	00	6C	00	00	00	FF	FF-FF	FF	00	00	FF	FF	FF	FF	a·l···ÿÿÿÿ··ÿÿÿÿ
1a0	E5	33	00	2E	00	70	00	64-00	66	00	0F	00	BB	2E	00	å3·.·p·d·f···».·
1b0	74	00	38	00	6A	00	35	00-61	00	00	00	64	00	73	00	t·8·j·5·a···d·s·
1c0	E5	77	00	69	00	6E	00	31-00	30	00	0F	00	BB	5F	00	åw·i·n·l·0···»_·
1d0	32	00	30	00	31	00	37	00-30	00	00	00	33	00	32	00	2·0·1·7·0···3·2·
1e0	E5	49	4E	31	30	5F	7E	31-50	41	52	20	00	67	16	68	åIN10_~1PAR ·g·h
1f0	5A	4B	5A	4B	00	00	14	68-5A	4B	0A	00	AE	05	08	00	ZKZK···hZK·®···
200	42	33	00	2E	00	70	00	64-00	66	00	0F	00	30	00	00	B3·.·p·d·f···0··
210	FF	FF	FF	FF	FF	FF	FF	FF-FF	FF	00	00	FF	FF	FF	FF	ÿÿÿÿÿÿÿÿÿÿ··ÿÿÿÿ
220	01	77	00	69	00	6E	00	31-00	30	00	0F	00	30	5F	00	·w·i·n·l·0···0_·
230	32	00	30	00	31	00	37	00-30	00	00	00	33	00	32	00	2·0·1·7·0···3·2·
240	57	49	4E	31	30	5F	7E	31-50	44	46	20	00	59	16	68	WIN10_~1PDF ·Y·h
250	5A	4B	5A	4B	00	00	14	68-5A	4B	0A	00	AE	05	08	00	ZKZK··hZK·®···

0x0000 - HIGH CLUSTER

Low Cluster Word (0x1A - 0x1B)

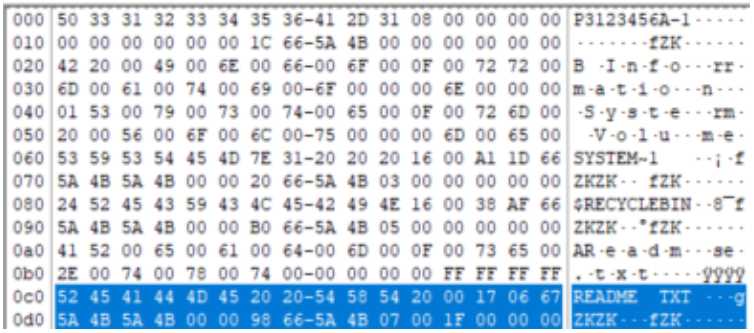


0x000A - LOW CLUSTER

Combine

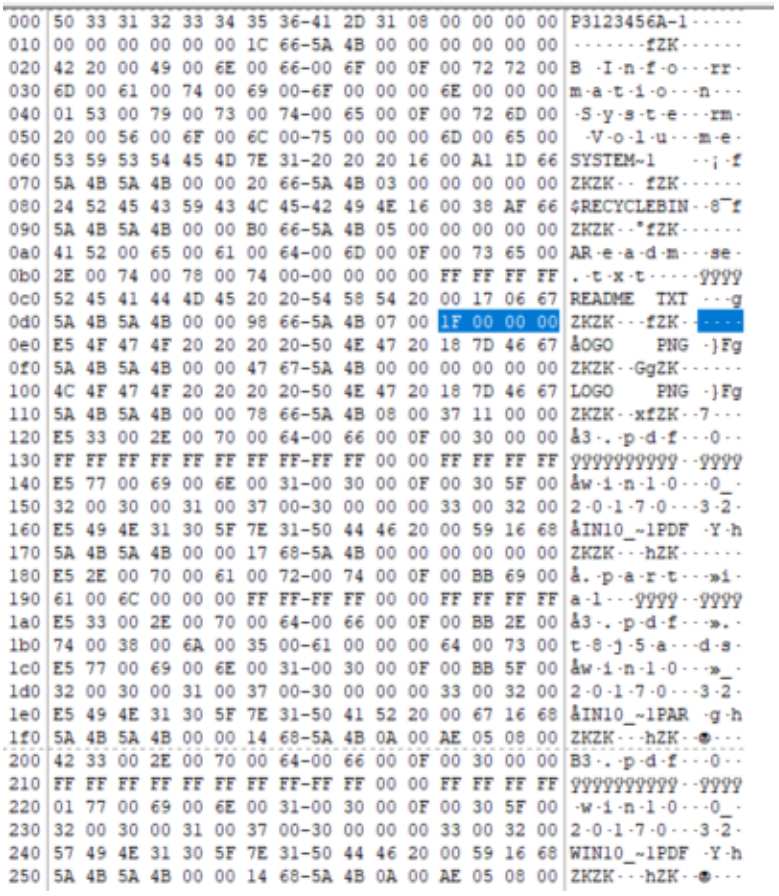
0x0000000A - HEX (First Cluster)

README.TXT



File Size

As per win10, always the last 4 bytes of the second row (after you find filename).

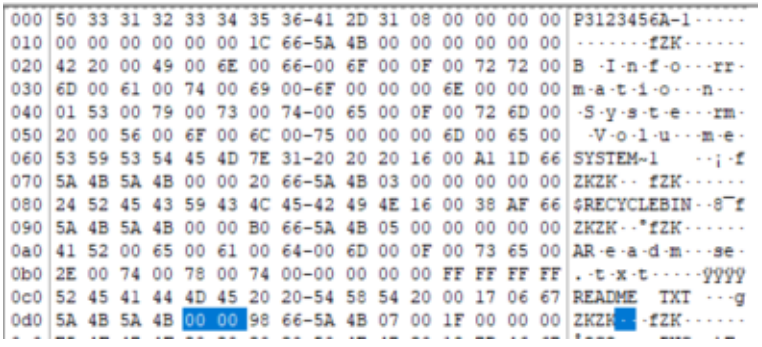


000	50	33	31	32	33	34	35	36-41	2D	31	08	00	00	00	00
010	00	00	00	00	00	00	1C	66-5A	4B	00	00	00	00	00	00
020	42	20	00	49	00	6E	00	66-00	6F	00	0F	00	72	72	00
030	6D	00	61	00	74	00	69	00-6F	00	00	00	6E	00	00	00
040	01	53	00	79	00	73	00	74-00	65	00	0F	00	72	6D	00
050	20	00	56	00	6F	00	6C	00-75	00	00	00	6D	00	65	00
060	53	59	53	54	45	4D	7E	31-20	20	20	16	00	A1	1D	66
070	5A	4B	5A	4B	00	00	20	66-5A	4B	03	00	00	00	00	00
080	24	52	45	43	59	43	4C	45-42	49	4E	16	00	38	AF	66
090	5A	4B	5A	4B	00	00	B0	66-5A	4B	05	00	00	00	00	00
0a0	41	52	00	65	00	61	00	64-00	6D	00	0F	00	73	65	00
0b0	2E	00	74	00	78	00	74	00-00	00	00	00	FF	FF	FF	FF
0c0	52	45	41	44	4D	45	20	20-54	58	54	20	00	17	06	67
0d0	5A	4B	5A	4B	00	00	98	66-5A	4B	07	00	1F	00	00	00
0e0	E5	4F	47	4F	20	20	20	20-50	4E	47	20	18	7D	46	67
0f0	5A	4B	5A	4B	00	00	47	67-5A	4B	00	00	00	00	00	00
100	4C	4F	47	4F	20	20	20	20-50	4E	47	20	18	7D	46	67
110	5A	4B	5A	4B	00	00	78	66-5A	4B	08	00	37	11	00	00
120	E5	33	00	2E	00	70	00	64-00	66	00	0F	00	30	00	00
130	FF	FF	FF	FF	FF	FF	FF	FF-FF	FF	00	00	FF	FF	FF	FF
140	E5	77	00	69	00	6E	00	31-00	30	00	0F	00	30	5F	00
150	32	00	30	00	31	00	37	00-30	00	00	00	33	00	32	00
160	E5	49	4E	31	30	5F	7E	31-50	44	46	20	00	59	16	68
170	5A	4B	5A	4B	00	00	17	68-5A	4B	00	00	00	00	00	00
180	E5	2E	00	70	00	61	00	72-00	74	00	0F	00	BB	69	00
190	61	00	6C	00	00	00	FF	FF-FF	FF	00	00	FF	FF	FF	FF
1a0	E5	33	00	2E	00	70	00	64-00	66	00	0F	00	BB	2E	00
1b0	74	00	38	00	6A	00	35	00-61	00	00	00	64	00	73	00
1c0	E5	77	00	69	00	6E	00	31-00	30	00	0F	00	BB	5F	00
1d0	32	00	30	00	31	00	37	00-30	00	00	00	33	00	32	00
1e0	E5	49	4E	31	30	5F	7E	31-50	41	52	20	00	67	16	68
1f0	5A	4B	5A	4B	00	00	14	68-5A	4B	0A	00	AE	05	08	00
200	42	33	00	2E	00	70	00	64-00	66	00	0F	00	30	00	00
210	FF	FF	FF	FF	FF	FF	FF	FF-FF	FF	00	00	FF	FF	FF	FF
220	01	77	00	69	00	6E	00	31-00	30	00	0F	00	30	5F	00
230	32	00	30	00	31	00	37	00-30	00	00	00	33	00	32	00
240	57	49	4E	31	30	5F	7E	31-50	44	46	20	00	59	16	68
250	5A	4B	5A	4B	00	00	14	68-5A	4B	0A	00	AE	05	08	00

In this case, 1F 00 00 00 is 0x0000001F which is decimal value is 31.

First Cluster

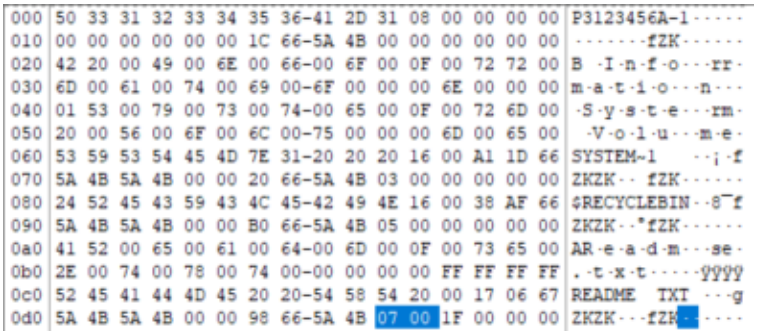
High Word Cluster



000	50	33	31	32	33	34	35	36-41	2D	31	08	00	00	00	00
010	00	00	00	00	00	00	1C	66-5A	4B	00	00	00	00	00	00
020	42	20	00	49	00	6E	00	66-00	6F	00	0F	00	72	72	00
030	6D	00	61	00	74	00	69	00-6F	00	00	00	6E	00	00	00
040	01	53	00	79	00	73	00	74-00	65	00	0F	00	72	6D	00
050	20	00	56	00	6F	00	6C	00-75	00	00	00	6D	00	65	00
060	53	59	53	54	45	4D	7E	31-20	20	20	16	00	A1	1D	66
070	5A	4B	5A	4B	00	00	20	66-5A	4B	03	00	00	00	00	00
080	24	52	45	43	59	43	4C	45-42	49	4E	16	00	38	AF	66
090	5A	4B	5A	4B	00	00	B0	66-5A	4B	05	00	00	00	00	00
0a0	41	52	00	65	00	61	00	64-00	6D	00	0F	00	73	65	00
0b0	2E	00	74	00	78	00	74	00-00	00	00	00	FF	FF	FF	FF
0c0	52	45	41	44	4D	45	20	20-54	58	54	20	00	17	06	67
0d0	5A	4B	5A	4B	00	00	98	66-5A	4B	07	00	1F	00	00	00

0x0000 - High Cluster

Low Word Cluster



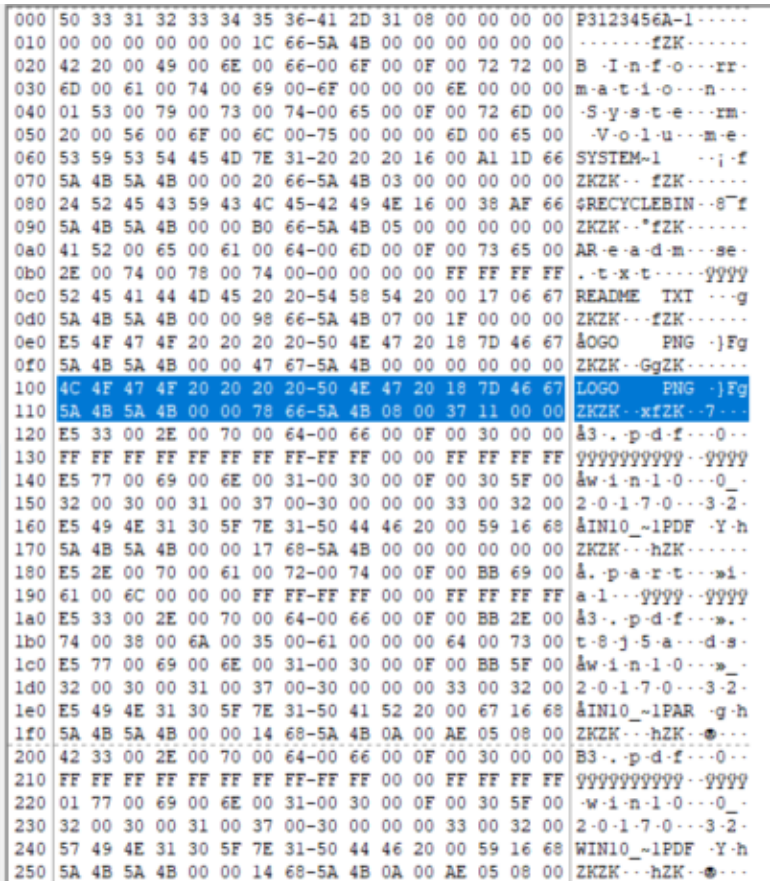
000	50	33	31	32	33	34	35	36-41	2D	31	08	00	00	00	00	P3123456A-1·····
010	00	00	00	00	00	00	1C	66-5A	4B	00	00	00	00	00	00	·····fZK·····
020	42	20	00	49	00	6E	00	66-00	6F	00	0F	00	72	72	00	B ·I·n·f·o···rr·
030	6D	00	61	00	74	00	69	00-6F	00	00	00	6E	00	00	00	m·a·t·i·o···n···
040	01	53	00	79	00	73	00	74-00	65	00	0F	00	72	6D	00	·S·y·s·t·e···rm·
050	20	00	56	00	6F	00	6C	00-75	00	00	00	6D	00	65	00	·V·o·l·u···m·e·
060	53	59	53	54	45	4D	7E	31-20	20	20	16	00	A1	1D	66	SYSTEM~1 ···¡·f
070	5A	4B	5A	4B	00	00	20	66-5A	4B	03	00	00	00	00	00	ZKZK·· fZK·····
080	24	52	45	43	59	43	4C	45-42	49	4E	16	00	38	AF	66	$RECYCLEBIN··8¯f
090	5A	4B	5A	4B	00	00	B0	66-5A	4B	05	00	00	00	00	00	ZKZK··°fZK·····
0a0	41	52	00	65	00	61	00	64-00	6D	00	0F	00	73	65	00	AR·e·a·d·m···se·
0b0	2E	00	74	00	78	00	74	00-00	00	00	00	FF	FF	FF	FF	.·t·x·t·····ÿÿÿÿ
0c0	52	45	41	44	4D	45	20	20-54	58	54	20	00	17	06	67	README TXT ···g
0d0	5A	4B	5A	4B	00	00	98	66-5A	4B	07	00	1F	00	00	00	ZKZK···fZK·····

0x0007 - Low Cluster

Combine

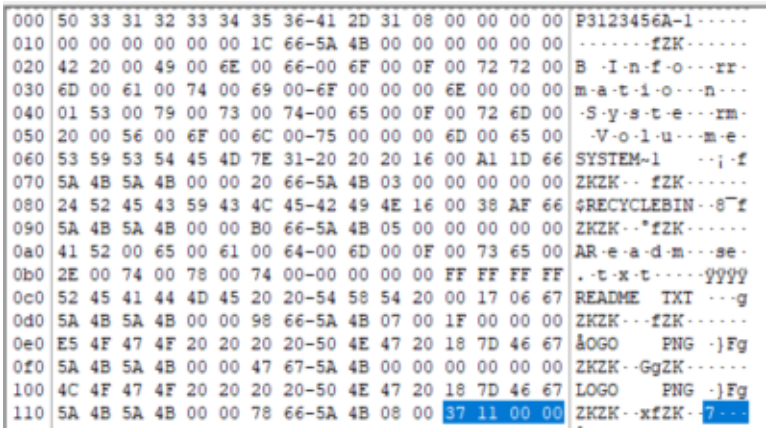
0x00000007

LOGO.PNG



000	50	33	31	32	33	34	35	36-41	2D	31	08	00	00	00	00	P3123456A-1·····
010	00	00	00	00	00	00	1C	66-5A	4B	00	00	00	00	00	00	·····fZK·····
020	42	20	00	49	00	6E	00	66-00	6F	00	0F	00	72	72	00	B ·I·n·f·o···rr·
030	6D	00	61	00	74	00	69	00-6F	00	00	00	6E	00	00	00	m·a·t·i·o···n···
040	01	53	00	79	00	73	00	74-00	65	00	0F	00	72	6D	00	·S·y·s·t·e···rm·
050	20	00	56	00	6F	00	6C	00-75	00	00	00	6D	00	65	00	·V·o·l·u···m·e·
060	53	59	53	54	45	4D	7E	31-20	20	20	16	00	A1	1D	66	SYSTEM~1 ···¡·f
070	5A	4B	5A	4B	00	00	20	66-5A	4B	03	00	00	00	00	00	ZKZK·· fZK·····
080	24	52	45	43	59	43	4C	45-42	49	4E	16	00	38	AF	66	$RECYCLEBIN··8¯f
090	5A	4B	5A	4B	00	00	B0	66-5A	4B	05	00	00	00	00	00	ZKZK··°fZK·····
0a0	41	52	00	65	00	61	00	64-00	6D	00	0F	00	73	65	00	AR·e·a·d·m···se·
0b0	2E	00	74	00	78	00	74	00-00	00	00	00	FF	FF	FF	FF	.·t·x·t·····ÿÿÿÿ
0c0	52	45	41	44	4D	45	20	20-54	58	54	20	00	17	06	67	README TXT ···g
0d0	5A	4B	5A	4B	00	00	98	66-5A	4B	07	00	1F	00	00	00	ZKZK···fZK·····
0e0	E5	4F	47	4F	20	20	20	20-50	4E	47	20	18	7D	46	67	åOGO PNG ·}Fg
0f0	5A	4B	5A	4B	00	00	47	67-5A	4B	00	00	00	00	00	00	ZKZK··GgZK·····
100	4C	4F	47	4F	20	20	20	20-50	4E	47	20	18	7D	46	67	LOGO PNG ·}Fg
110	5A	4B	5A	4B	00	00	78	66-5A	4B	08	00	37	11	00	00	ZKZK··xfZK··7···
120	E5	33	00	2E	00	70	00	64-00	66	00	0F	00	30	00	00	å3·.·p·d·f···0··
130	FF	FF	FF	FF	FF	FF	FF	FF-FF	FF	00	00	FF	FF	FF	FF	ÿÿÿÿÿÿÿÿÿ··ÿÿÿÿ
140	E5	77	00	69	00	6E	00	31-00	30	00	0F	00	30	5F	00	åw·i·n·1·0···0_·
150	32	00	30	00	31	00	37	00-30	00	00	00	33	00	32	00	2·0·1·7·0···3·2·
160	E5	49	4E	31	30	5F	7E	31-50	44	46	20	00	59	16	68	åIN10_~1PDF ·Y·h
170	5A	4B	5A	4B	00	00	17	68-5A	4B	00	00	00	00	00	00	ZKZK···hZK·····
180	E5	2E	00	70	00	61	00	72-00	74	00	0F	00	BB	69	00	å.·p·a·r·t···»i·
190	61	00	6C	00	00	00	FF	FF-FF	FF	00	00	FF	FF	FF	FF	a·l···ÿÿÿÿ··ÿÿÿÿ
1a0	E5	33	00	2E	00	70	00	64-00	66	00	0F	00	BB	2E	00	å3·.·p·d·f···».·
1b0	74	00	38	00	6A	00	35	00-61	00	00	00	64	00	73	00	t·8·j·5·a···d·s·
1c0	E5	77	00	69	00	6E	00	31-00	30	00	0F	00	BB	5F	00	åw·i·n·1·0···»_·
1d0	32	00	30	00	31	00	37	00-30	00	00	00	33	00	32	00	2·0·1·7·0···3·2·
1e0	E5	49	4E	31	30	5F	7E	31-50	41	52	20	00	67	16	68	åIN10_~1PAR ·g·h
1f0	5A	4B	5A	4B	00	00	14	68-5A	4B	0A	00	AE	05	08	00	ZKZK···hZK·®···
200	42	33	00	2E	00	70	00	64-00	66	00	0F	00	30	00	00	B3·.·p·d·f···0··
210	FF	FF	FF	FF	FF	FF	FF	FF-FF	FF	00	00	FF	FF	FF	FF	ÿÿÿÿÿÿÿÿÿ··ÿÿÿÿ
220	01	77	00	69	00	6E	00	31-00	30	00	0F	00	30	5F	00	·w·i·n·1·0···0_·
230	32	00	30	00	31	00	37	00-30	00	00	00	33	00	32	00	2·0·1·7·0···3·2·
240	57	49	4E	31	30	5F	7E	31-50	44	46	20	00	59	16	68	WIN10_~1PDF ·Y·h
250	5A	4B	5A	4B	00	00	14	68-5A	4B	0A	00	AE	05	08	00	ZKZK···hZK·®···

File Size

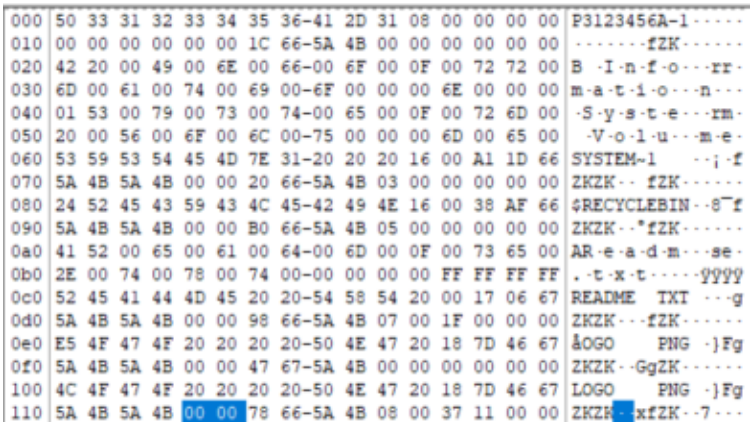


000	50	33	31	32	33	34	35	36-41	2D	31	08	00	00	00	00	P3123456A-1·····
010	00	00	00	00	00	00	1C	66-5A	4B	00	00	00	00	00	00	·····fZK·····
020	42	20	00	49	00	6E	00	66-00	6F	00	0F	00	72	72	00	B·I·n·f·o···rr·
030	6D	00	61	00	74	00	69	00-6F	00	00	00	6E	00	00	00	m·a·t·i·o···n···
040	01	53	00	79	00	73	00	74-00	65	00	0F	00	72	6D	00	·S·y·s·t·e···rm·
050	20	00	56	00	6F	00	6C	00-75	00	00	00	6D	00	65	00	·V·o·l·u···m·e·
060	53	59	53	54	45	4D	7E	31-20	20	20	16	00	A1	1D	66	SYSTEM~1 ···¡·f
070	5A	4B	5A	4B	00	00	20	66-5A	4B	03	00	00	00	00	00	ZKZK·· fZK·····
080	24	52	45	43	59	43	4C	45-42	49	4E	16	00	38	AF	66	$RECYCLEBIN··8¯f
090	5A	4B	5A	4B	00	00	B0	66-5A	4B	05	00	00	00	00	00	ZKZK··°fZK·····
0a0	41	52	00	65	00	61	00	64-00	6D	00	0F	00	73	65	00	AR·e·a·d·m···se·
0b0	2E	00	74	00	78	00	74	00-00	00	00	00	FF	FF	FF	FF	.·t·x·t·····ÿÿÿÿ
0c0	52	45	41	44	4D	45	20	20-54	58	54	20	00	17	06	67	README TXT ···g
0d0	5A	4B	5A	4B	00	00	98	66-5A	4B	07	00	1F	00	00	00	ZKZK···fZK·····
0e0	E5	4F	47	4F	20	20	20	20-50	4E	47	20	18	7D	46	67	åOGO PNG ·}Fg
0f0	5A	4B	5A	4B	00	00	47	67-5A	4B	00	00	00	00	00	00	ZKZK··GgZK·····
100	4C	4F	47	4F	20	20	20	20-50	4E	47	20	18	7D	46	67	LOGO PNG ·}Fg
110	5A	4B	5A	4B	00	00	78	66-5A	4B	08	00	37	11	00	00	ZKZK··xfZK··7···

0x00001137 - in decimal 4407.

Cluster

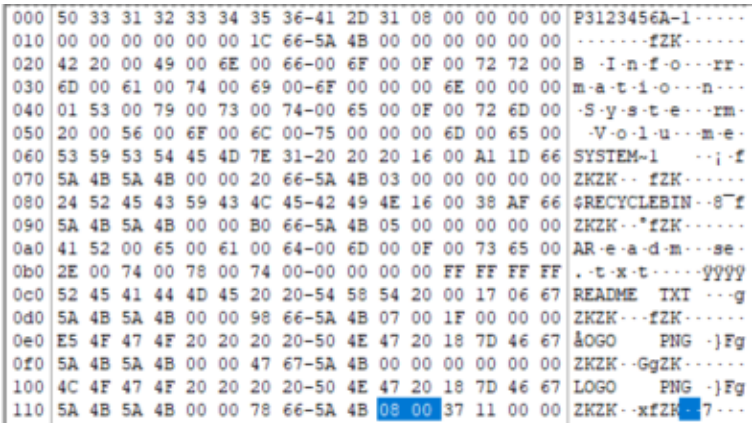
High Cluster Word



000	50	33	31	32	33	34	35	36-41	2D	31	08	00	00	00	00	P3123456A-1·····
010	00	00	00	00	00	00	1C	66-5A	4B	00	00	00	00	00	00	·····fZK·····
020	42	20	00	49	00	6E	00	66-00	6F	00	0F	00	72	72	00	B·I·n·f·o···rr·
030	6D	00	61	00	74	00	69	00-6F	00	00	00	6E	00	00	00	m·a·t·i·o···n···
040	01	53	00	79	00	73	00	74-00	65	00	0F	00	72	6D	00	·S·y·s·t·e···rm·
050	20	00	56	00	6F	00	6C	00-75	00	00	00	6D	00	65	00	·V·o·l·u···m·e·
060	53	59	53	54	45	4D	7E	31-20	20	20	16	00	A1	1D	66	SYSTEM~1 ···¡·f
070	5A	4B	5A	4B	00	00	20	66-5A	4B	03	00	00	00	00	00	ZKZK·· fZK·····
080	24	52	45	43	59	43	4C	45-42	49	4E	16	00	38	AF	66	$RECYCLEBIN··8¯f
090	5A	4B	5A	4B	00	00	B0	66-5A	4B	05	00	00	00	00	00	ZKZK··°fZK·····
0a0	41	52	00	65	00	61	00	64-00	6D	00	0F	00	73	65	00	AR·e·a·d·m···se·
0b0	2E	00	74	00	78	00	74	00-00	00	00	00	FF	FF	FF	FF	.·t·x·t·····ÿÿÿÿ
0c0	52	45	41	44	4D	45	20	20-54	58	54	20	00	17	06	67	README TXT ···g
0d0	5A	4B	5A	4B	00	00	98	66-5A	4B	07	00	1F	00	00	00	ZKZK···fZK·····
0e0	E5	4F	47	4F	20	20	20	20-50	4E	47	20	18	7D	46	67	åOGO PNG ·}Fg
0f0	5A	4B	5A	4B	00	00	47	67-5A	4B	00	00	00	00	00	00	ZKZK··GgZK·····
100	4C	4F	47	4F	20	20	20	20-50	4E	47	20	18	7D	46	67	LOGO PNG ·}Fg
110	5A	4B	5A	4B	00	00	78	66-5A	4B	08	00	37	11	00	00	ZKZK··xfZK··7···

0x0000 - High Cluster

Low Cluster Word



000	50	33	31	32	33	34	35	36-41	2D	31	08	00	00	00	00	P3123456A-1·····
010	00	00	00	00	00	00	1C	66-5A	4B	00	00	00	00	00	00	·····fZK·····
020	42	20	00	49	00	6E	00	66-00	6F	00	0F	00	72	72	00	B·I·n·f·o···rr·
030	6D	00	61	00	74	00	69	00-6F	00	00	00	6E	00	00	00	m·a·t·i·o···n···
040	01	53	00	79	00	73	00	74-00	65	00	0F	00	72	6D	00	·S·y·s·t·e···rm·
050	20	00	56	00	6F	00	6C	00-75	00	00	00	6D	00	65	00	·V·o·l·u···m·e·
060	53	59	53	54	45	4D	7E	31-20	20	20	16	00	A1	1D	66	SYSTEM~1 ···¡·f
070	5A	4B	5A	4B	00	00	20	66-5A	4B	03	00	00	00	00	00	ZKZK·· fZK·····
080	24	52	45	43	59	43	4C	45-42	49	4E	16	00	38	AF	66	$RECYCLEBIN··8¯f
090	5A	4B	5A	4B	00	00	B0	66-5A	4B	05	00	00	00	00	00	ZKZK··°fZK·····
0a0	41	52	00	65	00	61	00	64-00	6D	00	0F	00	73	65	00	AR·e·a·d·m···se·
0b0	2E	00	74	00	78	00	74	00-00	00	00	00	FF	FF	FF	FF	.·t·x·t·····ÿÿÿÿ
0c0	52	45	41	44	4D	45	20	20-54	58	54	20	00	17	06	67	README TXT ···g
0d0	5A	4B	5A	4B	00	00	98	66-5A	4B	07	00	1F	00	00	00	ZKZK···fZK·····
0e0	E5	4F	47	4F	20	20	20	20-50	4E	47	20	18	7D	46	67	åOGO PNG ·}Fg
0f0	5A	4B	5A	4B	00	00	47	67-5A	4B	00	00	00	00	00	00	ZKZK··GgZK·····
100	4C	4F	47	4F	20	20	20	20-50	4E	47	20	18	7D	46	67	LOGO PNG ·}Fg
110	5A	4B	5A	4B	00	00	78	66-5A	4B	08	00	37	11	00	00	ZKZK··xfZK··7···

0x0008 - Low Cluster

Combine

0x00000008

Calculating Last Cluster

Win10.PDF

METHOD 1 - CALCULATING - in Decimal (ofc)

1. Starting Cluster: 10 (0x000A)
2. File Size: 525,742
3. Cluster Size: 4096 bytes (8 sector/cluster x 512 bytes/sector)
4. Calculate total sectors:

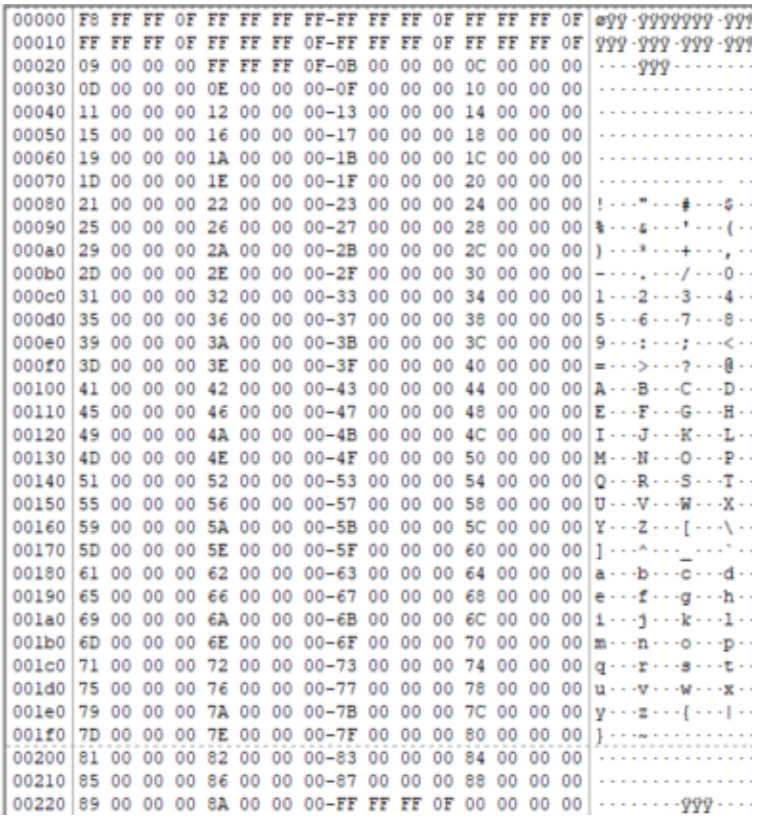
$$\frac{\mathbf{525,742}}{\mathbf{4,096}} = 128.355 \longrightarrow \mathbf{129}\ \text{clusters allocated}$$

5. Calculate Last Cluster:

$$10 + 129 - 1 = 138$$

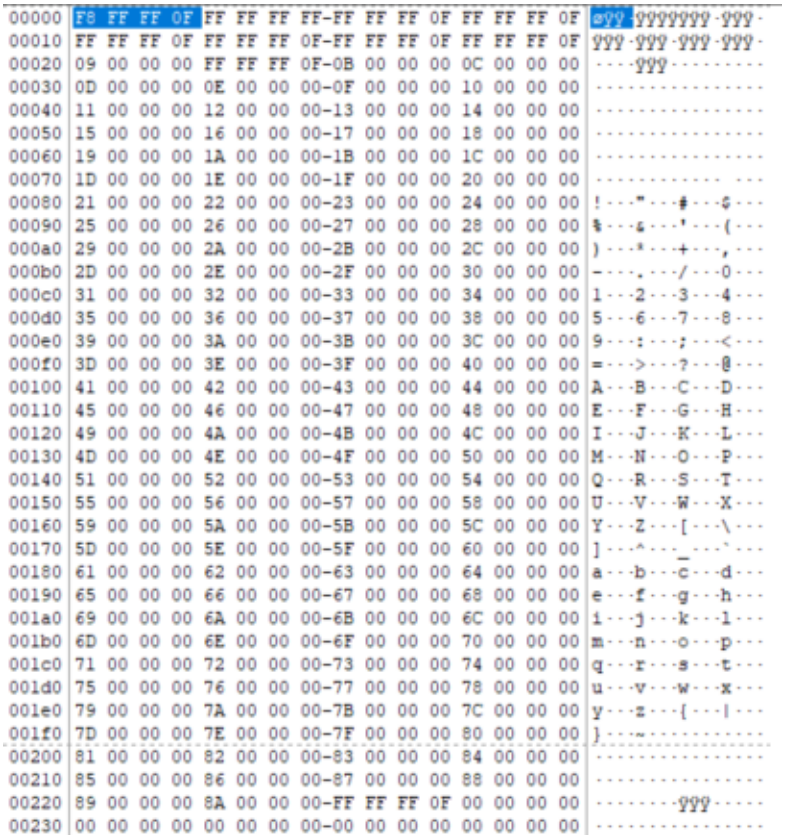
Method 2 - Tracing FAT TABLE in FTK Imager

> FAT1 Table in FTK Imager.



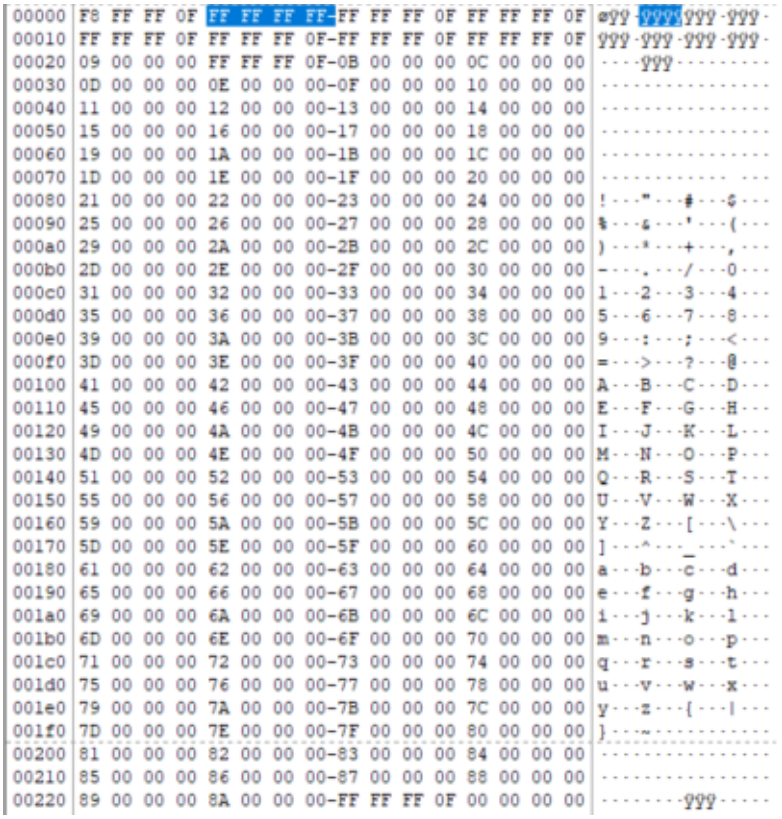
```
00000 F8 FF FF 0F FF FF FF FF-FF FF FF 0F FF FF FF 0F øÿÿ·ÿÿÿÿÿÿ·ÿÿÿ
00010 FF FF FF 0F FF FF FF 0F-FF FF FF 0F FF FF FF 0F ÿÿÿ·ÿÿÿ·ÿÿÿ·ÿÿÿ
00020 09 00 00 00 FF FF FF 0F-0B 00 00 00 0C 00 00 00 ····ÿÿÿ·····
00030 0D 00 00 00 0E 00 00 00-0F 00 00 00 10 00 00 00 ····
00040 11 00 00 00 12 00 00 00-13 00 00 00 14 00 00 00 ····
00050 15 00 00 00 16 00 00 00-17 00 00 00 18 00 00 00 ····
00060 19 00 00 00 1A 00 00 00-1B 00 00 00 1C 00 00 00 ····
00070 1D 00 00 00 1E 00 00 00-1F 00 00 00 20 00 00 00 ···· ···
00080 21 00 00 00 22 00 00 00-23 00 00 00 24 00 00 00 !···"···#···$···
00090 25 00 00 00 26 00 00 00-27 00 00 00 28 00 00 00 %···&···'···(···
000a0 29 00 00 00 2A 00 00 00-2B 00 00 00 2C 00 00 00 )···*···+···,···
000b0 2D 00 00 00 2E 00 00 00-2F 00 00 00 30 00 00 00 -···.···/···0···
000c0 31 00 00 00 32 00 00 00-33 00 00 00 34 00 00 00 1···2···3···4···
000d0 35 00 00 00 36 00 00 00-37 00 00 00 38 00 00 00 5···6···7···8···
000e0 39 00 00 00 3A 00 00 00-3B 00 00 00 3C 00 00 00 9···:···;···<···
000f0 3D 00 00 00 3E 00 00 00-3F 00 00 00 40 00 00 00 =···>···?···@···
00100 41 00 00 00 42 00 00 00-43 00 00 00 44 00 00 00 A···B···C···D···
00110 45 00 00 00 46 00 00 00-47 00 00 00 48 00 00 00 E···F···G···H···
00120 49 00 00 00 4A 00 00 00-4B 00 00 00 4C 00 00 00 I···J···K···L···
00130 4D 00 00 00 4E 00 00 00-4F 00 00 00 50 00 00 00 M···N···O···P···
00140 51 00 00 00 52 00 00 00-53 00 00 00 54 00 00 00 Q···R···S···T···
00150 55 00 00 00 56 00 00 00-57 00 00 00 58 00 00 00 U···V···W···X···
00160 59 00 00 00 5A 00 00 00-5B 00 00 00 5C 00 00 00 Y···Z···[···\···
00170 5D 00 00 00 5E 00 00 00-5F 00 00 00 60 00 00 00 ]···^···_···`···
00180 61 00 00 00 62 00 00 00-63 00 00 00 64 00 00 00 a···b···c···d···
00190 65 00 00 00 66 00 00 00-67 00 00 00 68 00 00 00 e···f···g···h···
001a0 69 00 00 00 6A 00 00 00-6B 00 00 00 6C 00 00 00 i···j···k···l···
001b0 6D 00 00 00 6E 00 00 00-6F 00 00 00 70 00 00 00 m···n···o···p···
001c0 71 00 00 00 72 00 00 00-73 00 00 00 74 00 00 00 q···r···s···t···
001d0 75 00 00 00 76 00 00 00-77 00 00 00 78 00 00 00 u···v···w···x···
001e0 79 00 00 00 7A 00 00 00-7B 00 00 00 7C 00 00 00 y···z···{···|···
001f0 7D 00 00 00 7E 00 00 00-7F 00 00 00 80 00 00 00 }···~···········
00200 81 00 00 00 82 00 00 00-83 00 00 00 84 00 00 00 ····
00210 85 00 00 00 86 00 00 00-87 00 00 00 88 00 00 00 ····
00220 89 00 00 00 8A 00 00 00-FF FF FF 0F 00 00 00 00 ····ÿÿÿ·····
```

Cluster 0



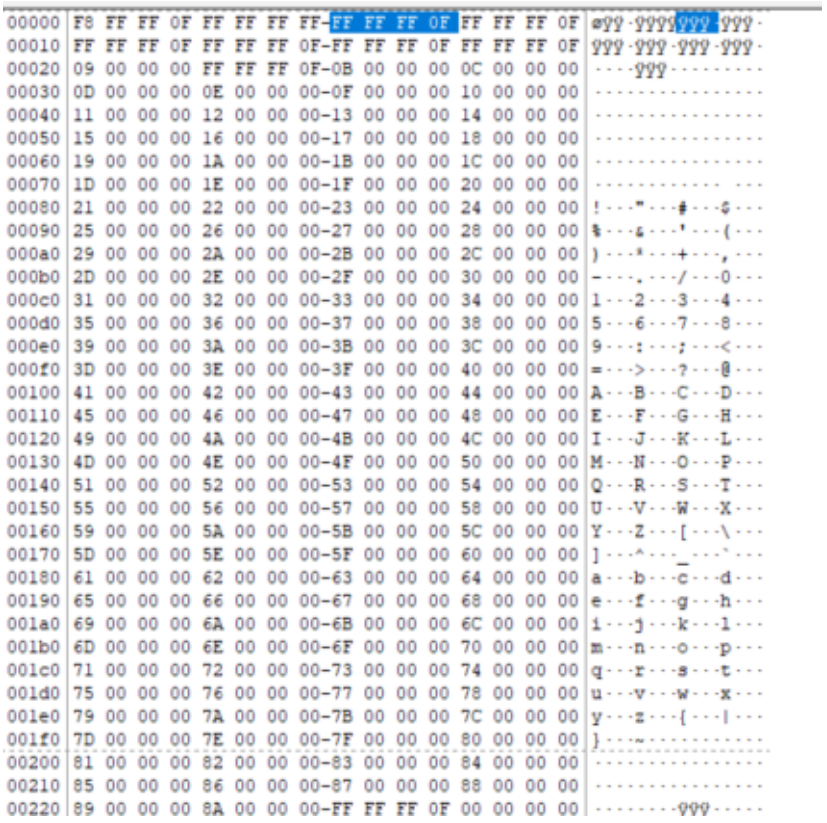
```
00000 F8 FF FF 0F FF FF FF FF-FF FF FF 0F FF FF FF 0F øÿÿ·ÿÿÿÿÿÿ·ÿÿÿ·
00010 FF FF FF 0F FF FF FF 0F-FF FF FF 0F FF FF FF 0F ÿÿÿ·ÿÿÿ·ÿÿÿ·ÿÿÿ·
00020 09 00 00 00 FF FF FF 0F-0B 00 00 00 0C 00 00 00 ····ÿÿÿ·····
00030 0D 00 00 00 0E 00 00 00-0F 00 00 00 10 00 00 00 ····
00040 11 00 00 00 12 00 00 00-13 00 00 00 14 00 00 00 ····
00050 15 00 00 00 16 00 00 00-17 00 00 00 18 00 00 00 ····
00060 19 00 00 00 1A 00 00 00-1B 00 00 00 1C 00 00 00 ····
00070 1D 00 00 00 1E 00 00 00-1F 00 00 00 20 00 00 00 ···· ···
00080 21 00 00 00 22 00 00 00-23 00 00 00 24 00 00 00 !···"···#···$···
00090 25 00 00 00 26 00 00 00-27 00 00 00 28 00 00 00 %···&···'···(···
000a0 29 00 00 00 2A 00 00 00-2B 00 00 00 2C 00 00 00 )···*···+···,···
000b0 2D 00 00 00 2E 00 00 00-2F 00 00 00 30 00 00 00 -···.···/···0···
000c0 31 00 00 00 32 00 00 00-33 00 00 00 34 00 00 00 1···2···3···4···
000d0 35 00 00 00 36 00 00 00-37 00 00 00 38 00 00 00 5···6···7···8···
000e0 39 00 00 00 3A 00 00 00-3B 00 00 00 3C 00 00 00 9···:···;···<···
000f0 3D 00 00 00 3E 00 00 00-3F 00 00 00 40 00 00 00 =···>···?···@···
00100 41 00 00 00 42 00 00 00-43 00 00 00 44 00 00 00 A···B···C···D···
00110 45 00 00 00 46 00 00 00-47 00 00 00 48 00 00 00 E···F···G···H···
00120 49 00 00 00 4A 00 00 00-4B 00 00 00 4C 00 00 00 I···J···K···L···
00130 4D 00 00 00 4E 00 00 00-4F 00 00 00 50 00 00 00 M···N···O···P···
00140 51 00 00 00 52 00 00 00-53 00 00 00 54 00 00 00 Q···R···S···T···
00150 55 00 00 00 56 00 00 00-57 00 00 00 58 00 00 00 U···V···W···X···
00160 59 00 00 00 5A 00 00 00-5B 00 00 00 5C 00 00 00 Y···Z···[···\···
00170 5D 00 00 00 5E 00 00 00-5F 00 00 00 60 00 00 00 ]···^···_···`···
00180 61 00 00 00 62 00 00 00-63 00 00 00 64 00 00 00 a···b···c···d···
00190 65 00 00 00 66 00 00 00-67 00 00 00 68 00 00 00 e···f···g···h···
001a0 69 00 00 00 6A 00 00 00-6B 00 00 00 6C 00 00 00 i···j···k···l···
001b0 6D 00 00 00 6E 00 00 00-6F 00 00 00 70 00 00 00 m···n···o···p···
001c0 71 00 00 00 72 00 00 00-73 00 00 00 74 00 00 00 q···r···s···t···
001d0 75 00 00 00 76 00 00 00-77 00 00 00 78 00 00 00 u···v···w···x···
001e0 79 00 00 00 7A 00 00 00-7B 00 00 00 7C 00 00 00 y···z···{···|···
001f0 7D 00 00 00 7E 00 00 00-7F 00 00 00 80 00 00 00 }···~···········
00200 81 00 00 00 82 00 00 00-83 00 00 00 84 00 00 00 ····
00210 85 00 00 00 86 00 00 00-87 00 00 00 88 00 00 00 ····
00220 89 00 00 00 8A 00 00 00-FF FF FF 0F 00 00 00 00 ····ÿÿÿ·····
00230 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 ····
```

Cluster 1



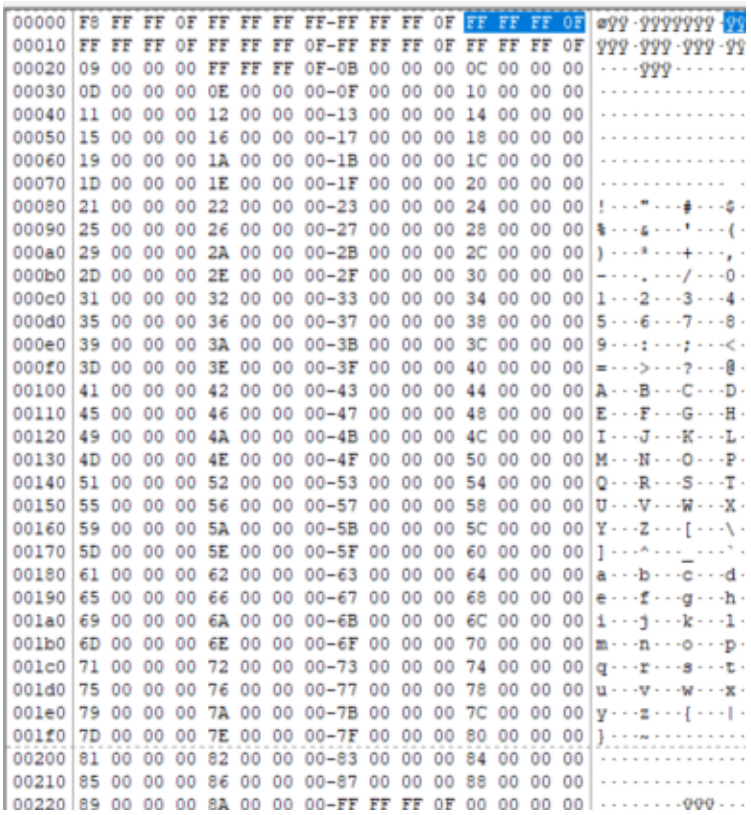
00000	F8	FF	FF	0F	FF	FF	FF	FF	FF	FF	FF	0F	FF	FF	FF	0F
00010	FF	FF	FF	0F	FF	FF	FF	0F	FF	FF	FF	0F	FF	FF	FF	0F
00020	09	00	00	00	FF	FF	FF	0F	0B	00	00	00	0C	00	00	00
00030	0D	00	00	00	0E	00	00	00	0F	00	00	00	10	00	00	00
00040	11	00	00	00	12	00	00	00	13	00	00	00	14	00	00	00
00050	15	00	00	00	16	00	00	00	17	00	00	00	18	00	00	00
00060	19	00	00	00	1A	00	00	00	1B	00	00	00	1C	00	00	00
00070	1D	00	00	00	1E	00	00	00	1F	00	00	00	20	00	00	00
00080	21	00	00	00	22	00	00	00	23	00	00	00	24	00	00	00
00090	25	00	00	00	26	00	00	00	27	00	00	00	28	00	00	00
000a0	29	00	00	00	2A	00	00	00	2B	00	00	00	2C	00	00	00
000b0	2D	00	00	00	2E	00	00	00	2F	00	00	00	30	00	00	00
000c0	31	00	00	00	32	00	00	00	33	00	00	00	34	00	00	00
000d0	35	00	00	00	36	00	00	00	37	00	00	00	38	00	00	00
000e0	39	00	00	00	3A	00	00	00	3B	00	00	00	3C	00	00	00
000f0	3D	00	00	00	3E	00	00	00	3F	00	00	00	40	00	00	00
00100	41	00	00	00	42	00	00	00	43	00	00	00	44	00	00	00
00110	45	00	00	00	46	00	00	00	47	00	00	00	48	00	00	00
00120	49	00	00	00	4A	00	00	00	4B	00	00	00	4C	00	00	00
00130	4D	00	00	00	4E	00	00	00	4F	00	00	00	50	00	00	00
00140	51	00	00	00	52	00	00	00	53	00	00	00	54	00	00	00
00150	55	00	00	00	56	00	00	00	57	00	00	00	58	00	00	00
00160	59	00	00	00	5A	00	00	00	5B	00	00	00	5C	00	00	00
00170	5D	00	00	00	5E	00	00	00	5F	00	00	00	60	00	00	00
00180	61	00	00	00	62	00	00	00	63	00	00	00	64	00	00	00
00190	65	00	00	00	66	00	00	00	67	00	00	00	68	00	00	00
001a0	69	00	00	00	6A	00	00	00	6B	00	00	00	6C	00	00	00
001b0	6D	00	00	00	6E	00	00	00	6F	00	00	00	70	00	00	00
001c0	71	00	00	00	72	00	00	00	73	00	00	00	74	00	00	00
001d0	75	00	00	00	76	00	00	00	77	00	00	00	78	00	00	00
001e0	79	00	00	00	7A	00	00	00	7B	00	00	00	7C	00	00	00
001f0	7D	00	00	00	7E	00	00	00	7F	00	00	00	80	00	00	00
00200	81	00	00	00	82	00	00	00	83	00	00	00	84	00	00	00
00210	85	00	00	00	86	00	00	00	87	00	00	00	88	00	00	00
00220	89	00	00	00	8A	00	00	00	FF	FF	FF	0F	00	00	00	00

Cluster 2



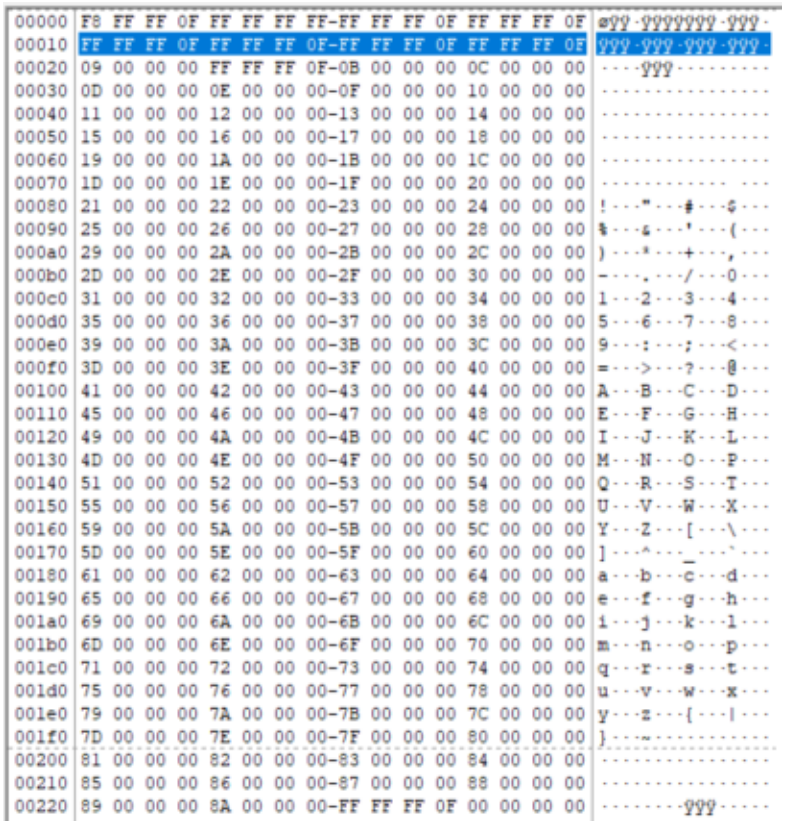
00000	F8	FF	FF	0F	FF	FF	FF	FF	FF	FF	FF	0F	FF	FF	FF	0F
00010	FF	FF	FF	0F	FF	FF	FF	0F	FF	FF	FF	0F	FF	FF	FF	0F
00020	09	00	00	00	FF	FF	FF	0F	0B	00	00	00	0C	00	00	00
00030	0D	00	00	00	0E	00	00	00	0F	00	00	00	10	00	00	00
00040	11	00	00	00	12	00	00	00	13	00	00	00	14	00	00	00
00050	15	00	00	00	16	00	00	00	17	00	00	00	18	00	00	00
00060	19	00	00	00	1A	00	00	00	1B	00	00	00	1C	00	00	00
00070	1D	00	00	00	1E	00	00	00	1F	00	00	00	20	00	00	00
00080	21	00	00	00	22	00	00	00	23	00	00	00	24	00	00	00
00090	25	00	00	00	26	00	00	00	27	00	00	00	28	00	00	00
000a0	29	00	00	00	2A	00	00	00	2B	00	00	00	2C	00	00	00
000b0	2D	00	00	00	2E	00	00	00	2F	00	00	00	30	00	00	00
000c0	31	00	00	00	32	00	00	00	33	00	00	00	34	00	00	00
000d0	35	00	00	00	36	00	00	00	37	00	00	00	38	00	00	00
000e0	39	00	00	00	3A	00	00	00	3B	00	00	00	3C	00	00	00
000f0	3D	00	00	00	3E	00	00	00	3F	00	00	00	40	00	00	00
00100	41	00	00	00	42	00	00	00	43	00	00	00	44	00	00	00
00110	45	00	00	00	46	00	00	00	47	00	00	00	48	00	00	00
00120	49	00	00	00	4A	00	00	00	4B	00	00	00	4C	00	00	00
00130	4D	00	00	00	4E	00	00	00	4F	00	00	00	50	00	00	00
00140	51	00	00	00	52	00	00	00	53	00	00	00	54	00	00	00
00150	55	00	00	00	56	00	00	00	57	00	00	00	58	00	00	00
00160	59	00	00	00	5A	00	00	00	5B	00	00	00	5C	00	00	00
00170	5D	00	00	00	5E	00	00	00	5F	00	00	00	60	00	00	00
00180	61	00	00	00	62	00	00	00	63	00	00	00	64	00	00	00
00190	65	00	00	00	66	00	00	00	67	00	00	00	68	00	00	00
001a0	69	00	00	00	6A	00	00	00	6B	00	00	00	6C	00	00	00
001b0	6D	00	00	00	6E	00	00	00	6F	00	00	00	70	00	00	00
001c0	71	00	00	00	72	00	00	00	73	00	00	00	74	00	00	00
001d0	75	00	00	00	76	00	00	00	77	00	00	00	78	00	00	00
001e0	79	00	00	00	7A	00	00	00	7B	00	00	00	7C	00	00	00
001f0	7D	00	00	00	7E	00	00	00	7F	00	00	00	80	00	00	00
00200	81	00	00	00	82	00	00	00	83	00	00	00	84	00	00	00
00210	85	00	00	00	86	00	00	00	87	00	00	00	88	00	00	00
00220	89	00	00	00	8A	00	00	00	FF	FF	FF	0F	00	00	00	00

Cluster 3



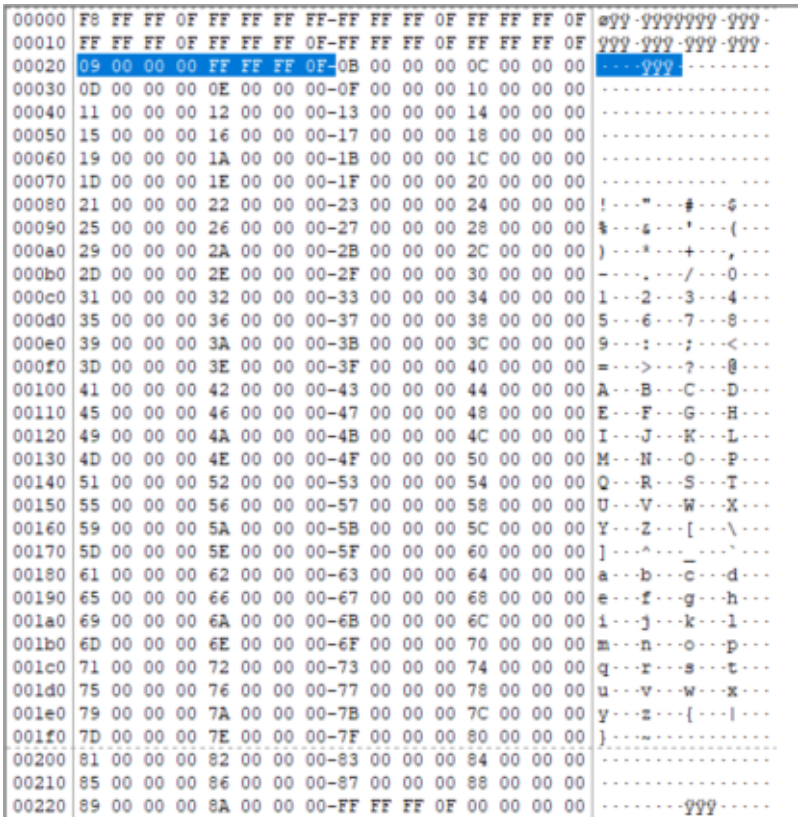
00000	F8	FF	FF	0F	FF	FF	FF	FF	FF	FF	FF	0F	FF	FF	FF	0F
00010	FF	FF	FF	0F	FF	FF	FF	0F	FF	FF	FF	0F	FF	FF	FF	0F
00020	09	00	00	00	FF	FF	FF	0F	0B	00	00	00	0C	00	00	00
00030	0D	00	00	00	0E	00	00	00	0F	00	00	00	10	00	00	00
00040	11	00	00	00	12	00	00	00	13	00	00	00	14	00	00	00
00050	15	00	00	00	16	00	00	00	17	00	00	00	18	00	00	00
00060	19	00	00	00	1A	00	00	00	1B	00	00	00	1C	00	00	00
00070	1D	00	00	00	1E	00	00	00	1F	00	00	00	20	00	00	00
00080	21	00	00	00	22	00	00	00	23	00	00	00	24	00	00	00
00090	25	00	00	00	26	00	00	00	27	00	00	00	28	00	00	00
000a0	29	00	00	00	2A	00	00	00	2B	00	00	00	2C	00	00	00
000b0	2D	00	00	00	2E	00	00	00	2F	00	00	00	30	00	00	00
000c0	31	00	00	00	32	00	00	00	33	00	00	00	34	00	00	00
000d0	35	00	00	00	36	00	00	00	37	00	00	00	38	00	00	00
000e0	39	00	00	00	3A	00	00	00	3B	00	00	00	3C	00	00	00
000f0	3D	00	00	00	3E	00	00	00	3F	00	00	00	40	00	00	00
00100	41	00	00	00	42	00	00	00	43	00	00	00	44	00	00	00
00110	45	00	00	00	46	00	00	00	47	00	00	00	48	00	00	00
00120	49	00	00	00	4A	00	00	00	4B	00	00	00	4C	00	00	00
00130	4D	00	00	00	4E	00	00	00	4F	00	00	00	50	00	00	00
00140	51	00	00	00	52	00	00	00	53	00	00	00	54	00	00	00
00150	55	00	00	00	56	00	00	00	57	00	00	00	58	00	00	00
00160	59	00	00	00	5A	00	00	00	5B	00	00	00	5C	00	00	00
00170	5D	00	00	00	5E	00	00	00	5F	00	00	00	60	00	00	00
00180	61	00	00	00	62	00	00	00	63	00	00	00	64	00	00	00
00190	65	00	00	00	66	00	00	00	67	00	00	00	68	00	00	00
001a0	69	00	00	00	6A	00	00	00	6B	00	00	00	6C	00	00	00
001b0	6D	00	00	00	6E	00	00	00	6F	00	00	00	70	00	00	00
001c0	71	00	00	00	72	00	00	00	73	00	00	00	74	00	00	00
001d0	75	00	00	00	76	00	00	00	77	00	00	00	78	00	00	00
001e0	79	00	00	00	7A	00	00	00	7B	00	00	00	7C	00	00	00
001f0	7D	00	00	00	7E	00	00	00	7F	00	00	00	80	00	00	00
00200	81	00	00	00	82	00	00	00	83	00	00	00	84	00	00	00
00210	85	00	00	00	86	00	00	00	87	00	00	00	88	00	00	00
00220	89	00	00	00	8A	00	00	00	FF	FF	FF	0F	00	00	00	00

Cluster 4-7



```
00000 F8 FF FF 0F FF FF FF FF-FF FF FF 0F FF FF FF 0F øÿÿ·ÿÿÿÿÿÿÿ·ÿÿÿ·
00010 FF FF FF 0F FF FF FF 0F-FF FF FF 0F FF FF FF 0F ÿÿÿ·ÿÿÿ·ÿÿÿ·ÿÿÿ·
00020 09 00 00 00 FF FF FF 0F-0B 00 00 00 0C 00 00 00 ····ÿÿÿ·········
00030 0D 00 00 00 0E 00 00 00-0F 00 00 00 10 00 00 00 ····************
00040 11 00 00 00 12 00 00 00-13 00 00 00 14 00 00 00 ····************
00050 15 00 00 00 16 00 00 00-17 00 00 00 18 00 00 00 ····************
00060 19 00 00 00 1A 00 00 00-1B 00 00 00 1C 00 00 00 ····************
00070 1D 00 00 00 1E 00 00 00-1F 00 00 00 20 00 00 00 ····************ ···
00080 21 00 00 00 22 00 00 00-23 00 00 00 24 00 00 00 !···"···#···$···
00090 25 00 00 00 26 00 00 00-27 00 00 00 28 00 00 00 %···&···'···(···
000a0 29 00 00 00 2A 00 00 00-2B 00 00 00 2C 00 00 00 )···*···+···,···
000b0 2D 00 00 00 2E 00 00 00-2F 00 00 00 30 00 00 00 -···.···/···0···
000c0 31 00 00 00 32 00 00 00-33 00 00 00 34 00 00 00 1···2···3···4···
000d0 35 00 00 00 36 00 00 00-37 00 00 00 38 00 00 00 5···6···7···8···
000e0 39 00 00 00 3A 00 00 00-3B 00 00 00 3C 00 00 00 9···:···;···<···
000f0 3D 00 00 00 3E 00 00 00-3F 00 00 00 40 00 00 00 =···>···?···@···
00100 41 00 00 00 42 00 00 00-43 00 00 00 44 00 00 00 A···B···C···D···
00110 45 00 00 00 46 00 00 00-47 00 00 00 48 00 00 00 E···F···G···H···
00120 49 00 00 00 4A 00 00 00-4B 00 00 00 4C 00 00 00 I···J···K···L···
00130 4D 00 00 00 4E 00 00 00-4F 00 00 00 50 00 00 00 M···N···O···P···
00140 51 00 00 00 52 00 00 00-53 00 00 00 54 00 00 00 Q···R···S···T···
00150 55 00 00 00 56 00 00 00-57 00 00 00 58 00 00 00 U···V···W···X···
00160 59 00 00 00 5A 00 00 00-5B 00 00 00 5C 00 00 00 Y···Z···[···\···
00170 5D 00 00 00 5E 00 00 00-5F 00 00 00 60 00 00 00 ]···^···_···`···
00180 61 00 00 00 62 00 00 00-63 00 00 00 64 00 00 00 a···b···c···d···
00190 65 00 00 00 66 00 00 00-67 00 00 00 68 00 00 00 e···f···g···h···
001a0 69 00 00 00 6A 00 00 00-6B 00 00 00 6C 00 00 00 i···j···k···l···
001b0 6D 00 00 00 6E 00 00 00-6F 00 00 00 70 00 00 00 m···n···o···p···
001c0 71 00 00 00 72 00 00 00-73 00 00 00 74 00 00 00 q···r···s···t···
001d0 75 00 00 00 76 00 00 00-77 00 00 00 78 00 00 00 u···v···w···x···
001e0 79 00 00 00 7A 00 00 00-7B 00 00 00 7C 00 00 00 y···z···{···|···
001f0 7D 00 00 00 7E 00 00 00-7F 00 00 00 80 00 00 00 }···~···········
00200 81 00 00 00 82 00 00 00-83 00 00 00 84 00 00 00 ····************
00210 85 00 00 00 86 00 00 00-87 00 00 00 88 00 00 00 ····************
00220 89 00 00 00 8A 00 00 00-FF FF FF 0F 00 00 00 00 ····ÿÿÿ·····
```

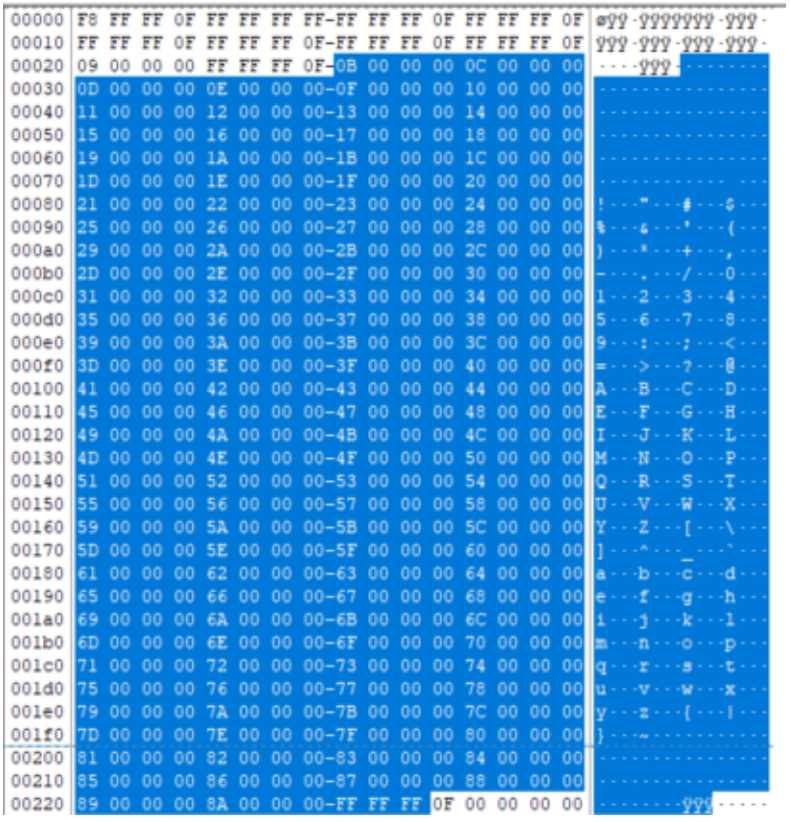
Cluster 8-9



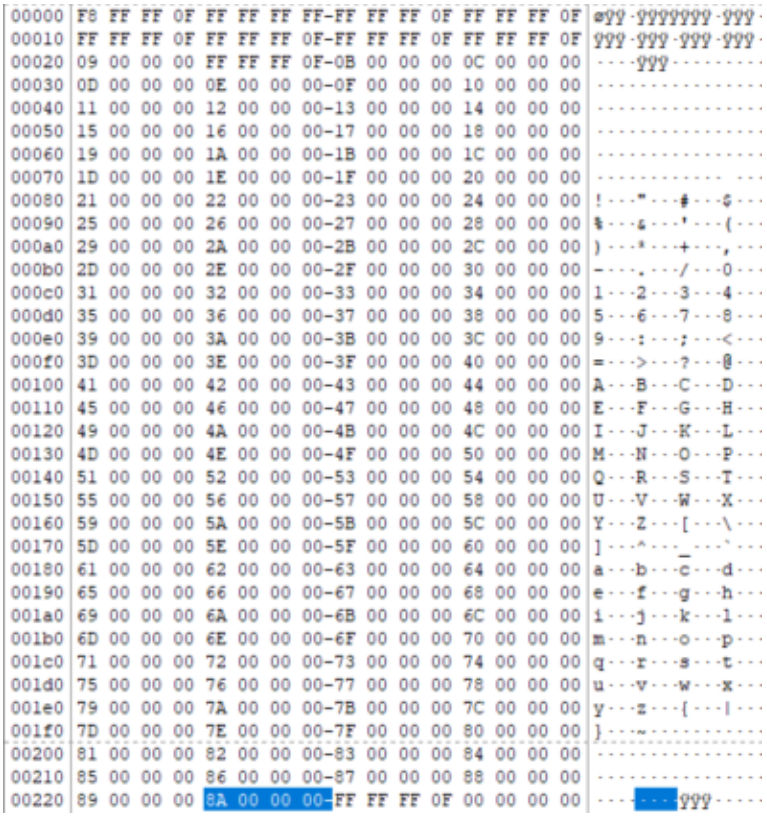
00000	F8	FF	FF	0F	FF	FF	FF	FF	FF	FF	FF	0F	FF	FF	FF	0F
00010	FF	FF	FF	0F	FF	FF	FF	0F	FF	FF	FF	0F	FF	FF	FF	0F
00020	09	00	00	00	FF	FF	FF	0F	0B	00	00	00	0C	00	00	00
00030	0D	00	00	00	0E	00	00	00	0F	00	00	00	10	00	00	00
00040	11	00	00	00	12	00	00	00	13	00	00	00	14	00	00	00
00050	15	00	00	00	16	00	00	00	17	00	00	00	18	00	00	00
00060	19	00	00	00	1A	00	00	00	1B	00	00	00	1C	00	00	00
00070	1D	00	00	00	1E	00	00	00	1F	00	00	00	20	00	00	00
00080	21	00	00	00	22	00	00	00	23	00	00	00	24	00	00	00
00090	25	00	00	00	26	00	00	00	27	00	00	00	28	00	00	00
000a0	29	00	00	00	2A	00	00	00	2B	00	00	00	2C	00	00	00
000b0	2D	00	00	00	2E	00	00	00	2F	00	00	00	30	00	00	00
000c0	31	00	00	00	32	00	00	00	33	00	00	00	34	00	00	00
000d0	35	00	00	00	36	00	00	00	37	00	00	00	38	00	00	00
000e0	39	00	00	00	3A	00	00	00	3B	00	00	00	3C	00	00	00
000f0	3D	00	00	00	3E	00	00	00	3F	00	00	00	40	00	00	00
00100	41	00	00	00	42	00	00	00	43	00	00	00	44	00	00	00
00110	45	00	00	00	46	00	00	00	47	00	00	00	48	00	00	00
00120	49	00	00	00	4A	00	00	00	4B	00	00	00	4C	00	00	00
00130	4D	00	00	00	4E	00	00	00	4F	00	00	00	50	00	00	00
00140	51	00	00	00	52	00	00	00	53	00	00	00	54	00	00	00
00150	55	00	00	00	56	00	00	00	57	00	00	00	58	00	00	00
00160	59	00	00	00	5A	00	00	00	5B	00	00	00	5C	00	00	00
00170	5D	00	00	00	5E	00	00	00	5F	00	00	00	60	00	00	00
00180	61	00	00	00	62	00	00	00	63	00	00	00	64	00	00	00
00190	65	00	00	00	66	00	00	00	67	00	00	00	68	00	00	00
001a0	69	00	00	00	6A	00	00	00	6B	00	00	00	6C	00	00	00
001b0	6D	00	00	00	6E	00	00	00	6F	00	00	00	70	00	00	00
001c0	71	00	00	00	72	00	00	00	73	00	00	00	74	00	00	00
001d0	75	00	00	00	76	00	00	00	77	00	00	00	78	00	00	00
001e0	79	00	00	00	7A	00	00	00	7B	00	00	00	7C	00	00	00
001f0	7D	00	00	00	7E	00	00	00	7F	00	00	00	80	00	00	00
00200	81	00	00	00	82	00	00	00	83	00	00	00	84	00	00	00
00210	85	00	00	00	86	00	00	00	87	00	00	00	88	00	00	00
00220	89	00	00	00	8A	00	00	00	FF	FF	FF	0F	00	00	00	00

Sequence to find LC

After cluster 10, find the End-of-File Marker (EOF). It usually ends with a FF FF FF.



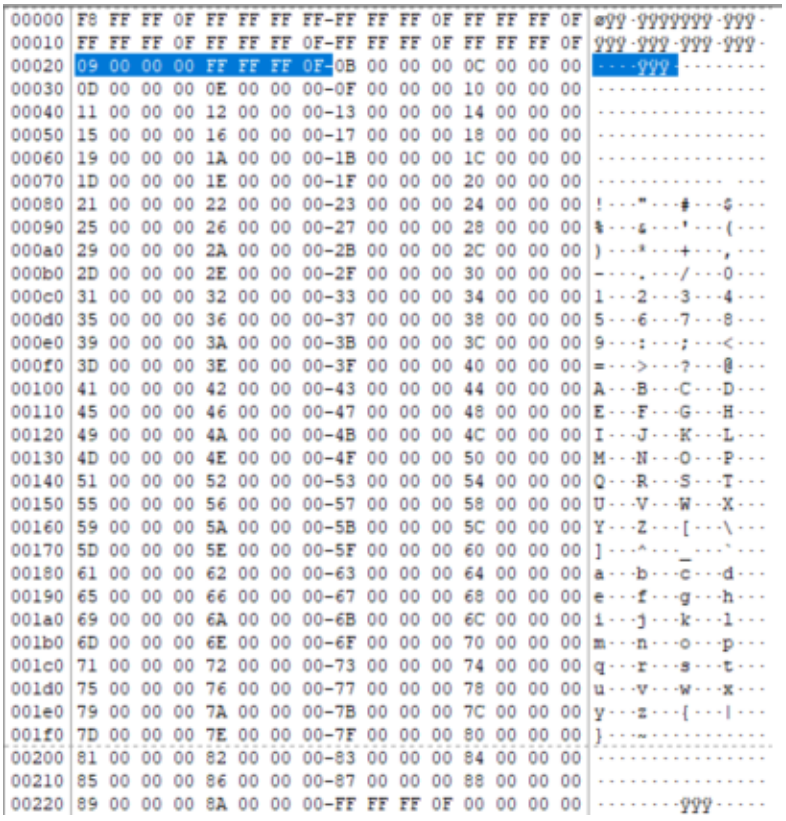
Last Cluster - 4 bytes before EOF



8A 00 00 00 in LE - 0x0000008A

LOGO.PNG

> FAT1 TABLE.

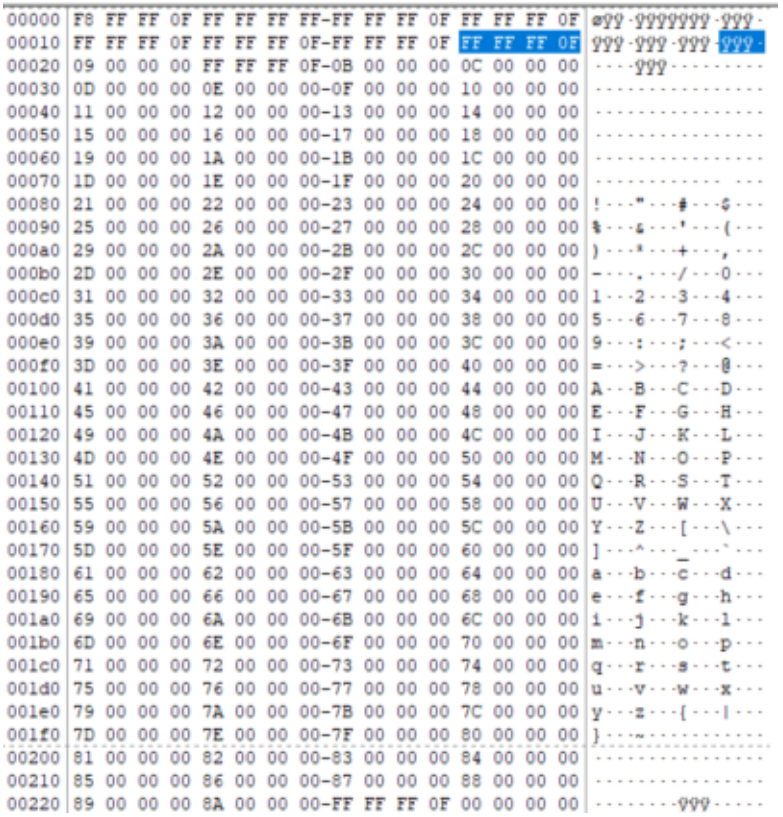


00000	F8	FF	FF	0F	FF	FF	FF	FF	FF	FF	FF	0F	FF	FF	FF	0F
00010	FF	FF	FF	0F	FF	FF	FF	0F	FF	FF	FF	0F	FF	FF	FF	0F
00020	09	00	00	00	FF	FF	FF	0F	0B	00	00	00	0C	00	00	00
00030	0D	00	00	00	0E	00	00	00	0F	00	00	00	10	00	00	00
00040	11	00	00	00	12	00	00	00	13	00	00	00	14	00	00	00
00050	15	00	00	00	16	00	00	00	17	00	00	00	18	00	00	00
00060	19	00	00	00	1A	00	00	00	1B	00	00	00	1C	00	00	00
00070	1D	00	00	00	1E	00	00	00	1F	00	00	00	20	00	00	00
00080	21	00	00	00	22	00	00	00	23	00	00	00	24	00	00	00
00090	25	00	00	00	26	00	00	00	27	00	00	00	28	00	00	00
000a0	29	00	00	00	2A	00	00	00	2B	00	00	00	2C	00	00	00
000b0	2D	00	00	00	2E	00	00	00	2F	00	00	00	30	00	00	00
000c0	31	00	00	00	32	00	00	00	33	00	00	00	34	00	00	00
000d0	35	00	00	00	36	00	00	00	37	00	00	00	38	00	00	00
000e0	39	00	00	00	3A	00	00	00	3B	00	00	00	3C	00	00	00
000f0	3D	00	00	00	3E	00	00	00	3F	00	00	00	40	00	00	00
00100	41	00	00	00	42	00	00	00	43	00	00	00	44	00	00	00
00110	45	00	00	00	46	00	00	00	47	00	00	00	48	00	00	00
00120	49	00	00	00	4A	00	00	00	4B	00	00	00	4C	00	00	00
00130	4D	00	00	00	4E	00	00	00	4F	00	00	00	50	00	00	00
00140	51	00	00	00	52	00	00	00	53	00	00	00	54	00	00	00
00150	55	00	00	00	56	00	00	00	57	00	00	00	58	00	00	00
00160	59	00	00	00	5A	00	00	00	5B	00	00	00	5C	00	00	00
00170	5D	00	00	00	5E	00	00	00	5F	00	00	00	60	00	00	00
00180	61	00	00	00	62	00	00	00	63	00	00	00	64	00	00	00
00190	65	00	00	00	66	00	00	00	67	00	00	00	68	00	00	00
001a0	69	00	00	00	6A	00	00	00	6B	00	00	00	6C	00	00	00
001b0	6D	00	00	00	6E	00	00	00	6F	00	00	00	70	00	00	00
001c0	71	00	00	00	72	00	00	00	73	00	00	00	74	00	00	00
001d0	75	00	00	00	76	00	00	00	77	00	00	00	78	00	00	00
001e0	79	00	00	00	7A	00	00	00	7B	00	00	00	7C	00	00	00
001f0	7D	00	00	00	7E	00	00	00	7F	00	00	00	80	00	00	00
00200	81	00	00	00	82	00	00	00	83	00	00	00	84	00	00	00
00210	85	00	00	00	86	00	00	00	87	00	00	00	88	00	00	00
00220	89	00	00	00	8A	00	00	00	FF	FF	FF	0F	00	00	00	00

Cluster 8 (09 00 00 00) den EOF Marker (FF FF FF 0F).

0x00000009 - Last Cluster for LOGO.PNG.

README.TXT



```
00000 F8 FF FF 0F FF FF FF FF-FF FF FF 0F FF FF FF 0F øÿÿ·ÿÿÿÿÿÿ·ÿÿÿ·
00010 FF FF FF 0F FF FF FF FF-0F FF FF 0F FF FF FF 0F ÿÿÿ·ÿÿÿ·ÿÿÿ·ÿÿÿ·
00020 09 00 00 00 FF FF FF 0F-0B 00 00 00 0C 00 00 00 ····ÿÿÿ·········
00030 0D 00 00 00 0E 00 00 00-0F 00 00 00 10 00 00 00 ····************
00040 11 00 00 00 12 00 00 00-13 00 00 00 14 00 00 00 ····************
00050 15 00 00 00 16 00 00 00-17 00 00 00 18 00 00 00 ····************
00060 19 00 00 00 1A 00 00 00-1B 00 00 00 1C 00 00 00 ····************
00070 1D 00 00 00 1E 00 00 00-1F 00 00 00 20 00 00 00 ············ ···
00080 21 00 00 00 22 00 00 00-23 00 00 00 24 00 00 00 !···"···#···$···
00090 25 00 00 00 26 00 00 00-27 00 00 00 28 00 00 00 %···&···'···(···
000a0 29 00 00 00 2A 00 00 00-2B 00 00 00 2C 00 00 00 )···*···+···,···
000b0 2D 00 00 00 2E 00 00 00-2F 00 00 00 30 00 00 00 -···.···/···0···
000c0 31 00 00 00 32 00 00 00-33 00 00 00 34 00 00 00 1···2···3···4···
000d0 35 00 00 00 36 00 00 00-37 00 00 00 38 00 00 00 5···6···7···8···
000e0 39 00 00 00 3A 00 00 00-3B 00 00 00 3C 00 00 00 9···:···;···<···
000f0 3D 00 00 00 3E 00 00 00-3F 00 00 00 40 00 00 00 =···>···?···@···
00100 41 00 00 00 42 00 00 00-43 00 00 00 44 00 00 00 A···B···C···D···
00110 45 00 00 00 46 00 00 00-47 00 00 00 48 00 00 00 E···F···G···H···
00120 49 00 00 00 4A 00 00 00-4B 00 00 00 4C 00 00 00 I···J···K···L···
00130 4D 00 00 00 4E 00 00 00-4F 00 00 00 50 00 00 00 M···N···O···P···
00140 51 00 00 00 52 00 00 00-53 00 00 00 54 00 00 00 Q···R···S···T···
00150 55 00 00 00 56 00 00 00-57 00 00 00 58 00 00 00 U···V···W···X···
00160 59 00 00 00 5A 00 00 00-5B 00 00 00 5C 00 00 00 Y···Z···[···\···
00170 5D 00 00 00 5E 00 00 00-5F 00 00 00 60 00 00 00 ]···^···_···`···
00180 61 00 00 00 62 00 00 00-63 00 00 00 64 00 00 00 a···b···c···d···
00190 65 00 00 00 66 00 00 00-67 00 00 00 68 00 00 00 e···f···g···h···
001a0 69 00 00 00 6A 00 00 00-6B 00 00 00 6C 00 00 00 i···j···k···l···
001b0 6D 00 00 00 6E 00 00 00-6F 00 00 00 70 00 00 00 m···n···o···p···
001c0 71 00 00 00 72 00 00 00-73 00 00 00 74 00 00 00 q···r···s···t···
001d0 75 00 00 00 76 00 00 00-77 00 00 00 78 00 00 00 u···v···w···x···
001e0 79 00 00 00 7A 00 00 00-7B 00 00 00 7C 00 00 00 y···z···{···|···
001f0 7D 00 00 00 7E 00 00 00-7F 00 00 00 80 00 00 00 }···~···········
00200 81 00 00 00 82 00 00 00-83 00 00 00 84 00 00 00 ····************
00210 85 00 00 00 86 00 00 00-87 00 00 00 88 00 00 00 ····************
00220 89 00 00 00 8A 00 00 00-FF FF FF 0F 00 00 00 00 ········ÿÿÿ·····
```

This file is a VERY SPECIAL FILE. As you notice cluster 7, already has the EOC (End-of-Chain) marker. The file chain stops IMMEDIATELY. It only occupies one cluster. Hence it’s last cluster is the same as the start cluster.

0x00000007 - LAST CLUSTER